Malware Analysis and Prevention Strategy

Final project



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Tools and Techniques Showcase



WannaCry



Wazuh



Malware Analysis Lab

رواد مصر الرقمية

وزارة الاتصالات وتكنولوجيا المعلومات

1. Executive Summary

A combined static and dynamic analysis was conducted on a 64-bit Windows executable identified as a modified variant of the WannaCry ransomware. The analysis was performed within an isolated VMware Workstation environment on 2021-04-28 and re-evaluated in 2026.

Unlike the original WannaCry outbreak of May 2017 — which propagated autonomously via the EternalBlue SMB exploit (MS17-010) and demanded ransom via Bitcoin — this variant presents a Chinese-language graphical user interface (GUI) and instructs victims to contact the threat actor directly at the QQ email address 3905960807@qq.com, demanding a ransom of USD $300. The absence of self-propagation mechanisms in the observed execution confirms this variant was delivered and executed manually.

Proof Of Concept Malware Analysis

https://youtube.com/playlist?list=PLO1VSSKnwZUgGaiDZXU-mKuh8CUZx-gAd&si=stv5G8pJOGF5HWrc

1.1 Key Findings

- The sample is a PE32+ (64-bit) executable, unobfuscated, and compiled with Microsoft Visual C/C++ 14.44.
- Windows Cryptography APIs (CryptEncrypt, CryptAcquireContextW, CryptCreateHash) are imported and executed to perform file encryption.
- The malware spawns @WanaDecryptor@.exe as a persistent GUI process to display the ransom note and countdown timer.
- The desktop wallpaper is programmatically replaced via SystemParametersInfoW to present the ransom message upon reboot.
- A child process (cmd.exe) is launched to interact with Windows Registry keys related to Internet Settings cache and notification data.
- A temporary artifact (ICO5452.tmp) is dropped in %APPDATA%\Local\Temp\.
- No kill-switch domain query was observed during the dynamic analysis session. This may be due to the variant being patched or the network simulation configuration.
- No evidence of SMB lateral movement was captured; the sample appears designed for targeted, manual delivery rather than worm-style propagation.

1.2 Risk Assessment Summary

Category	Assessment
Overall Threat Level	CRITICAL
Confidentiality Impact	High — user files encrypted and rendered inaccessible
Integrity Impact	High — file system and registry modified; wallpaper altered
Availability Impact	Critical — files encrypted; system effectively held hostage
Propagation Risk	Medium — no worm propagation observed; manual delivery assumed
Attribution	Unconfirmed — Chinese-language GUI and QQ contact suggest regional targeting

2. Sample Information

Property	Value
File Name	b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe
File Type	PE32+ executable (GUI) — x86-64 (AMD64)
File Size	0x00218800 bytes (2,197,504 bytes / ~2.10 MB)
File signature	Microsoft Linker 14.44
MD5	70920f342814ca5c662cbbed00d60eef
SHA1	3e96e1f6a2d7cc6b72473d141e1962260325a0e7
SHA256	b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf
entropy	2.910 (Not Packed)
Compile Timestamp	0x69A97D6E (raw TimeDateStamp — interpreted value not verified)
Compiler	Microsoft Visual C/C++ 14.44 (2025 toolchain, per DIE)
Packer / Obfuscation	None detected — entropy: 2.90978 (consistent with uncompressed PE)
Analysis Environment	VMware Workstation — isolated, non-networked sandbox
Source / Submission	Analyzed locally; VirusTotal and Hybrid-Analysis references provided
VirusTotal Reference	https://shorturl.at/Wm9Pk
Hybrid-Analysis Reference	https://shorturl.at/0vvb8
Creation Time	2026-03-05 12:56:14 UTC
First Seen In The Wild	2026-03-09 04:01:13 UTC
First Submission	2026-03-06 10:52:34 UTC
Other Names:	• WannaCry.exe • b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe • xl63jblz.exe • 2026-03-06_70920f342814ca5c662cbbed00d60eef_icedid_luca-stealer_njrat_petya_stealc_wannacry • _b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe • main_Cry.exe

Process Tree

- **main_Cry.exe** (2068)
  - **@WanaDecryptor@.exe** (5440)
- **services.exe** (668)
  - **svchost.exe** (800) -k DcomLaunch -p
    - **dllhost.exe** (872) /Processid:{AB8902B4-09CA-4BB6-B78D-A8F59079A8D5}

Malware Family :

Tags: exe filecoder Ransom:Win32/Filecoder!MSR WannaCry

Ransomware

- Changes the desktop background picture
  Detected indicator that file is ransomware

Spyware

- Found a string that may be used as part of an injection method
  Tries to read/open stored key files

Persistence

- Writes data to a remote process

Fingerprint

- Queries kernel debugger information
  Queries sensitive IE security settings
  Queries the display settings of system associated file extensions

Evasive

- Marks file for deletion
  Reads VM-specific registry key

YARA Signatures

- DebuggerCheck__API
- DetectEncryptedVariants
- golang_bin_JCorn_CSC846
- pe_detect_tls_callbacks
- RANSOMWARE

2. 3omda Malware Analysis Playbook

Malware Analysis Playbook (Updated)

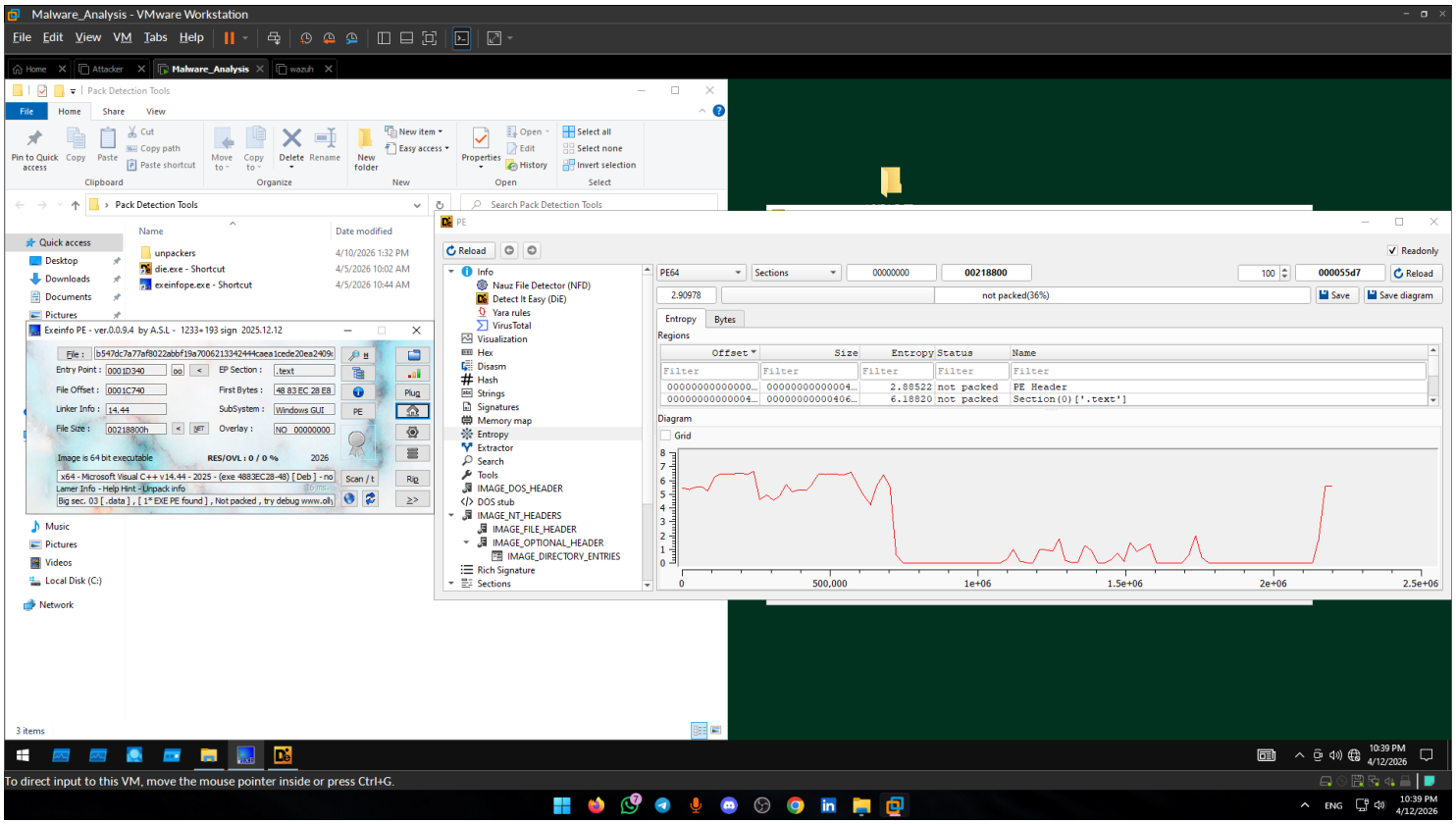
Table 1: Malware Analysis Methodology and Key Findings

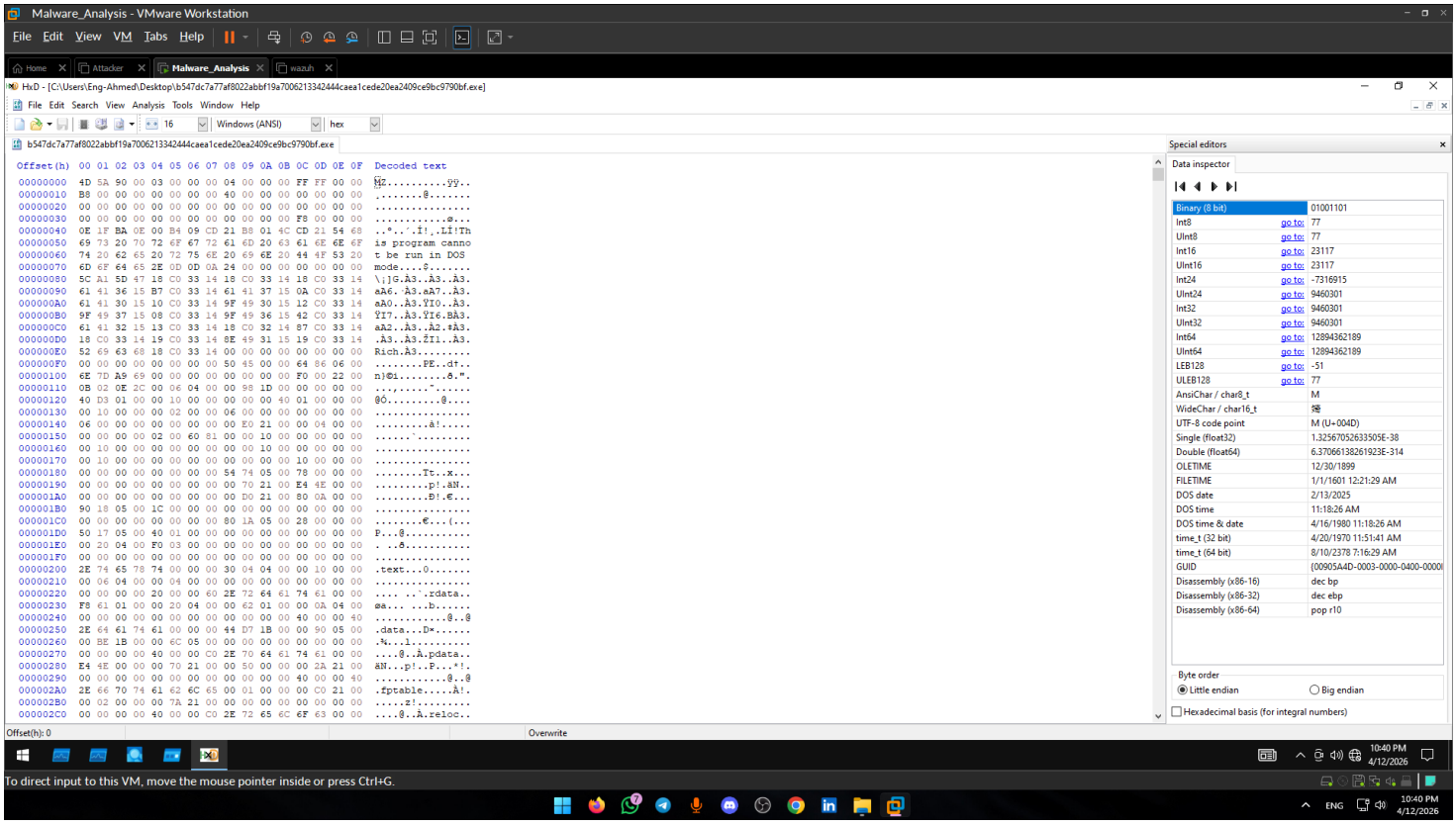
Phase	Task	Tools	Expected / Observed Key Findings
1. OSINT & Automated Sandboxing	Querying file hashes to gather threat intelligence, threat scores, and automated behavioral reports.	VirusTotal, Hybrid Analysis	Initial detection rates, community insights, network indicators, sandbox behavioral tagging.
2. String & Obfuscation Analysis	Extracting readable and obfuscated strings to identify embedded artifacts.	FLOSS, Strings	Hardcoded QQ email (3905960807@qq.com), registry paths, file artifacts (e.g., Forever_AES_KEY).
3. Basic Static Analysis & Packer Detection	Identifying architecture, compiler information, and packer status.	Detect It Easy, EXEInfo	64-bit Windows executable, not packed (Entropy: 2.90978), compiled using MSVC.
4. Hex & Raw Data Analysis	Verifying magic bytes and inspecting raw binary structure.	HxD	Valid PE structure confirmed (MZ header 4D 5A, DOS stub intact).
5. Advanced Static Analysis (PE Structure)	Examining PE headers, sections, and imported APIs.	PEStudio, CFF Explorer	Cryptographic APIs, filesystem traversal, UI manipulation (SystemParametersInfoW).
6. Dynamic Analysis & Behavioral Monitoring	Executing sample in sandbox and monitoring system, registry, and network behavior.	Process Hacker, Process Monitor, FolderChangesView, RegShot, Wireshark, FakeNet / Fake DNS	Ransomware GUI execution, file encryption activity, registry modifications, network traffic simulation, 100k+ system events.
7. Behavioral Graphing & Visualization	Reconstructing execution flow using collected telemetry.	ProcDot	Process tree mapping, cmd.exe spawning, temp file creation, registry changes.

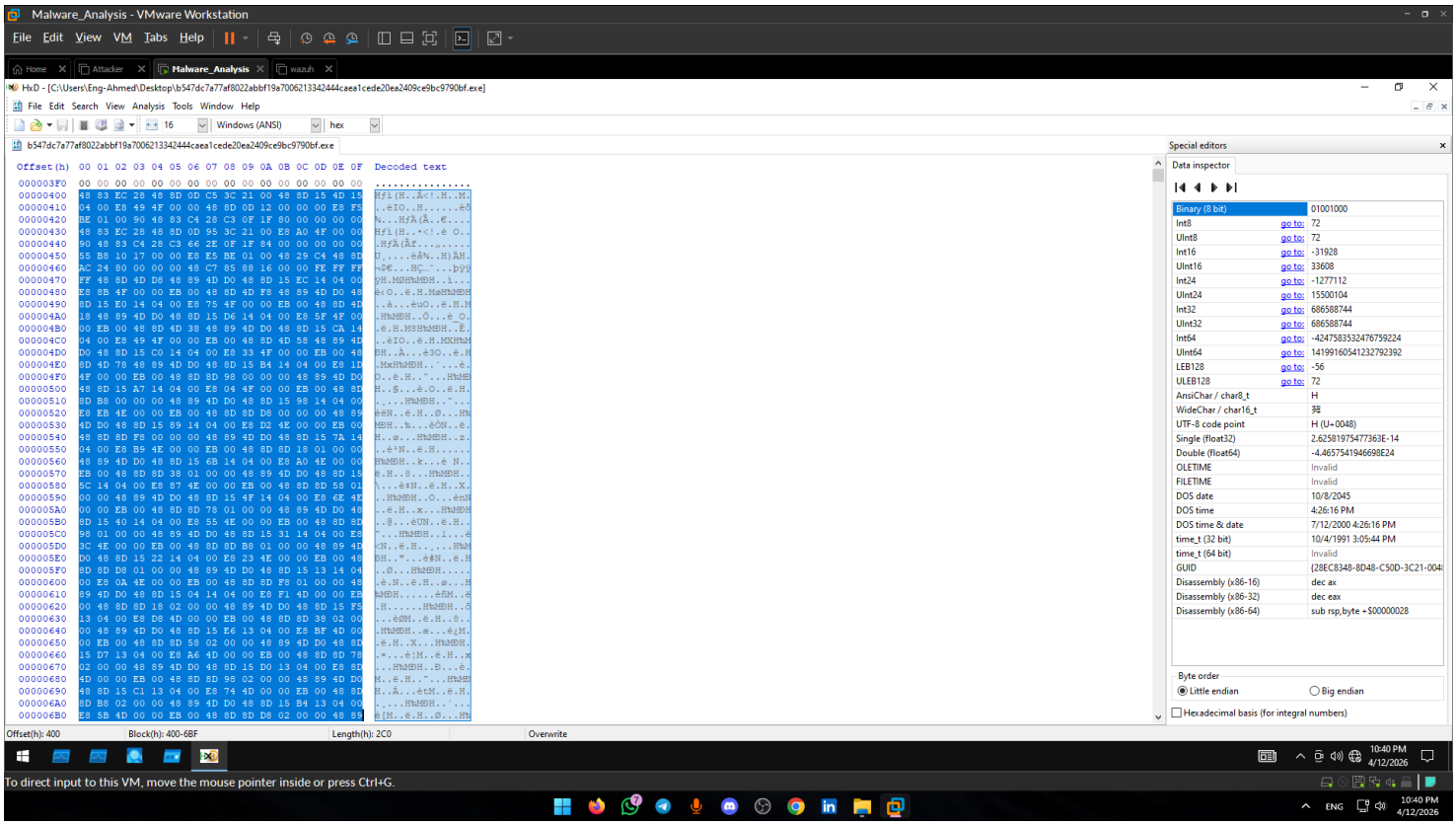
3. Static Analysis

3.1 File Structure Observations

- File format confirmed as Portable Executable (PE) — MZ magic bytes 0x5A4D at offset 0x00.
- Architecture: 64-bit (AMD64); GUI subsystem (not console).
- Low entropy (2.91) across the binary indicates the absence of packing, encryption, or heavy compression. All strings and imports are available in plaintext.
- Compiler fingerprint: Microsoft Visual C/C++ version 14.44 (2025 toolchain), as identified by Detect It Easy (DIE).
- Compile timestamp field: 0x69A97D6E. Note: PE timestamps can be forged; independent corroboration via other artefacts is advisable.



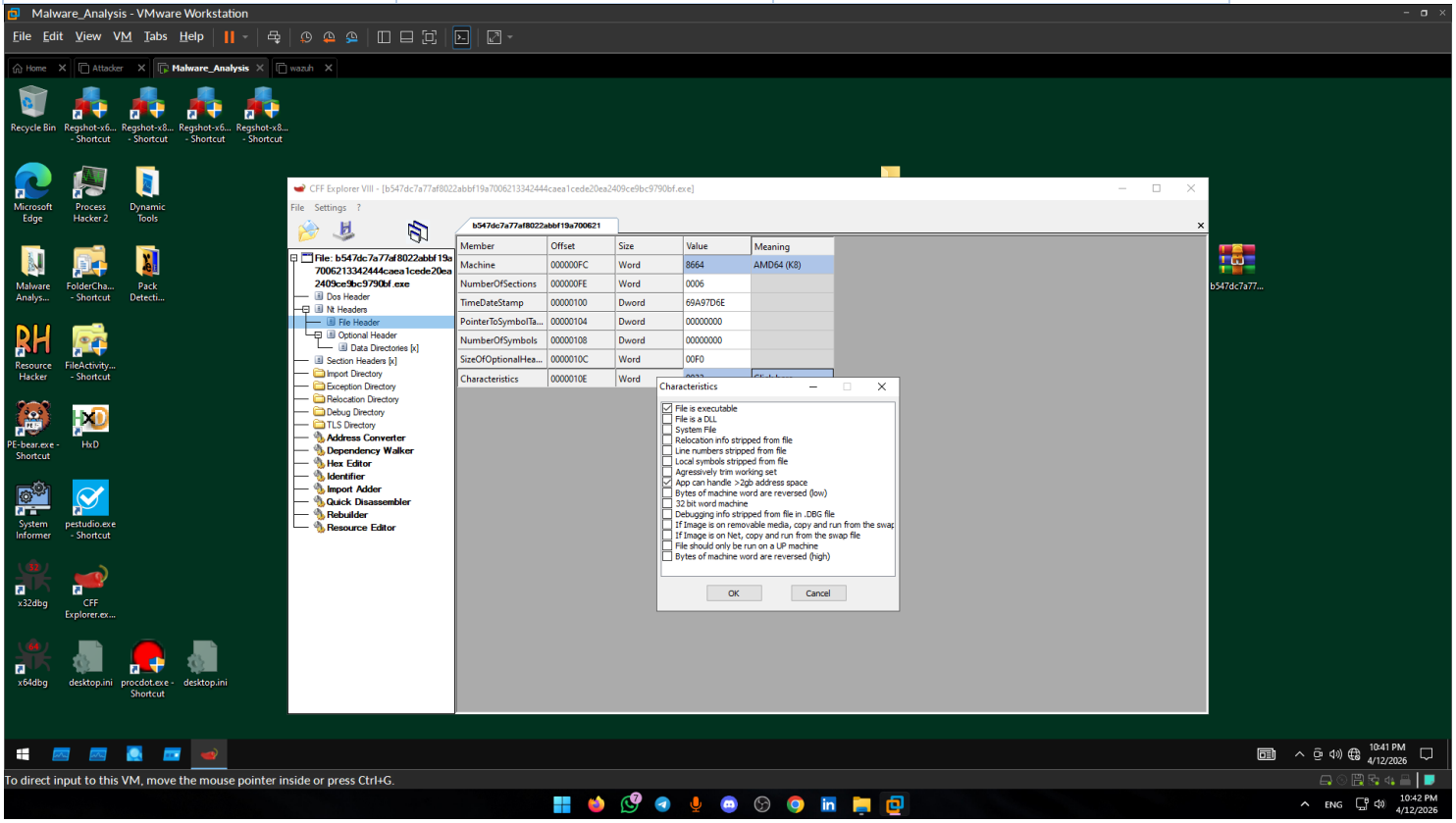

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
HxD - [C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe]
b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe
Special editors
Data inspector
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

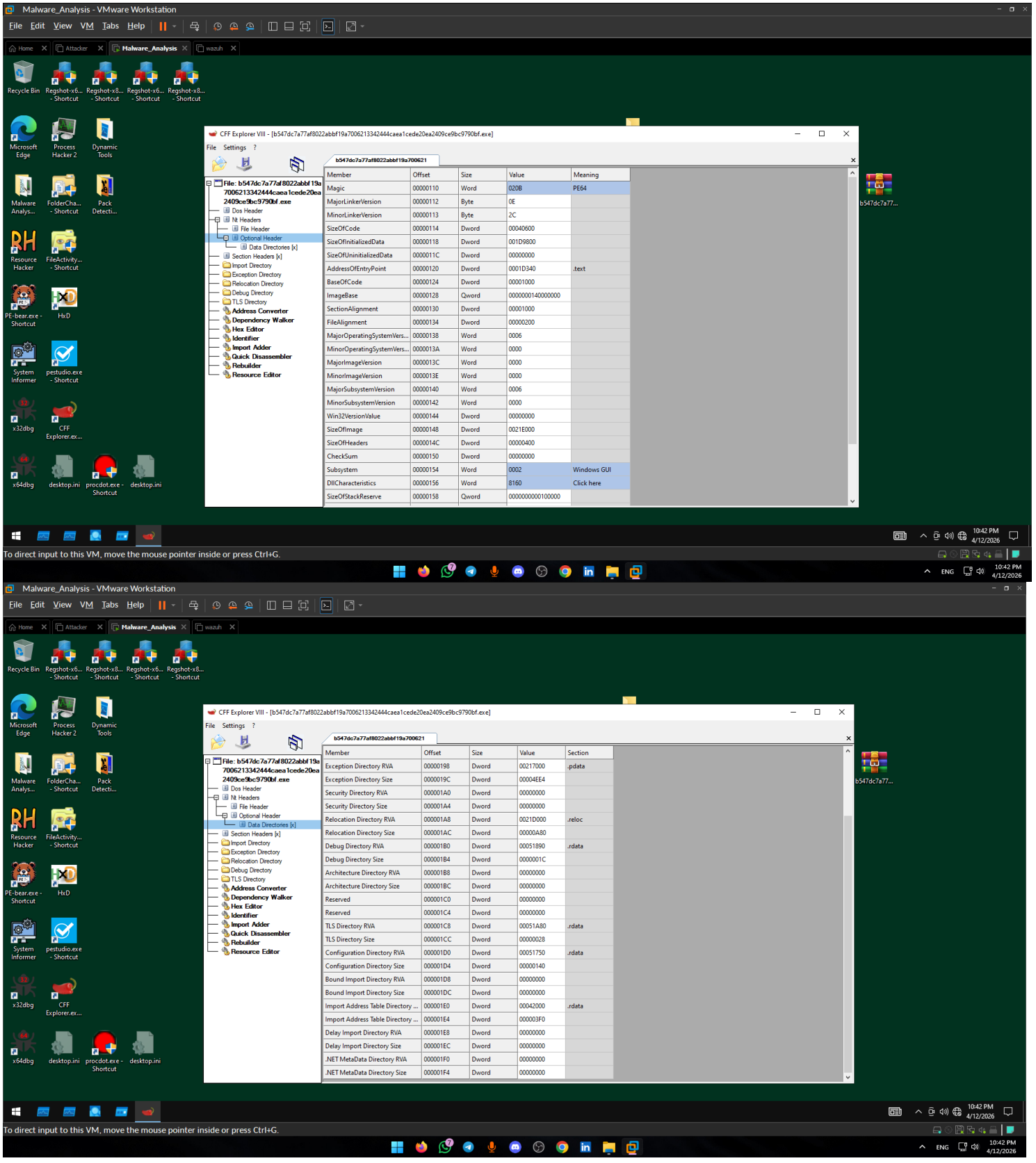

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3.2 Imported Windows APIs

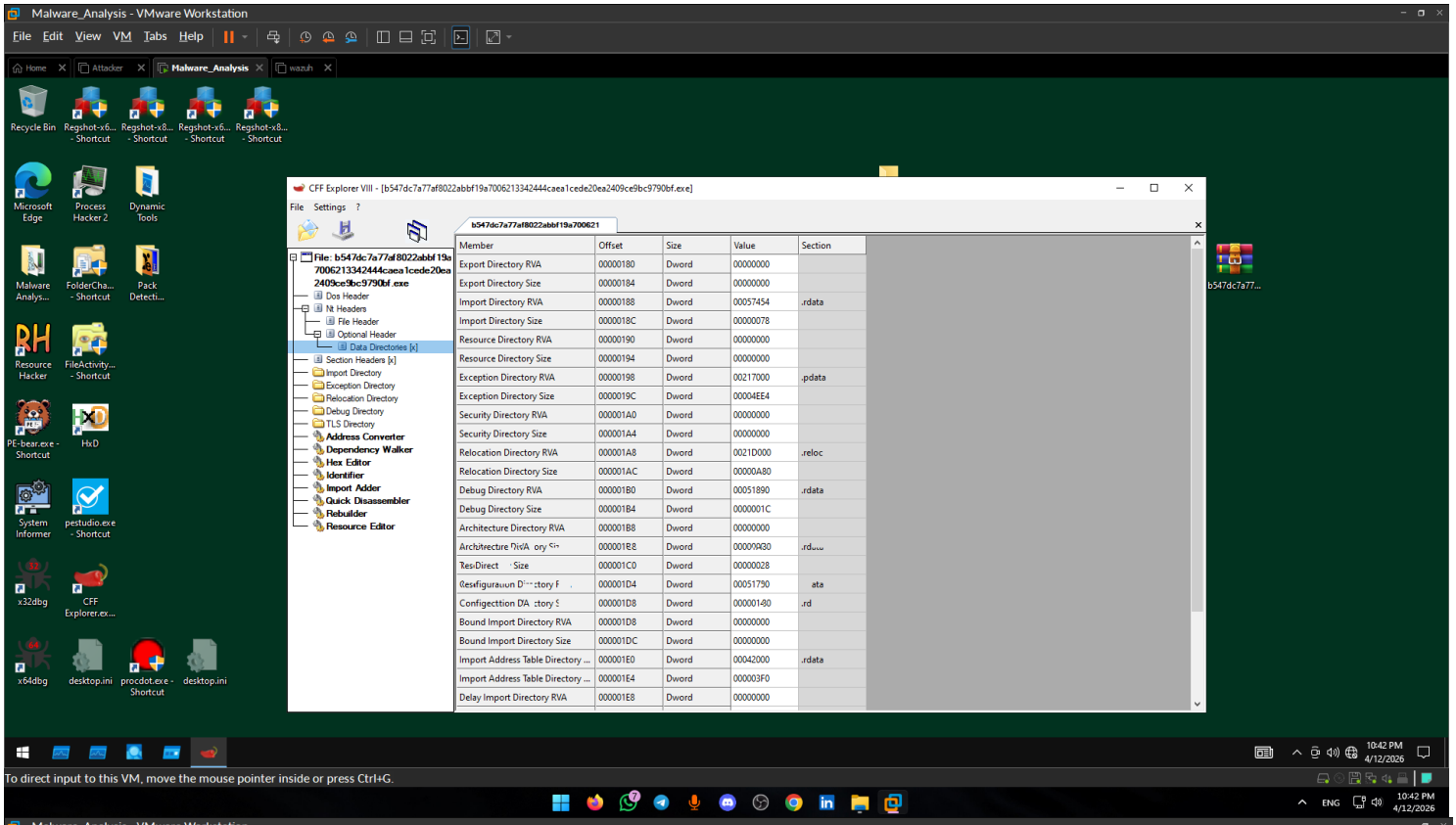
The following Windows API functions were identified in the import table. Their presence constitutes strong static indicators of ransomware behavior:

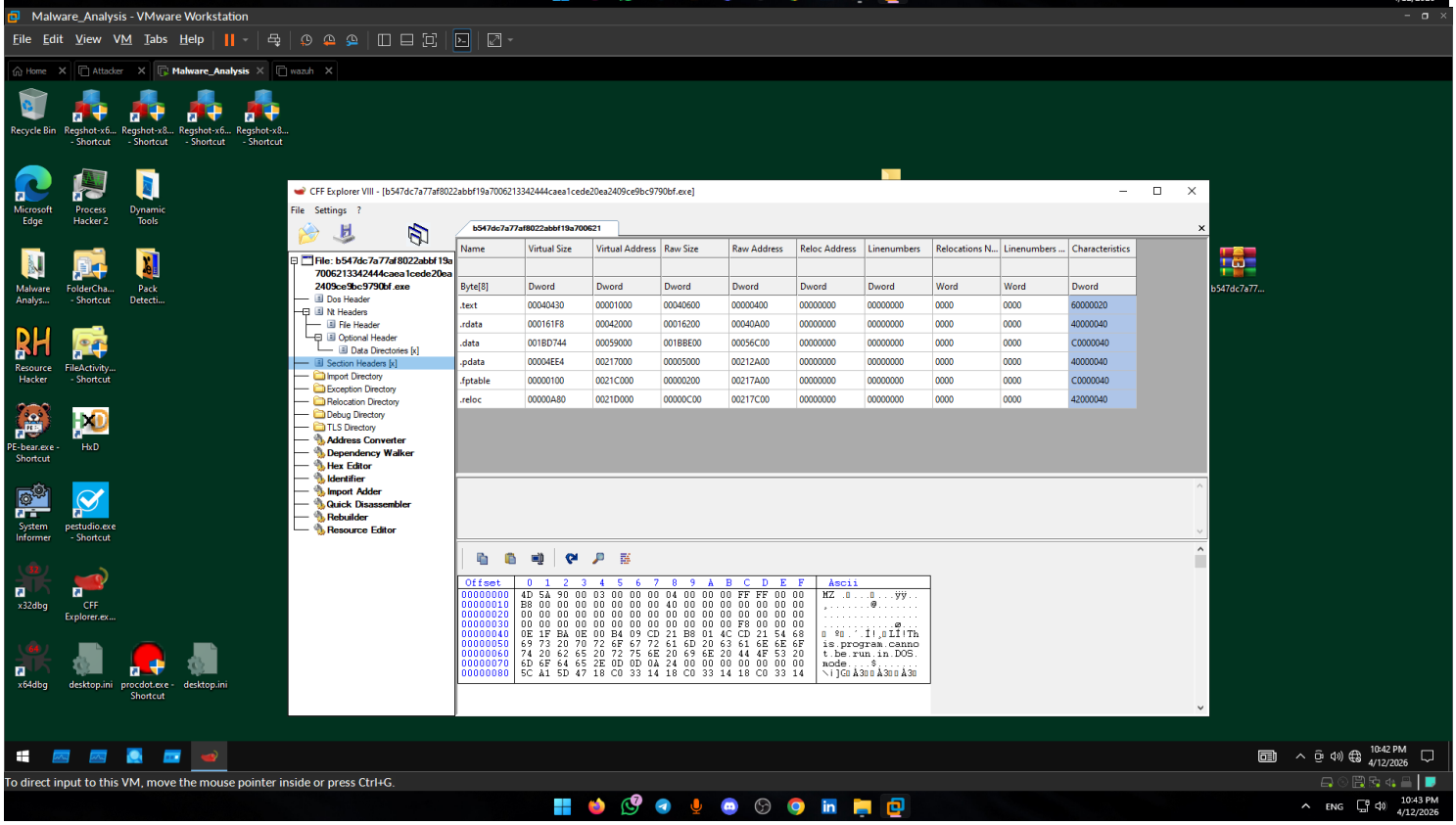
API Function	Category	Observed Purpose
CryptAcquireContextW	Cryptography (wincrypt.h)	Acquire a cryptographic service provider (CSP) context
CryptCreateHash	Cryptography (wincrypt.h)	Initialize a hashing object for key derivation
CryptEncrypt	Cryptography (wincrypt.h)	Encrypt file data blocks using the acquired CSP
CryptDecrypt	Cryptography (wincrypt.h)	Decrypt functionality (likely used by decryptor GUI)
SystemParametersInfoW	User32.dll	Modify desktop wallpaper to display ransom note
CreateFileW / WriteFile	Kernel32.dll	Create and write to files (encryption output, temp files)
CreateProcessW / ShellExecuteW	Kernel32.dll / Shell32.dll	Spawn child processes (@WanaDecryptor@.exe, cmd.exe)
RegSetValueExW / RegOpenKeyExW	Advapi32.dll	Write and query Windows Registry keys

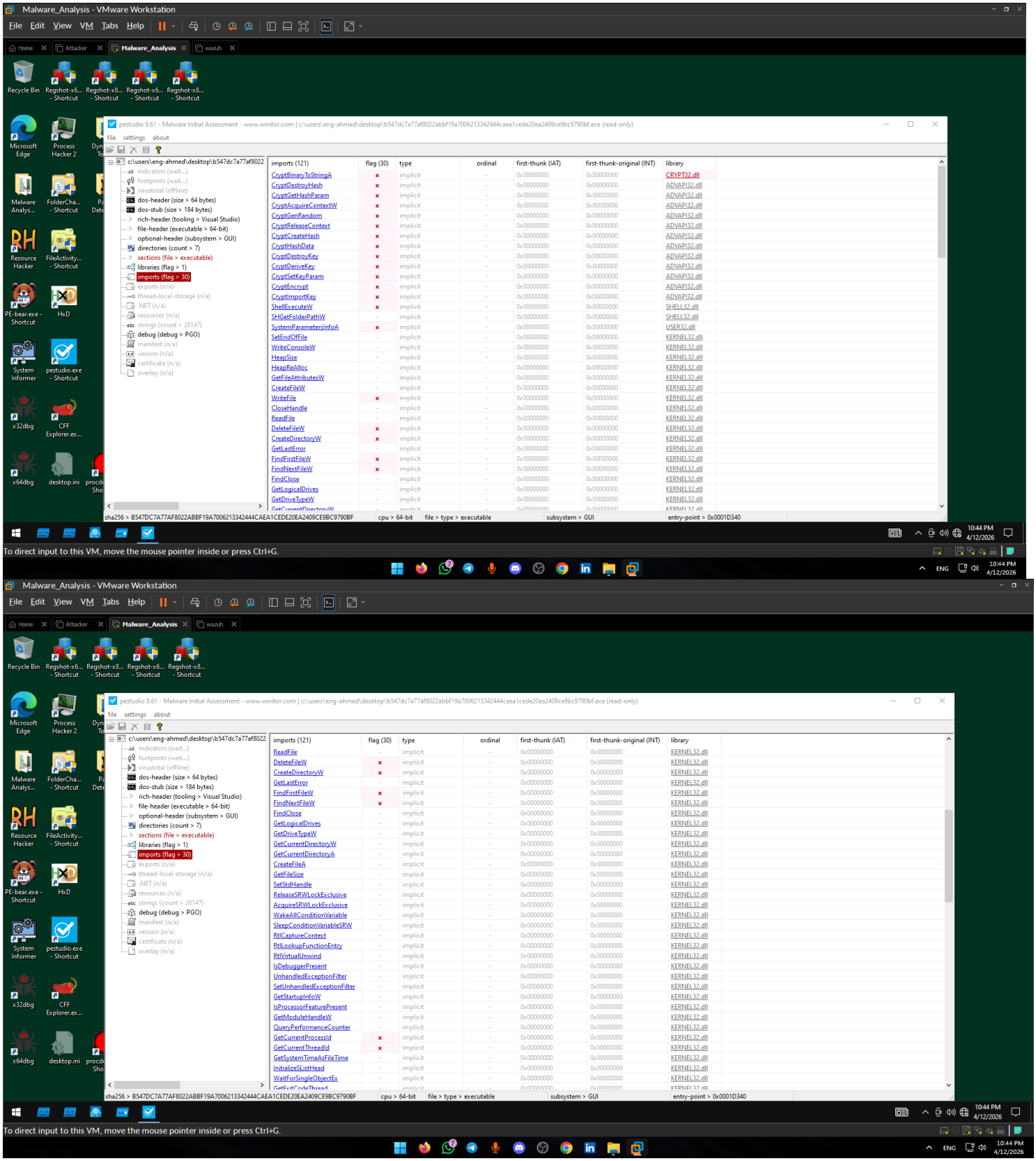



Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
CFF Explorer VIII - [b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe]
File Settings ?
b547dc7a77af8022abbf19a700621
File: b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe
Dos Header
Nt Headers
File Header
Optional Header
Data Directories [x]
Section Headers [x]
Import Directory
Exception Directory
Relocation Directory
Debug Directory
TLS Directory
Address Converter
Dependency Walker
Hex Editor
Identifier
Import Adder
Quick Disassembler
Rebuilder
Resource Editor
Member Offset Size Value Meaning
Magic 00000110 Word 020B PE64
MajorLinkerVersion 00000112 Byte 0E
MinorLinkerVersion 00000113 Byte 2C
SizeOfCode 00000114 Dword 00040600
SizeOfInitializedData 00000118 Dword 001D9800
SizeOfUninitializedData 0000011C Dword 00000000
AddressOfEntryPoint 00000120 Dword 0001D340 .text
BaseOfCode 00000124 Dword 00001000
ImageBase 00000128 Qword 0000000140000000
SectionAlignment 00000130 Dword 00001000
FileAlignment 00000134 Dword 00000200
MajorOperatingSystemVers... 00000138 Word 0006
MinorOperatingSystemVers... 0000013A Word 0000
MajorImageVersion 0000013C Word 0000
MinorImageVersion 0000013E Word 0000
MajorSubsystemVersion 00000140 Word 0006
MinorSubsystemVersion 00000142 Word 0000
Win32VersionValue 00000144 Dword 00000000
SizeOfImage 00000148 Dword 0021E000
SizeOfHeaders 0000014C Dword 00000400
CheckSum 00000150 Dword 00000000
Subsystem 00000154 Word 0002 Windows GUI
DllCharacteristics 00000156 Word 8160 Click here
SizeOfStackReserve 00000158 Qword 0000000000100000
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

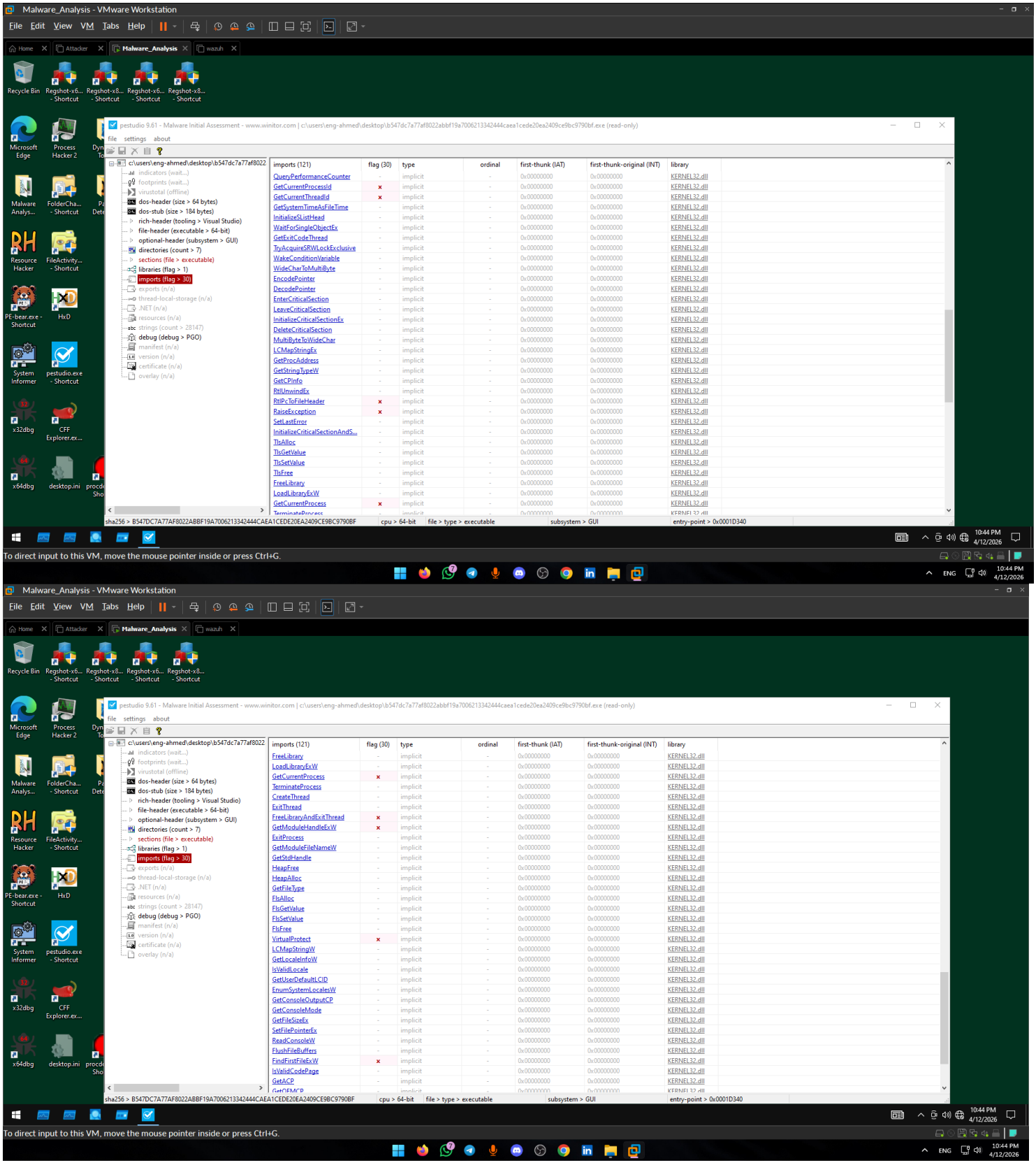
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Import Directory
Exception Directory
Relocation Directory
Debug Directory
TLS Directory
Address Converter
Dependency Walker
Hex Editor
Identifier
Import Adder
Quick Disassembler
Rebuilder
Resource Editor
Member Offset Size Value Section
Exception Directory RVA 00000198 Dword 00217000 .pdata
Exception Directory Size 0000019C Dword 00004EE4
Security Directory RVA 000001A0 Dword 00000000
Security Directory Size 000001A4 Dword 00000000
Relocation Directory RVA 000001A8 Dword 0021D000 .reloc
Relocation Directory Size 000001AC Dword 00000A80
Debug Directory RVA 000001B0 Dword 00051890 .rdata
Debug Directory Size 000001B4 Dword 0000001C
Architecture Directory RVA 000001B8 Dword 00000000
Architecture Directory Size 000001BC Dword 00000000
Reserved 000001C0 Dword 00000000
Reserved 000001C4 Dword 00000000
TLS Directory RVA 000001C8 Dword 00051A80 .rdata
TLS Directory Size 000001CC Dword 00000028
Configuration Directory RVA 000001D0 Dword 00051750 .rdata
Configuration Directory Size 000001D4 Dword 00000140
Bound Import Directory RVA 000001D8 Dword 00000000
Bound Import Directory Size 000001DC Dword 00000000
Import Address Table Directory ... 000001E0 Dword 00042000 .rdata
Import Address Table Directory ... 000001E4 Dword 000003F0
Delay Import Directory RVA 000001E8 Dword 00000000
Delay Import Directory Size 000001EC Dword 00000000
.NET MetaData Directory RVA 000001F0 Dword 00000000
.NET MetaData Directory Size 000001F4 Dword 00000000
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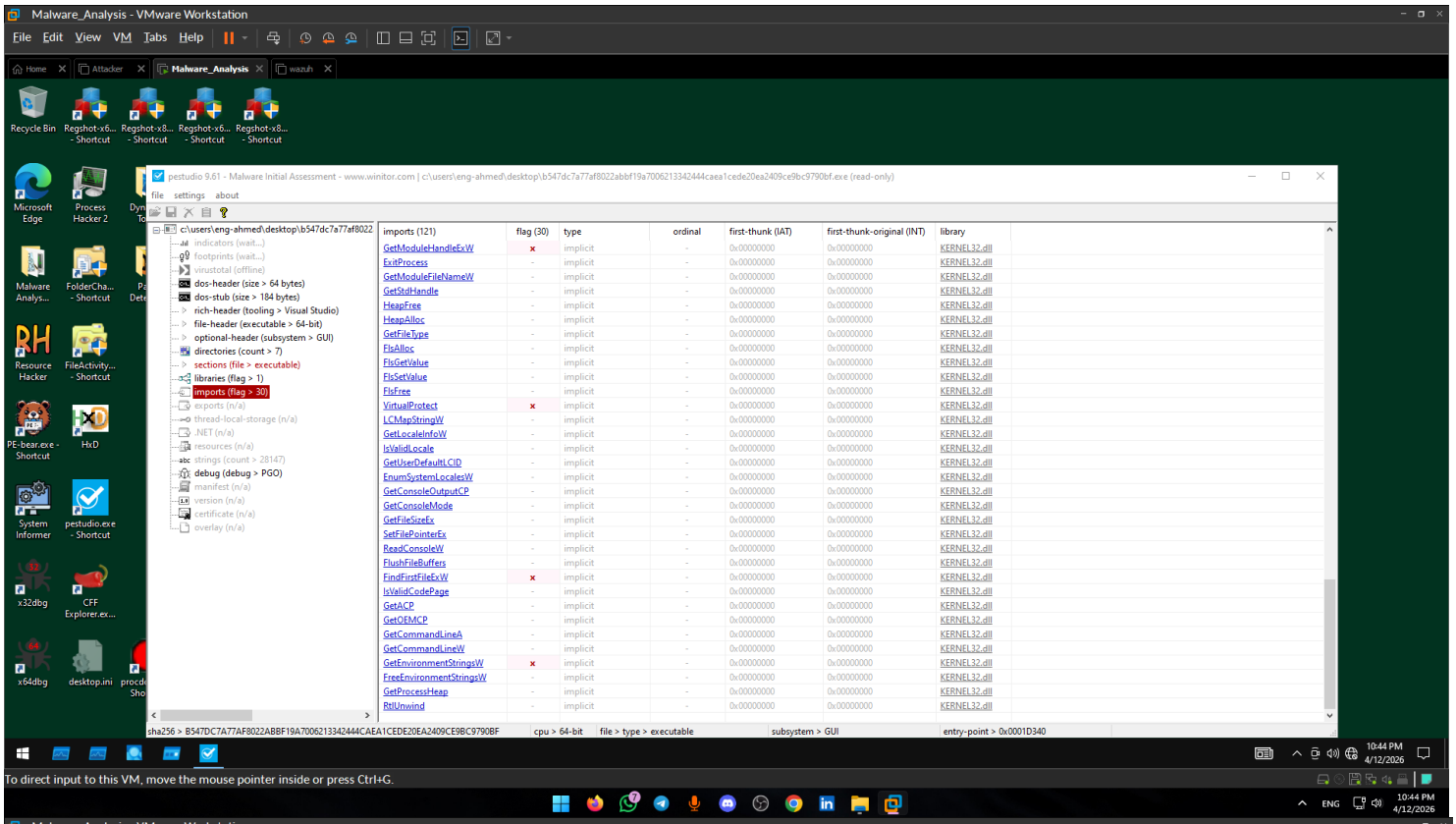

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Dependency Walker
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Quick Disassembler
Rebuilder
Resource Editor
Member Offset Size Value Section
Export Directory RVA 00000180 Dword 00000000
Export Directory Size 00000184 Dword 00000000
Import Directory RVA 00000188 Dword 00057454 .rdata
Import Directory Size 0000018C Dword 00000078
Resource Directory RVA 00000190 Dword 00000000
Resource Directory Size 00000194 Dword 00000000
Exception Directory RVA 00000198 Dword 00217000 .pdata
Exception Directory Size 0000019C Dword 00004EE4
Security Directory RVA 000001A0 Dword 00000000
Security Directory Size 000001A4 Dword 00000000
Relocation Directory RVA 000001A8 Dword 0021D000 .reloc
Relocation Directory Size 000001AC Dword 00000A80
Debug Directory RVA 000001B0 Dword 00051890 .rdata
Debug Directory Size 000001B4 Dword 0000001C
Architecture Directory RVA 000001B8 Dword 00000000
000001C0 Dword 00000028
000001D4 Dword 00051790 ata
000001D8 Dword 000001A0 .rd
Bound Import Directory RVA 000001D8 Dword 00000000
Bound Import Directory Size 000001DC Dword 00000000
Import Address Table Directory ... 000001E0 Dword 00042000 .rdata
Import Address Table Directory ... 000001E4 Dword 000003F0
Delay Import Directory RVA 000001E8 Dword 00000000
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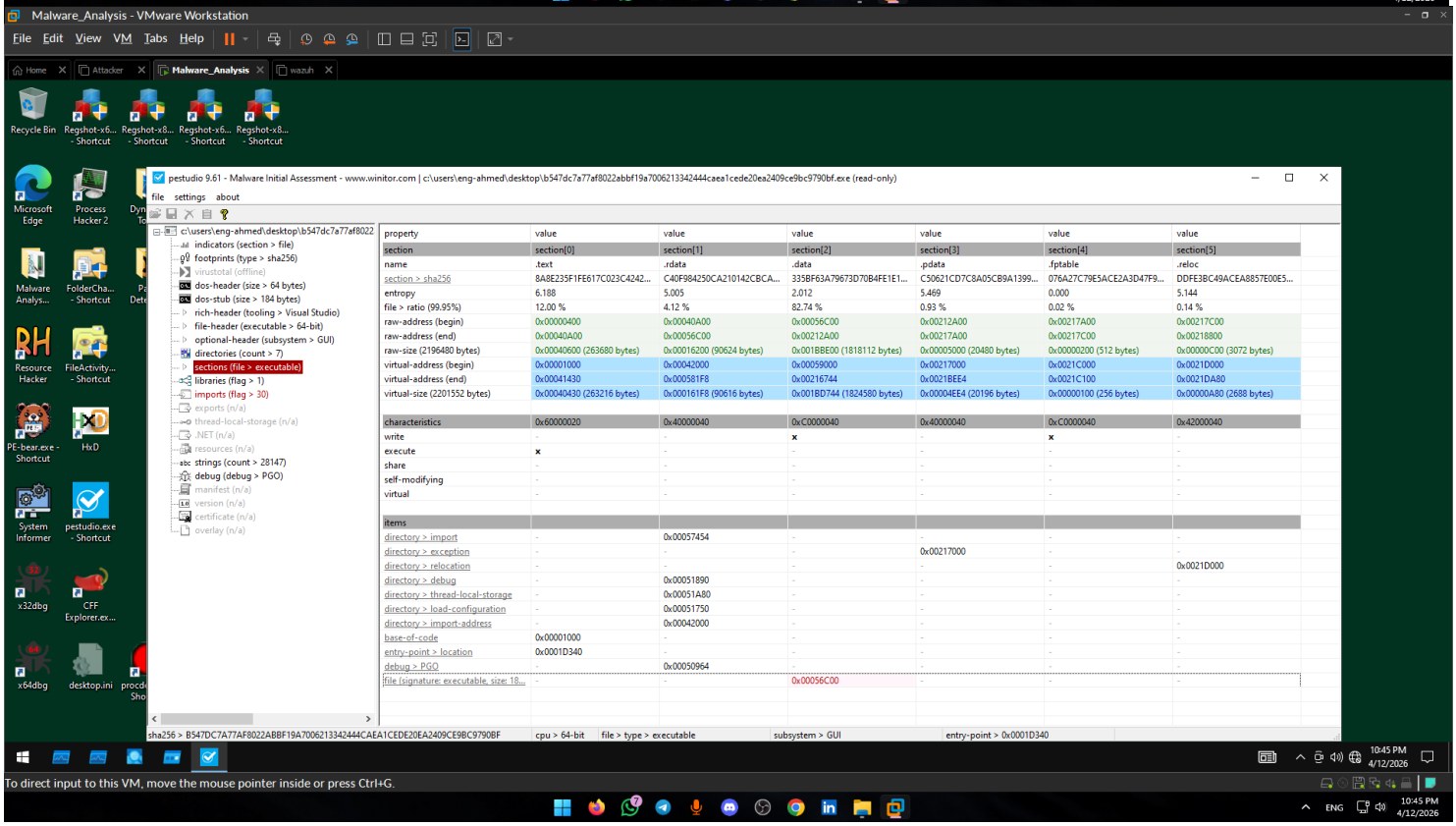

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TLS Directory
Address Converter
Dependency Walker
Hex Editor
Identifier
Import Adder
Quick Disassembler
Rebuilder
Resource Editor
Name Virtual Size Virtual Address Raw Size Raw Address Reloc Address Linenumbers Relocations N... Linenumbers ... Characteristics
Byte[8] Dword Dword Dword Dword Dword Dword Word Word Dword
.text 00040430 00001000 00040600 00000400 00000000 00000000 0000 0000 60000020
.rdata 000161F8 00042000 00016200 00040A00 00000000 00000000 0000 0000 40000040
.data 001BD744 00059000 001BBE00 00056C00 00000000 00000000 0000 0000 C0000040
.pdata 00004EE4 00217000 00005000 00212A00 00000000 00000000 0000 0000 40000040
.fptable 00000100 0021C000 00000200 00217A00 00000000 00000000 0000 0000 C0000040
.reloc 00000A80 0021D000 00000C00 00217C00 00000000 00000000 0000 0000 42000040
Offset 0 1 2 3 4 5 6 7 8 9 A B C D E F Ascii
00000000 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ..........ÿÿ..
00000010 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 ....@.......
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ....
00000030 00 00 00 00 00 00 00 00 00 00 00 00 F8 00 00 00 ....ø...
00000040 0E 1F BA 0E 00 B4 09 CD 21 B8 01 4C CD 21 54 68 ..º..´.Í!¸.LÍ!Th
00000050 69 73 20 70 72 6F 67 72 61 6D 20 63 61 6E 6E 6F is program canno
00000060 74 20 62 65 20 72 75 6E 20 69 6E 20 44 4F 53 20 t be run in DOS
00000070 6D 6F 64 65 2E 0D 0D 0A 24 00 00 00 00 00 00 00 mode....$.......
00000080 5C A1 5D 47 18 C0 33 14 18 C0 33 14 18 C0 33 14 \¡]G.À3..À3..À3.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe (read-only)
file settings about
imports (121) flag (30) type ordinal first-thunk (IAT) first-thunk-original (INT) library
sha256 > B547DC7A77AF8022ABBF19A7006213342444CAEA1CEDE20EA2409CE9BC9790BF
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x0001D340
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe (read-only)
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file settings about
imports (121) flag (30) type ordinal first-thunk (IAT) first-thunk-original (INT) library
QueryPerformanceCounter implicit KERNEL32.dll
GetCurrentProcessId x implicit KERNEL32.dll
GetCurrentThreadId x implicit KERNEL32.dll
GetSystemTimeAsFileTime implicit KERNEL32.dll
InitializeSListHead implicit KERNEL32.dll
WaitForSingleObjectEx implicit KERNEL32.dll
GetExitCodeThread implicit KERNEL32.dll
TryAcquireSRWLockExclusive implicit KERNEL32.dll
WakeConditionVariable implicit KERNEL32.dll
WideCharToMultiByte implicit KERNEL32.dll
EncodePointer implicit KERNEL32.dll
DecodePointer implicit KERNEL32.dll
EnterCriticalSection implicit KERNEL32.dll
LeaveCriticalSection implicit KERNEL32.dll
InitializeCriticalSectionEx implicit KERNEL32.dll
DeleteCriticalSection implicit KERNEL32.dll
MultiByteToWideChar implicit KERNEL32.dll
LCMapStringEx implicit KERNEL32.dll
GetProcAddress implicit KERNEL32.dll
GetStringTypeW implicit KERNEL32.dll
GetCPInfo implicit KERNEL32.dll
RtlUnwindEx implicit KERNEL32.dll
RtlPcToFileHeader x implicit KERNEL32.dll
RaiseException x implicit KERNEL32.dll
SetLastError implicit KERNEL32.dll
InitializeCriticalSectionAndS... implicit KERNEL32.dll
TlsAlloc implicit KERNEL32.dll
TlsGetValue implicit KERNEL32.dll
TlsSetValue implicit KERNEL32.dll
TlsFree implicit KERNEL32.dll
FreeLibrary implicit KERNEL32.dll
LoadLibraryExW implicit KERNEL32.dll
GetCurrentProcess x implicit KERNEL32.dll
TerminateProcess implicit KERNEL32.dll
CreateThread implicit KERNEL32.dll
ExitThread implicit KERNEL32.dll
FreeLibraryAndExitThread x implicit KERNEL32.dll
GetModuleHandleExW x implicit KERNEL32.dll
ExitProcess implicit KERNEL32.dll
GetModuleFileNameW implicit KERNEL32.dll
GetStdHandle implicit KERNEL32.dll
HeapFree implicit KERNEL32.dll
HeapAlloc implicit KERNEL32.dll
GetFileType implicit KERNEL32.dll
FlsAlloc implicit KERNEL32.dll
FlsGetValue implicit KERNEL32.dll
FlsSetValue implicit KERNEL32.dll
FlsFree implicit KERNEL32.dll
VirtualProtect x implicit KERNEL32.dll
LCMapStringW implicit KERNEL32.dll
GetLocaleInfoW implicit KERNEL32.dll
IsValidLocale implicit KERNEL32.dll
GetUserDefaultLCID implicit KERNEL32.dll
EnumSystemLocalesW implicit KERNEL32.dll
GetConsoleOutputCP implicit KERNEL32.dll
GetConsoleMode implicit KERNEL32.dll
GetFileSizeEx implicit KERNEL32.dll
SetFilePointerEx implicit KERNEL32.dll
ReadConsoleW implicit KERNEL32.dll
FlushFileBuffers implicit KERNEL32.dll
FindFirstFileExW x implicit KERNEL32.dll
IsValidCodePage implicit KERNEL32.dll
GetACP implicit KERNEL32.dll
sha256 > B547DC7A77AF8022ABBF19A7006213342444CAEA1CEDE20EA2409CE9BC9790BF
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x0001D340
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Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe (read-only)
file settings about
imports (121) | flag (30) | type | ordinal | first-thunk (IAT) | first-thunk-original (INT) | library
GetModuleHandleExW | x | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
ExitProcess | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetModuleFileNameW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetStdHandle | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
HeapFree | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
HeapAlloc | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetFileType | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FlsAlloc | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FlsGetValue | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FlsSetValue | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FlsFree | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
VirtualProtect | x | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
LCMapStringW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetLocaleInfoW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
IsValidLocale | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetUserDefaultLCID | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
EnumSystemLocalesW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetConsoleOutputCP | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetConsoleMode | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetFileSizeEx | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
SetFilePointerEx | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
ReadConsoleW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FlushFileBuffers | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FindFirstFileExW | x | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
IsValidCodePage | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetACP | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetOEMCP | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetCommandLineA | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetCommandLineW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetEnvironmentStringsW | x | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
FreeEnvironmentStringsW | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
GetProcessHeap | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
RtlUnwind | - | implicit | - | 0x00000000 | 0x00000000 | KERNEL32.dll
sha256 > B547DC7A77AF8022ABBF19A7006213342444CAEA1CEDE20EA2409CE9BC9790BF
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x0001D340
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe (read-only)
file settings about
property | value | value | value | value | value | value
section | section[0] | section[1] | section[2] | section[3] | section[4] | section[5]
name | .text | .rdata | .data | .pdata | .fptable | .reloc
section > sha256 | 8A8E235F1FE617C023C4242... | C40F984250CA210142CBCA... | 335BF63A79673D70B4FE1E1... | C50621CD7C8A05CB9A1399... | 076A27C79E5ACE2A3D47F9... | DDFE3BC49ACEA8857E00E5...
entropy | 6.188 | 5.005 | 2.012 | 5.469 | 0.000 | 5.144
file > ratio (99.95%) | 12.00 % | 4.12 % | 82.74 % | 0.93 % | 0.02 % | 0.14 %
raw-address (begin) | 0x00000400 | 0x00040A00 | 0x00056C00 | 0x00212A00 | 0x00217A00 | 0x00217C00
raw-address (end) | 0x00040A00 | 0x00056C00 | 0x00212A00 | 0x00217A00 | 0x00217C00 | 0x00218800
raw-size (2196480 bytes) | 0x00040600 (263680 bytes) | 0x00016200 (90624 bytes) | 0x001BBE00 (1818112 bytes) | 0x00005000 (20480 bytes) | 0x00000200 (512 bytes) | 0x00000C00 (3072 bytes)
virtual-address (begin) | 0x00001000 | 0x00042000 | 0x00059000 | 0x00217000 | 0x0021C000 | 0x0021D000
virtual-address (end) | 0x00041430 | 0x000581F8 | 0x00216744 | 0x0021BEE4 | 0x0021C100 | 0x0021DA80
virtual-size (2201552 bytes) | 0x00040430 (263216 bytes) | 0x00161F8 (90616 bytes) | 0x001BD744 (1824580 bytes) | 0x00004EE4 (20196 bytes) | 0x00000100 (256 bytes) | 0x00000A80 (2688 bytes)
characteristics | 0x60000020 | 0x40000040 | 0xC0000040 | 0x40000040 | 0xC0000040 | 0x42000040
write | - | - | x | - | x | -
execute | x | - | - | - | - | -
share | - | - | - | - | - | -
self-modifying | - | - | - | - | - | -
virtual | - | - | - | - | - | -
items
directory > import | - | 0x00057454 | - | - | - | -
directory > exception | - | - | - | 0x00217000 | - | -
directory > relocation | - | - | - | - | - | 0x0021D000
directory > debug | - | 0x00051890 | - | - | - | -
directory > thread-local-storage | - | 0x00051A80 | - | - | - | -
directory > load-configuration | - | 0x00051750 | - | - | - | -
directory > import-address | - | 0x00042000 | - | - | - | -
base-of-code | 0x00001000 | - | - | - | - | -
entry-point > location | 0x0001D340 | - | - | - | - | -
debug > PGO | - | 0x00050964 | - | - | - | -
file (signature: executable, size: 18... | - | - | 0x00056C00 | - | - | -
sha256 > B547DC7A77AF8022ABBF19A7006213342444CAEA1CEDE20EA2409CE9BC9790BF
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x0001D340
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3.3 Strings Analysis

Significant plaintext strings extracted from the binary (via static analysis) include:

String	Significance
@WanaDecryptor@	Name of the ransom GUI executable spawned by the dropper
3905960807@qq.com	Threat actor contact email — primary IOC
Forever_AES_KEY	AES key storage directory / artifact name — primary IOC
ICO5452.tmp	Temporary file dropped during execution
Bilibili	Reference to Chinese video platform — supports regional targeting hypothesis
CryptEncrypt / CryptDecrypt	Cryptographic API string references visible in IAT
AppData\Local\Temp	Target path for temporary artifact drop

3.4 FLOSS / Obfuscated String Analysis

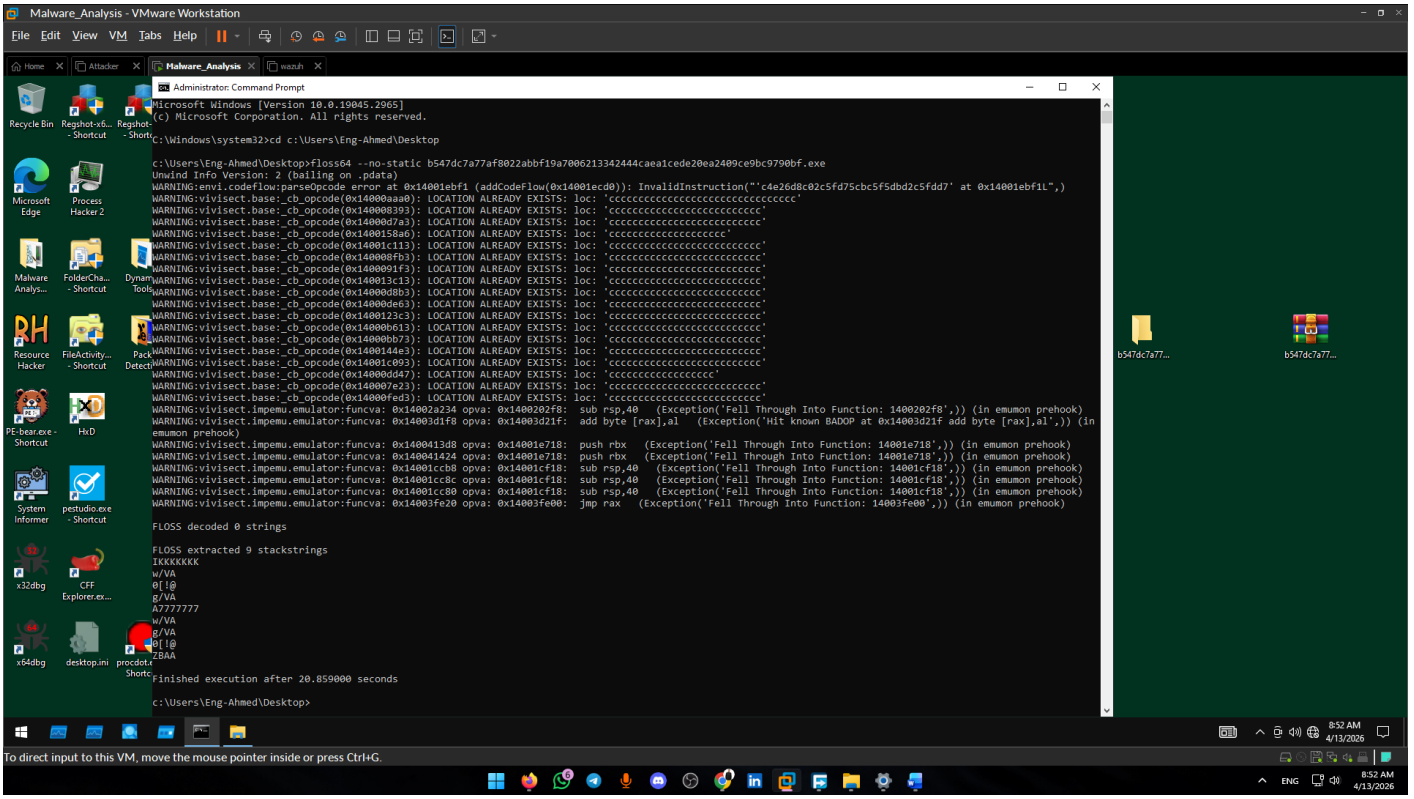
The FLOSS (FireEye Labs Obfuscated String Solver) tool was executed with the --no-static flag to target dynamically constructed strings. The analysis completed in approximately 20.8 seconds.

- Decoded obfuscated strings: 0 (none identified).
- Stack strings extracted: 9.
- The 9 extracted stack strings consisted entirely of short, non-actionable character sequences (e.g., "1KKKKKKK", "A7777777", "W/VA"), which are likely alignment padding or compiler artefacts rather than intentionally obfuscated content.

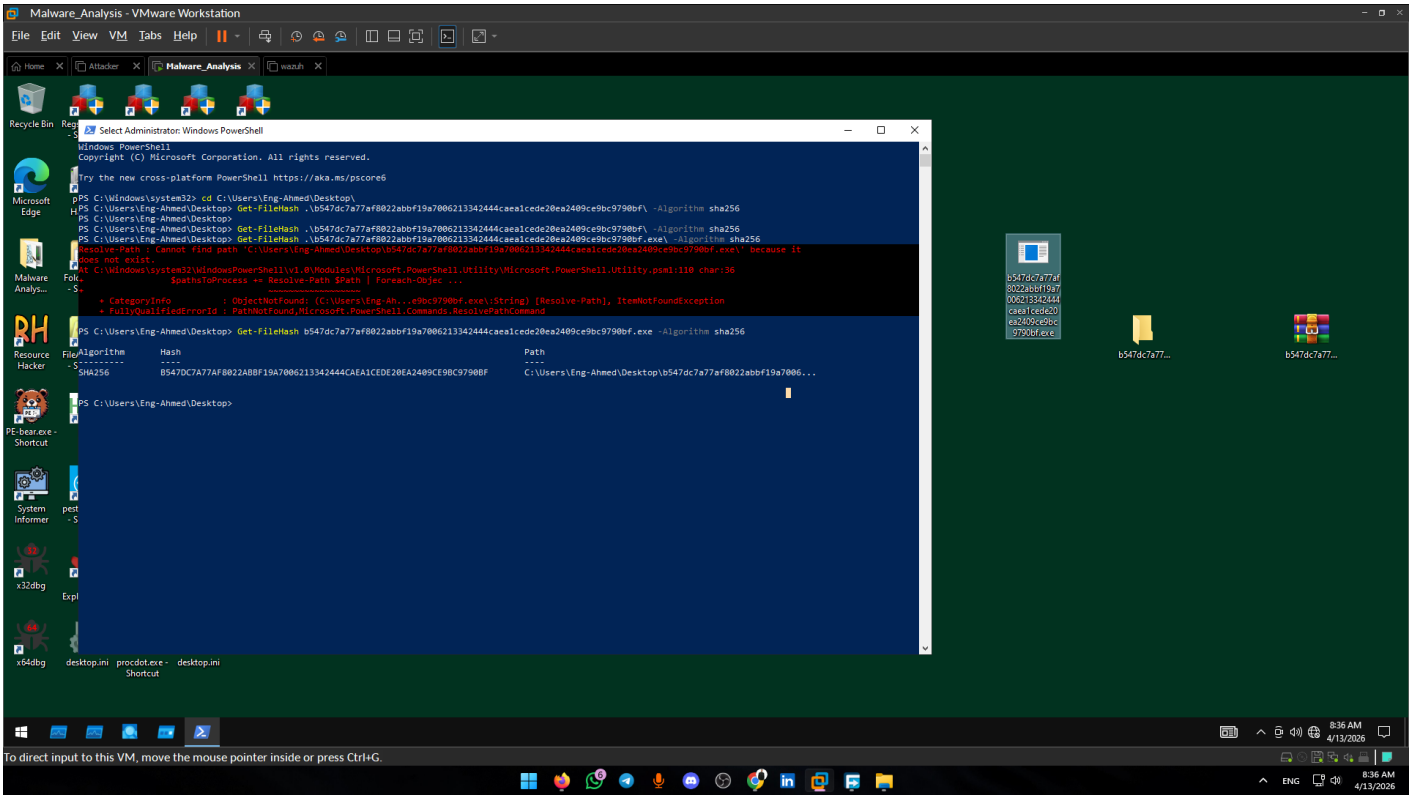
> *Analyst Interpretation: The absence of obfuscated strings is consistent with the low entropy observed across the binary. This sample does not appear to employ runtime string decryption or self-modifying code, which simplifies detection via static YARA signatures.*

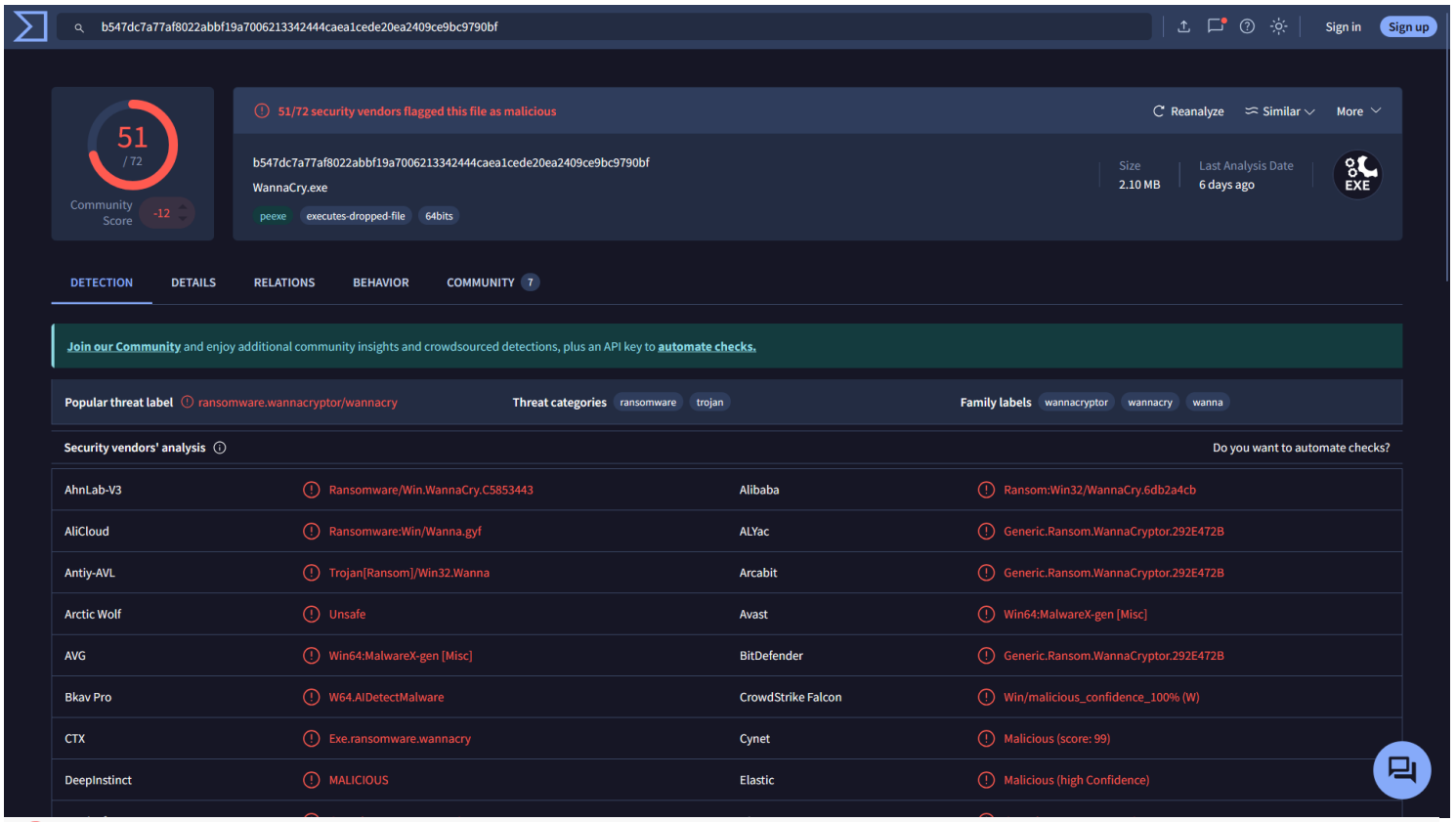
3.5 Packing / Obfuscation Assessment

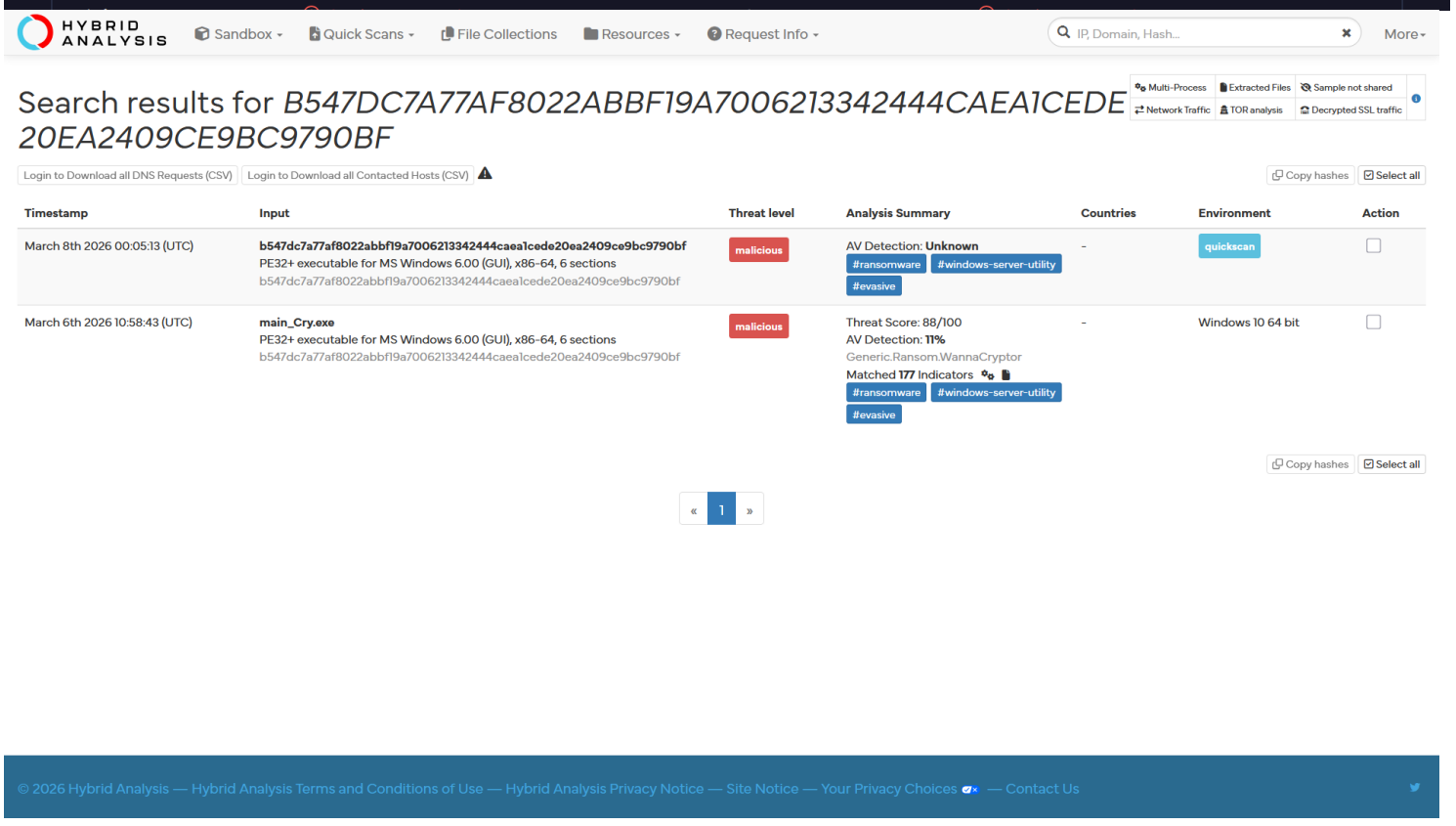
- Packing: Not detected.
- Entropy: 2.91 (well below the threshold of ~6.8–7.2 that indicates packing or encryption).
- Anti-analysis techniques: No evidence of VM detection, debugger evasion, or timing-based anti-analysis in the observed static artefacts.



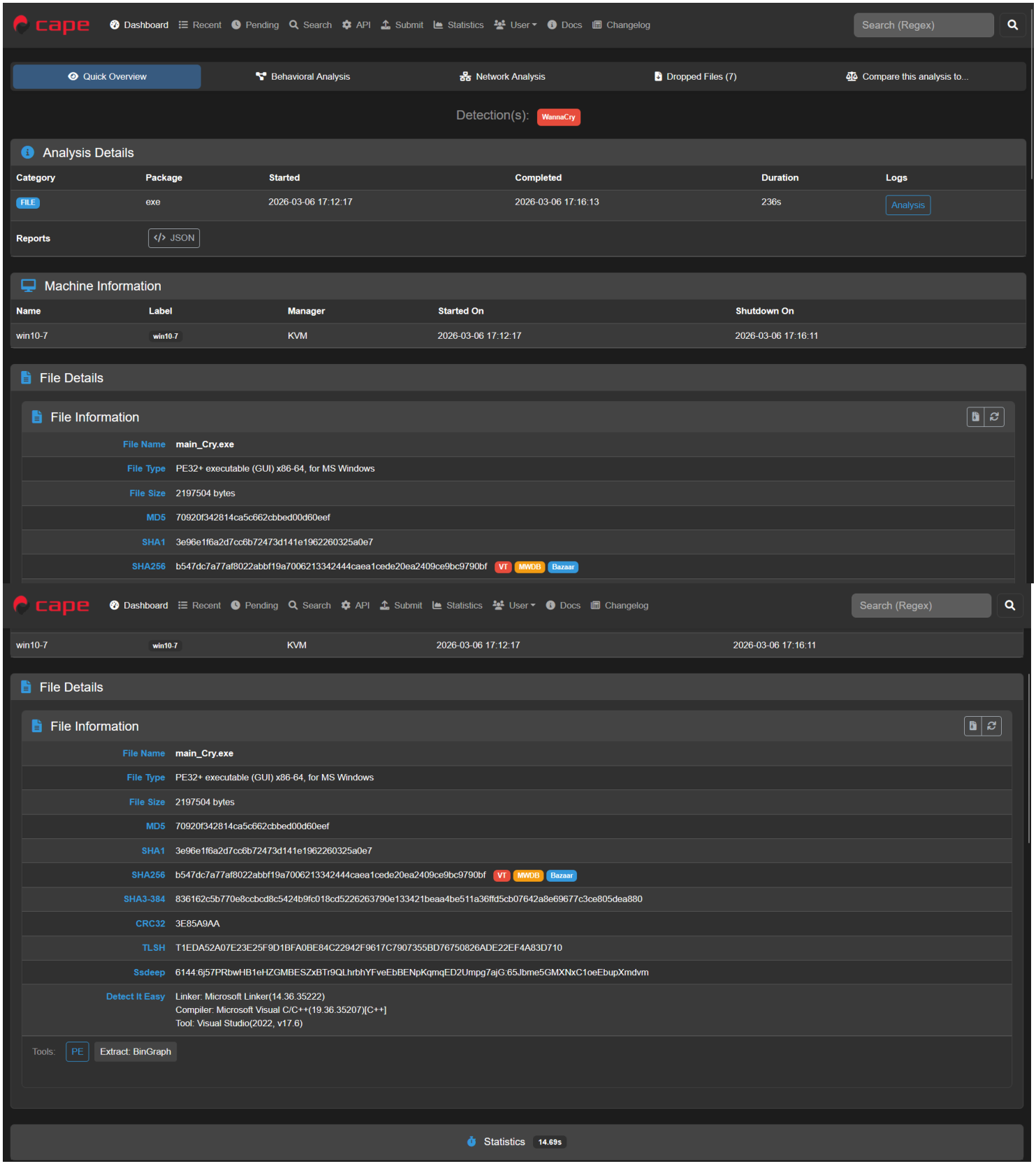
3. VirusTotal & HybridAnalysis Report

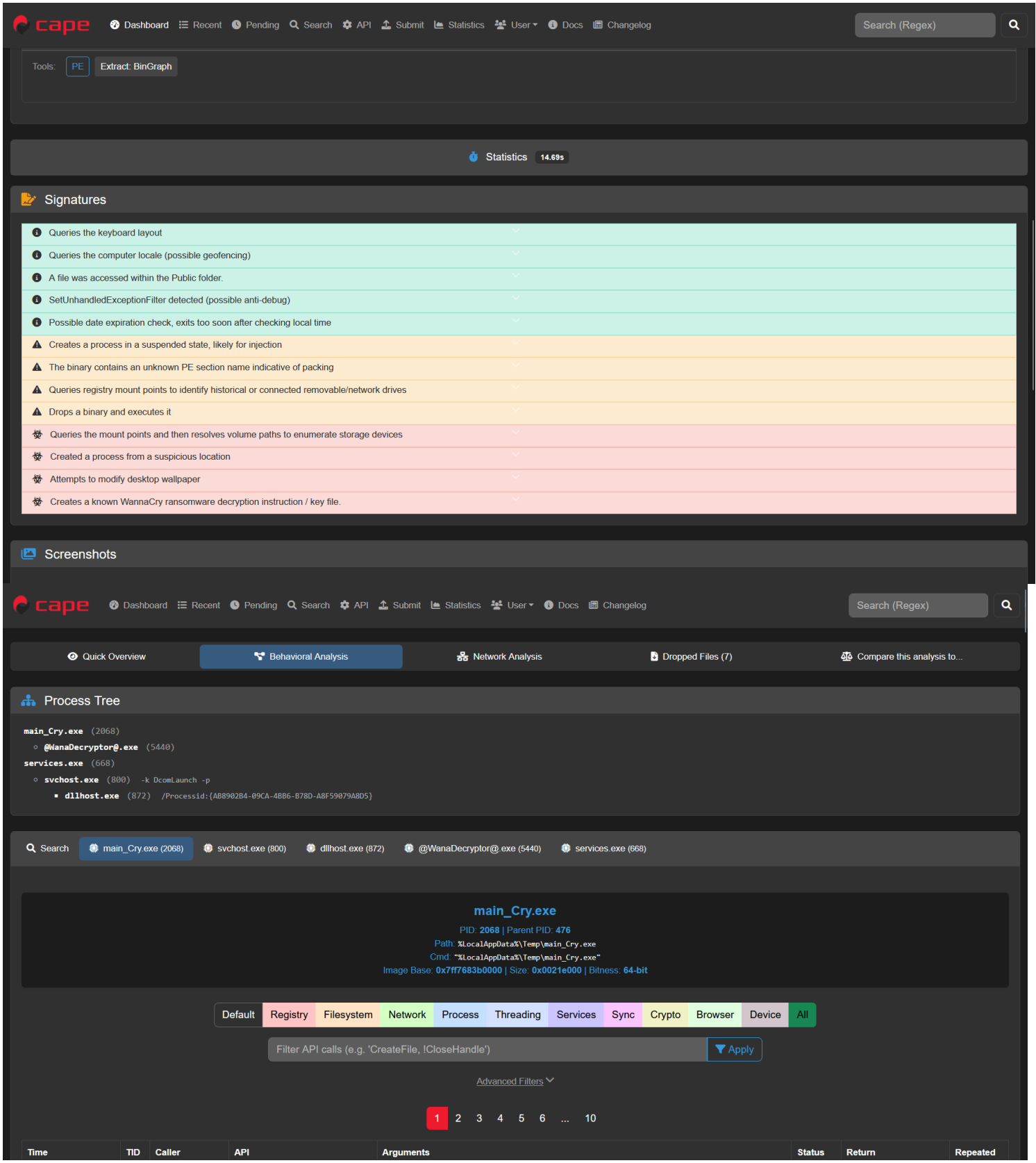






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cape
Dashboard
Recent
Pending
Search
API
Submit
Statistics
User
Docs
Changelog
Search (Regex)
Quick Overview
Behavioral Analysis
Network Analysis
Dropped Files (7)
Compare this analysis to...
Detection(s):
WannaCry
Analysis Details
Category
Package
Started
Completed
Duration
Logs
FILE
exe
2026-03-06 17:12:17
2026-03-06 17:16:13
236s
Analysis
Reports
JSON
Machine Information
Name
Label
Manager
Started On
Shutdown On
win10-7
win10-7
KVM
2026-03-06 17:12:17
2026-03-06 17:16:11
File Details
File Information
File Name
main_Cry.exe
File Type
PE32+ executable (GUI) x86-64, for MS Windows
File Size
2197504 bytes
MD5
70920f342814ca5c662cbbed00d60eef
SHA1
3e96e1f6a2d7cc6b72473d141e1962260325a0e7
SHA256
b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf
VT
MWDB
Bazaar
SHA3-384
836162c5b770e8ccbcd8c5424b9fc018cd5226263790e133421beaa4be511a36ffd5cb07642a8e69677c3ce805dea880
CRC32
3E85A9AA
TLSH
T1EDA52A07E23E25F9D1BFA0BE84C22942F9617C7907355BD76750826ADE22EF4A83D710
Ssdeep
6144:6j57PRbwHB1eHZGMBESZxBTr9QLhrbhYFveEbBENpKqmqED2Umpg7ajG:65Jbme5GMXNxC1oeEbupXmdvm
Detect It Easy
Linker: Microsoft Linker(14.36.35222)
Compiler: Microsoft Visual C/C++(19.36.35207)[C++]
Tool: Visual Studio(2022, v17.6)
Tools:
PE
Extract: BinGraph
Statistics
14.69s


cape
Dashboard
Recent
Pending
Search
API
Submit
Statistics
User
Docs
Changelog
Search (Regex)
Tools:
PE
Extract: BinGraph
Statistics
14.69s
Signatures
Queries the keyboard layout
Queries the computer locale (possible geofencing)
A file was accessed within the Public folder.
SetUnhandledExceptionFilter detected (possible anti-debug)
Possible date expiration check, exits too soon after checking local time
Creates a process in a suspended state, likely for injection
The binary contains an unknown PE section name indicative of packing
Queries registry mount points to identify historical or connected removable/network drives
Drops a binary and executes it
Queries the mount points and then resolves volume paths to enumerate storage devices
Created a process from a suspicious location
Attempts to modify desktop wallpaper
Creates a known WannaCry ransomware decryption instruction / key file.
Screenshots
cape
Dashboard
Recent
Pending
Search
API
Submit
Statistics
User
Docs
Changelog
Search (Regex)
Quick Overview
Behavioral Analysis
Network Analysis
Dropped Files (7)
Compare this analysis to...
Process Tree
main_Cry.exe (2068)
@WanaDecryptor@.exe (5440)
services.exe (668)
svchost.exe (800) -k DcomLaunch -p
dllhost.exe (872) /Processid:{AB8902B4-09CA-4BB6-B78D-A8F59079A8D5}
Search
main_Cry.exe (2068)
svchost.exe (800)
dllhost.exe (872)
@WanaDecryptor@.exe (5440)
services.exe (668)
main_Cry.exe
PID: 2068 | Parent PID: 476
Path: %LocalAppData%\Temp\main_Cry.exe
Cmd: "%LocalAppData%\Temp\main_Cry.exe"
Image Base: 0x7ff7683b0000 | Size: 0x0021e000 | Bitness: 64-bit
Default
Registry
Filesystem
Network
Process
Threading
Services
Sync
Crypto
Browser
Device
All
Filter API calls (e.g. 'CreateFile, !CloseHandle')
Apply
Advanced Filters
1
2
3
4
5
6
...
10
Time
TID
Caller
API
Arguments
Status
Return
Repeated

4. Dynamic Analysis

All dynamic analysis was conducted within an isolated VMware Workstation environment. FakeNet-NG was used for network simulation. Process Monitor (ProcMon) and ProcDOT were used for behavioral logging and visualization. The sample was executed manually; no exploit delivery chain was triggered.

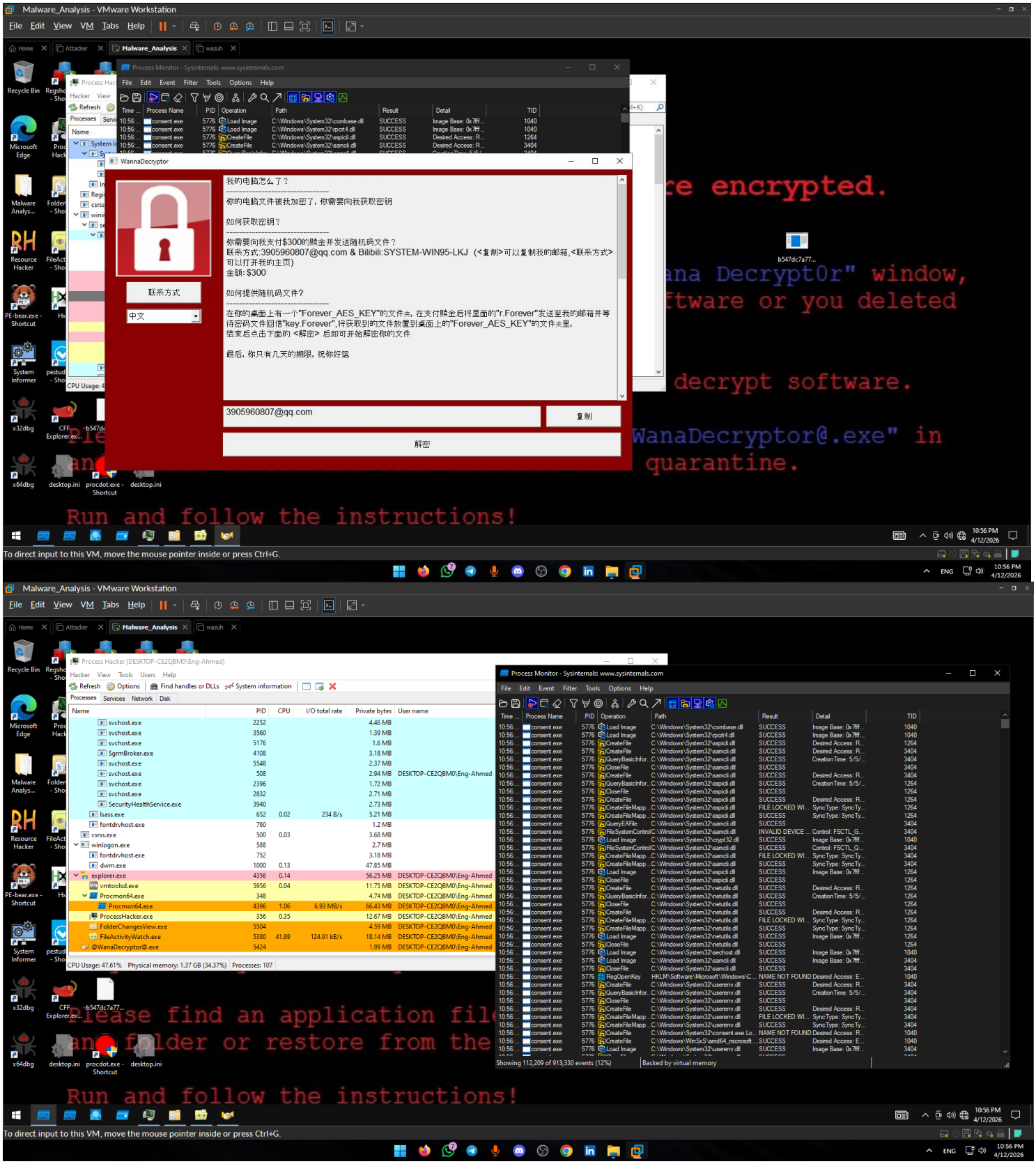
4.1 Behavioral Summary

Upon execution, the malware establishes a multi-stage operational hierarchy:

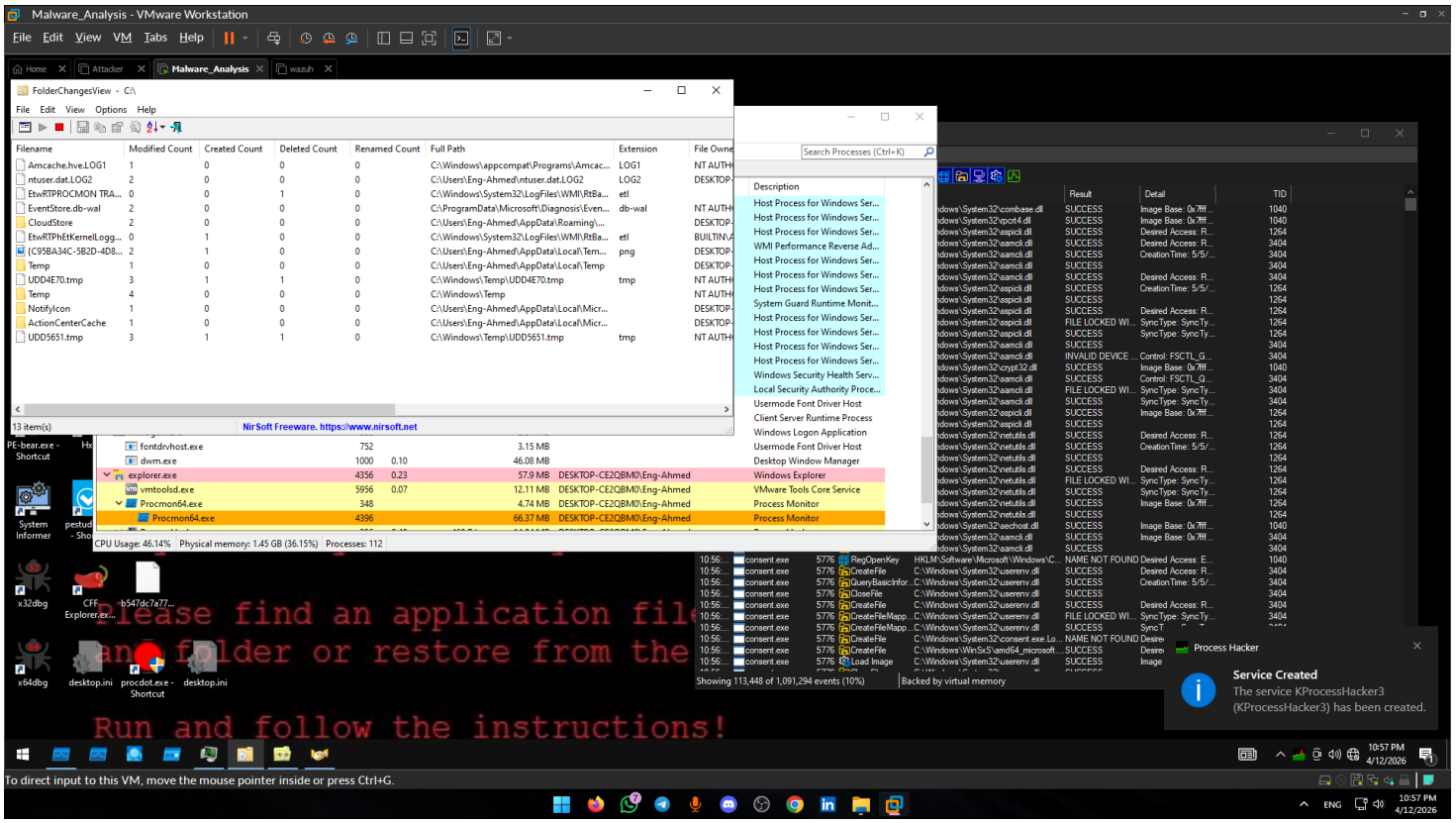
- Stage 1 — Initial execution: The primary binary launches @WanaDecryptor@.exe (observed as PID 3304 and PID 5424 across analysis runs).
- Stage 2 — File encryption: Cryptographic routines are invoked via the Windows CryptoAPI to encrypt user files. The target file scope was not fully enumerated in the observed execution window.
- Stage 3 — Artifact drop: Thread 4008 (spawned by PID 3304) creates the temporary file ICO5452.tmp in %APPDATA%\Local\Temp\, likely carrying icon or UI resources for the ransom GUI.
- Stage 4 — Child process execution: cmd.exe (PID 5948) is spawned, in turn initiating conhost.exe (PID 2072). This process tree is used to interact with registry keys governing Internet Settings cache and notification data.
- Stage 5 — Ransom display: SystemParametersInfoW is called to replace the desktop wallpaper with a ransom notice, and the @WanaDecryptor@.exe GUI window is displayed to the victim with the ransom demand and countdown timer.

4.2 File System Activity

Activity Type	Path / Value	Notes
File Created	%APPDATA%\Local\Temp\ICO5452.tmp	Icon/resource artifact for ransom GUI
File Written	User files (specific paths not enumerated)	Encryption output — scope not fully observed
Key/Directory Reference	Forever_AES_KEY	AES key storage artifact; exact path not resolved in report


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Process Monitor - Sysinternals: www.sysinternals.com
WannaDecryptor
我的电脑怎么了？
你的电脑文件被我加密了，你需要向我获取密钥
如何获取密钥？
你需要向我支付$300的赎金并发送随机码文件？
联系方式,3905960807@qq.com & Bilibili:SYSTEM-WIN95-LKJ （<复制>可以复制我的邮箱,<联系方式>可以打开我的主页）
金额:$300
如何提供随机码文件？
在你的桌面上有一个"Forever_AES_KEY"的文件夹，在支付赎金后将里面的"r.Forever"发送至我的邮箱并等待密码文件回信"key.Forever",将获取到的文件放置到桌面上的"Forever_AES_KEY"的文件夹里，
结束后点击下面的 <解密> 后即可开始解密你的文件
最后，你只有几天的期限，祝你好运
联系方式
中文
3905960807@qq.com
复制
解密
encrypted.
Decryptor" window,
or you deleted
decrypt software.
quarantine.
Run and follow the instructions!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Malware_Analysis - VMware Workstation
Process Hacker [DESKTOP-CE2QBM0\Eng-Ahmed]
Process Monitor - Sysinternals: www.sysinternals.com
Run and follow the instructions!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

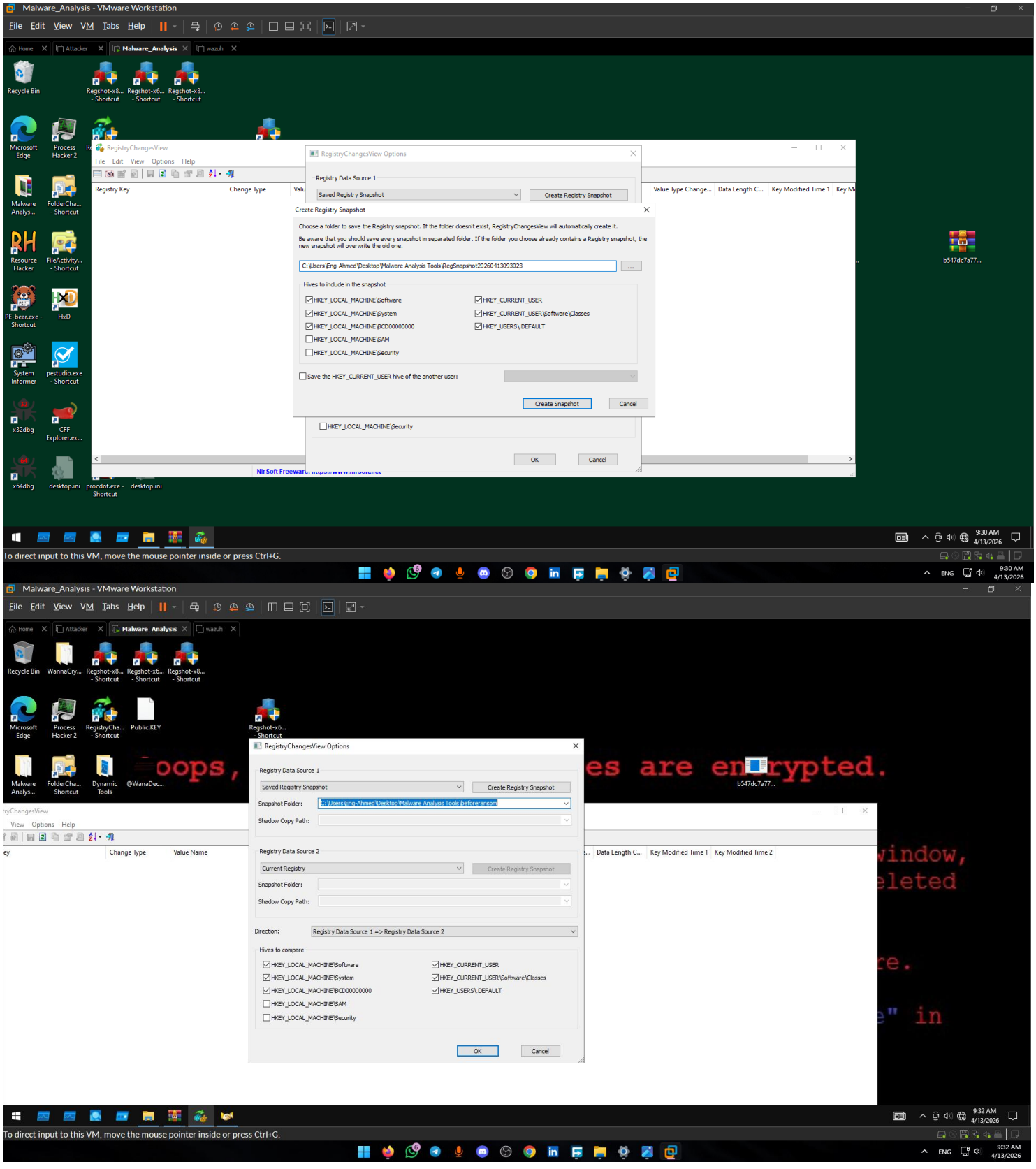


4.3 Registry Modifications

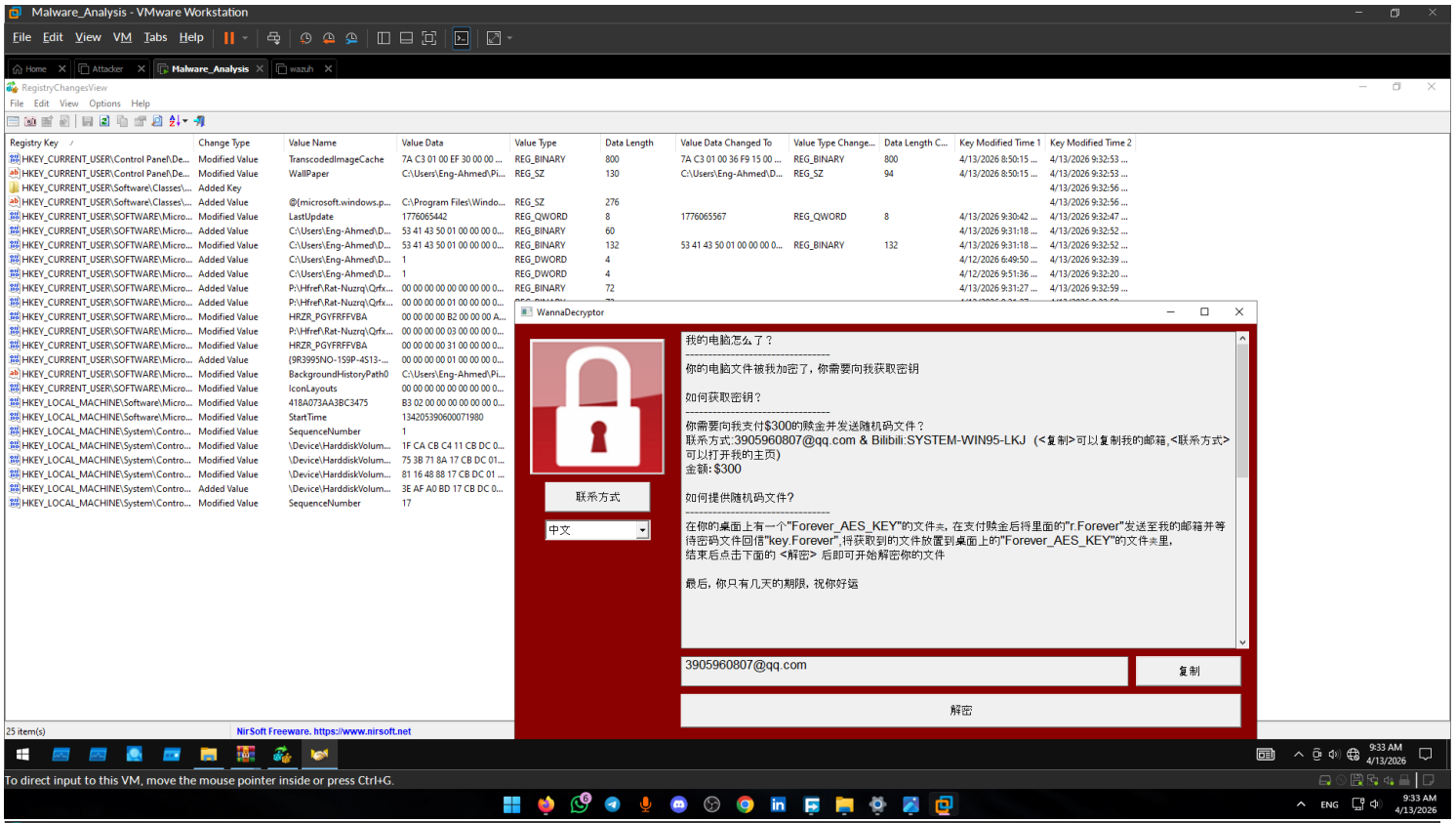
Registry activity was categorized into two groups — malicious modifications and OS-generated noise:

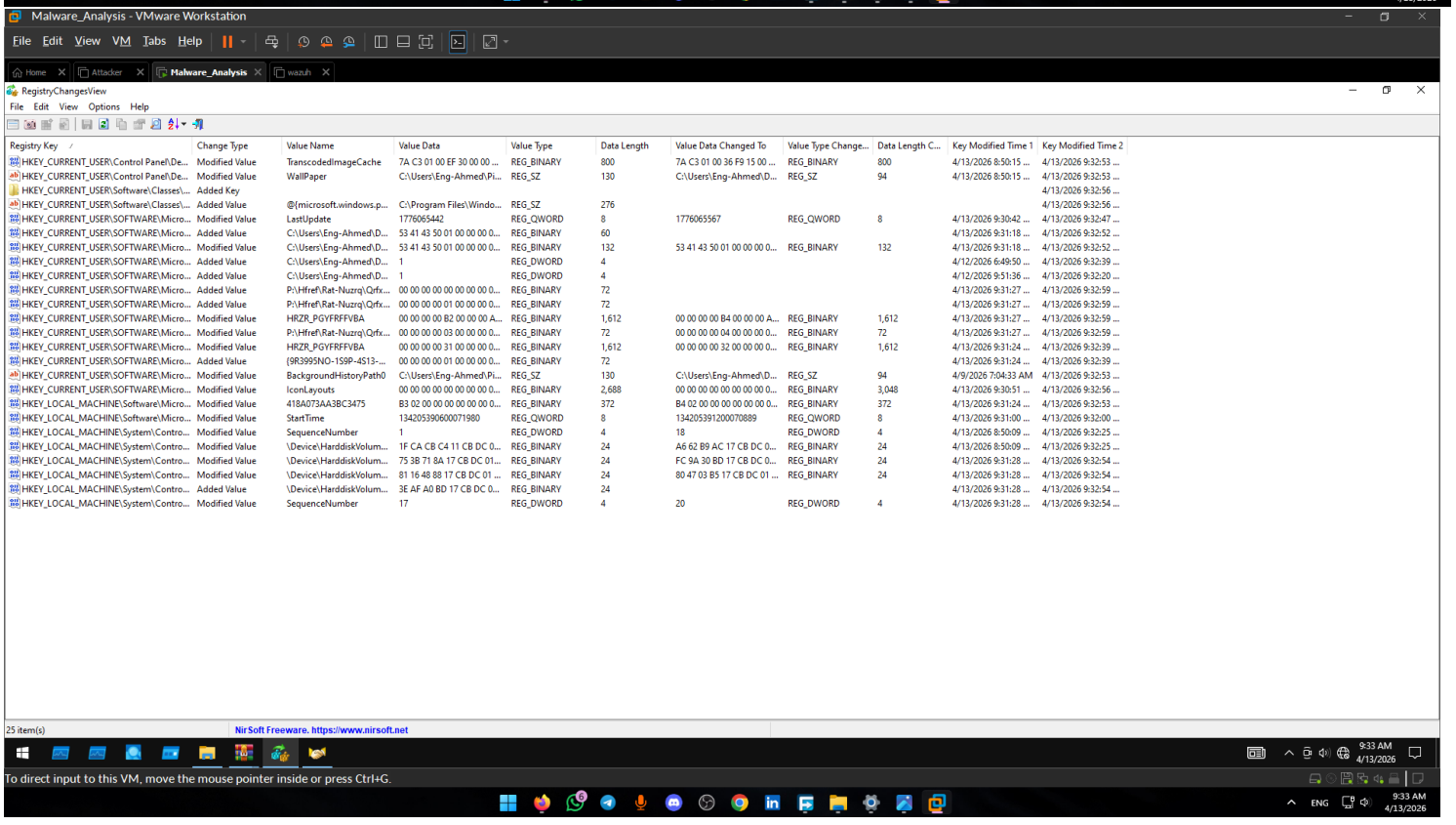
4.3.1 Malicious / High-Value Modifications

Registry Key	Value Name	Significance
HKCU\Control Panel\Desktop	WallPaper	Wallpaper replaced with ransom note — high-confidence IOC
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\...	(Execution record)	BAM key records binary execution — confirms payload ran


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
wazuh
Recycle Bin
Regshot-x8... - Shortcut
Regshot-x6... - Shortcut
Regshot-x8... - Shortcut
Microsoft Edge
Process Hacker 2
Malware Analys...
FolderCha... - Shortcut
Resource Hacker
FileActivity... - Shortcut
PE-bear.exe - Shortcut
HxD
System Informer
pestudio.exe - Shortcut
x32dbg
CFF Explorer.ex...
x64dbg
desktop.ini
procdot.exe - Shortcut
desktop.ini
b547dc7a77...
RegistryChangesView
File Edit View Options Help
Registry Key
Change Type
Value Type Change...
Data Length C...
Key Modified Time 1
RegistryChangesView Options
Registry Data Source 1
Saved Registry Snapshot
Create Registry Snapshot
Create Registry Snapshot
Choose a folder to save the Registry snapshot. If the folder doesn't exist, RegistryChangesView will automatically create it.
Be aware that you should save every snapshot in separated folder. If the folder you choose already contains a Registry snapshot, the new snapshot will overwrite the old one.
C:\Users\Eng-Ahmed\Desktop\Malware Analysis Tools\RegSnapshot20260413093023
Hives to include in the snapshot
HKEY_LOCAL_MACHINE\Software
HKEY_LOCAL_MACHINE\System
HKEY_LOCAL_MACHINE\BCD00000000
HKEY_LOCAL_MACHINE\SAM
HKEY_LOCAL_MACHINE\Security
HKEY_CURRENT_USER
HKEY_CURRENT_USER\Software\Classes
HKEY_USERS\.DEFAULT
Save the HKEY_CURRENT_USER hive of the another user:
Create Snapshot
Cancel
OK
Cancel
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
9:30 AM
4/13/2026

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Recycle Bin
WannaCry...
Regshot-x8... - Shortcut
Regshot-x6... - Shortcut
Regshot-x8... - Shortcut
Microsoft Edge
Process Hacker 2
RegistryCha... - Shortcut
Public.KEY
Regshot-x6... - Shortcut
Malware Analys...
FolderCha... - Shortcut
Dynamic Tools
@WanaDec...
b547dc7a77...
oops,
es are encrypted.
RegistryChangesView Options
Registry Data Source 1
Saved Registry Snapshot
Create Registry Snapshot
Snapshot Folder:
C:\Users\Eng-Ahmed\Desktop\Malware Analysis Tools\beforeransom
Shadow Copy Path:
Registry Data Source 2
Current Registry
Create Registry Snapshot
Snapshot Folder:
Shadow Copy Path:
Direction:
Registry Data Source 1 => Registry Data Source 2
Hives to compare
HKEY_LOCAL_MACHINE\Software
HKEY_LOCAL_MACHINE\System
HKEY_LOCAL_MACHINE\BCD00000000
HKEY_LOCAL_MACHINE\SAM
HKEY_LOCAL_MACHINE\Security
HKEY_CURRENT_USER
HKEY_CURRENT_USER\Software\Classes
HKEY_USERS\.DEFAULT
OK
Cancel
Change Type
Value Name
Data Length C...
Key Modified Time 1
Key Modified Time 2
window,
eleted
re.
e" in
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
9:32 AM
4/13/2026


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
RegistryChangesView
File Edit View Options Help
WannaDecryptor
我的电脑怎么了？
你的电脑文件被我加密了，你需要向我获取密钥
如何获取密钥？
你需要向我支付$300的赎金并发送随机码文件？
联系方式:3905960807@qq.com & Bilibili:SYSTEM-WIN95-LKJ （<复制>可以复制我的邮箱,<联系方式>可以打开我的主页）
金额: $300
如何提供随机码文件?
在你的桌面上有一个"Forever_AES_KEY"的文件夹，在支付赎金后将里面的"r.Forever"发送至我的邮箱并等待密码文件回信"key.Forever",将获取到的文件放置到桌面上的"Forever_AES_KEY"的文件夹里，
结束后点击下面的 <解密> 后即可开始解密你的文件
最后，你只有几天的期限，祝你好运
联系方式
中文
3905960807@qq.com
复制
解密
NirSoft Freeware. https://www.nirsoft.net
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
RegistryChangesView
File Edit View Options Help
NirSoft Freeware. https://www.nirsoft.net
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

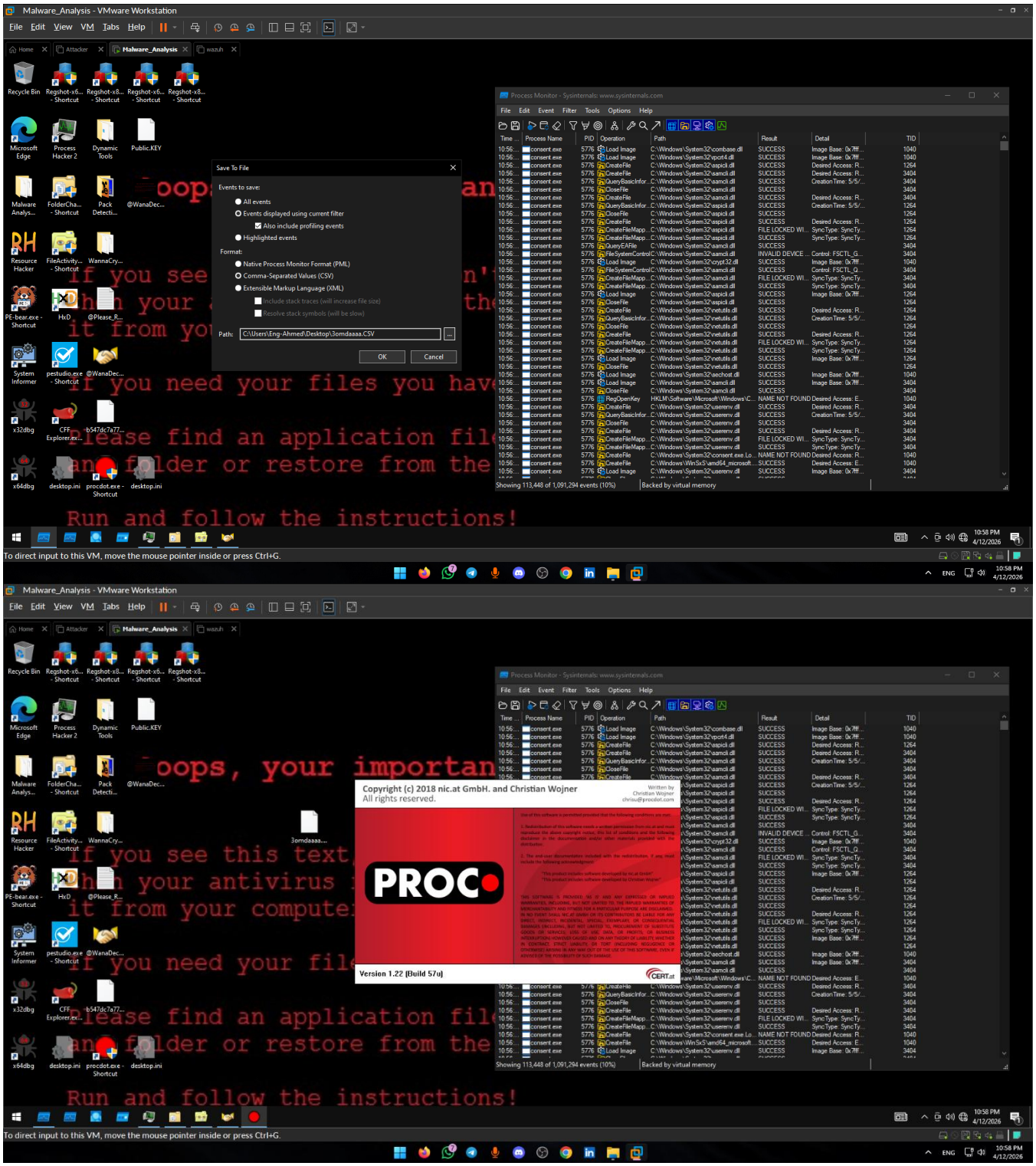
4.3.2 System-Generated Noise (Low Analytical Value)

Registry Key Area	Assessment
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Content	Normal OS cache update triggered by process activity
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Cookies	Normal OS cache update — not malicious
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\History	Normal OS cache update — not malicious
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\	System notification data — generated by OS in response to disk I/O
IconLayouts, HarddiskVolume entries, Sequence numbers	Standard Windows OS response to heavy file system activity — discard

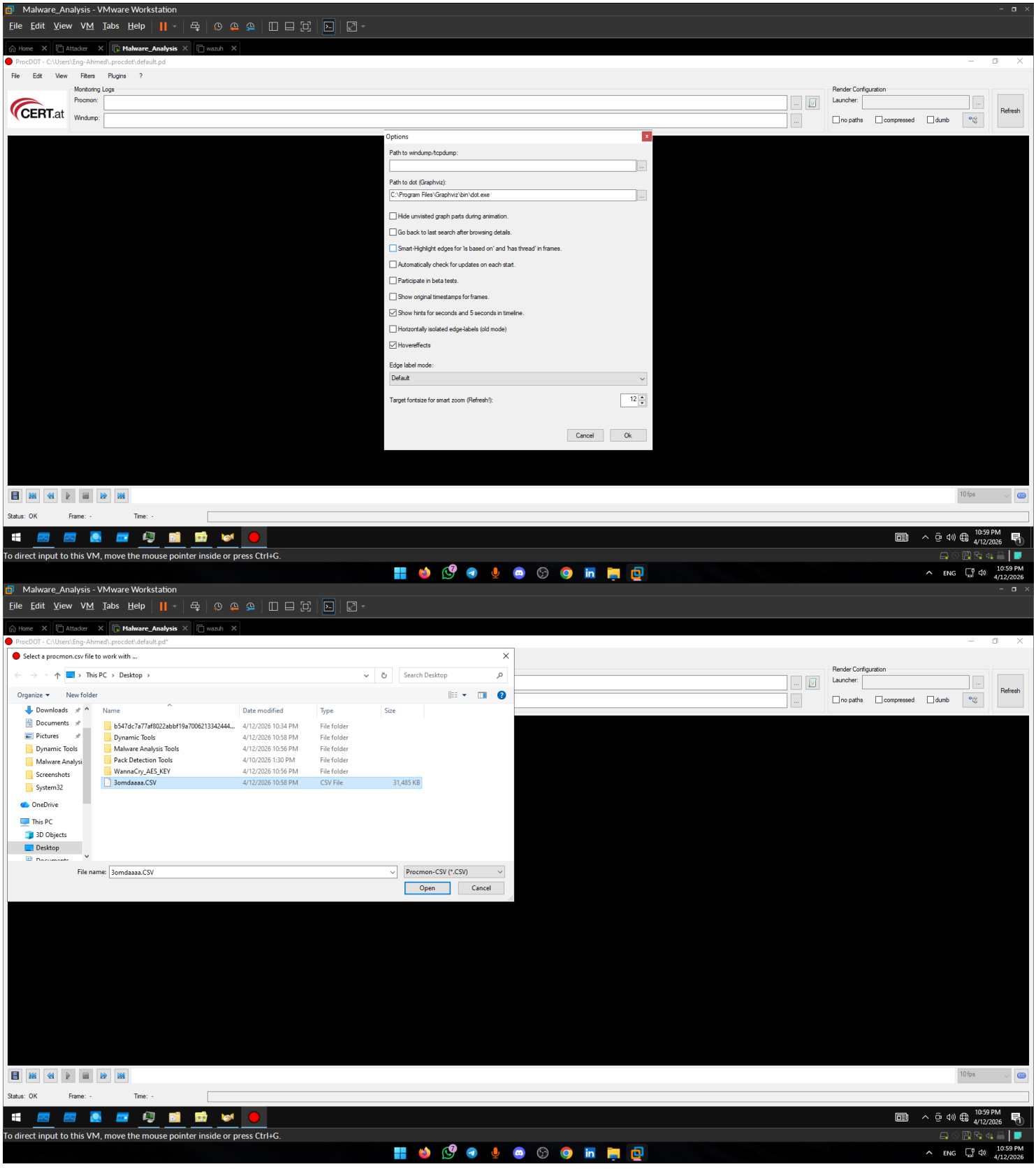
4.4 Process Injection & Persistence Mechanisms

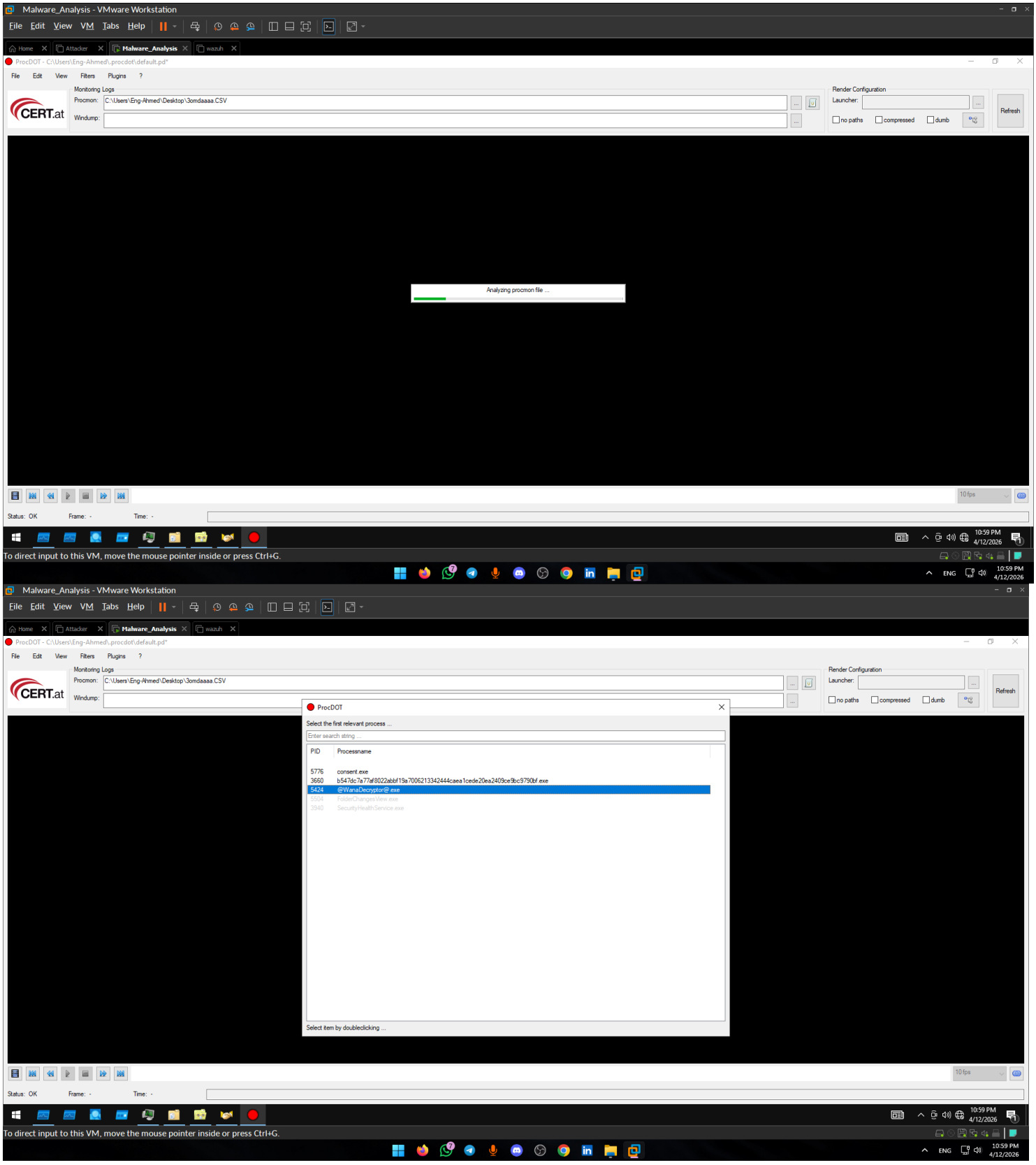
- Persistence via BAM registry key: Execution artefacts recorded in HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\... confirm the binary was executed. However, this key is a historical execution record, not a persistence mechanism.
- Auto-start persistence (Run keys, scheduled tasks, malicious services): No evidence observed during the dynamic execution timeframe analyzed.
- Process spawning: @WanaDecryptor@.exe is spawned as a persistent child process. This GUI process maintains operator interaction with the victim and functions as an extortion interface.

> *i Analyst Note: The absence of observed Run key persistence does not conclusively rule out persistence. The analysis window may not have captured all execution stages. In the original WannaCry, persistence was achieved via a Windows service installation.*

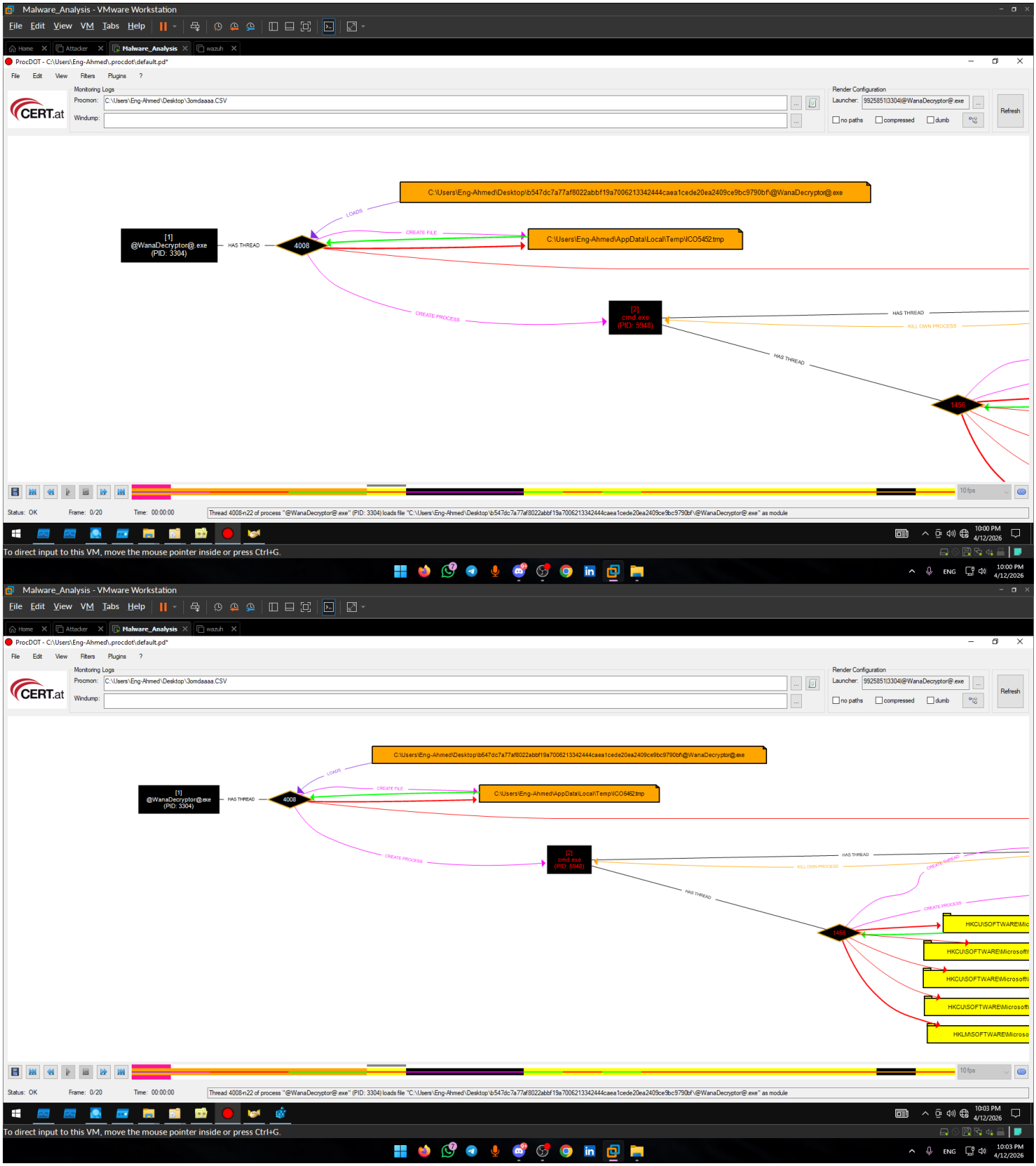

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Save To File
Events to save:
All events
Events displayed using current filter
Also include profiling events
Highlighted events
Format:
Native Process Monitor Format (PML)
Comma-Separated Values (CSV)
Extensible Markup Language (XML)
Include stack traces (will increase file size)
Resolve stack symbols (will be slow)
Path: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
OK
Cancel
Process Monitor - Sysinternals: www.sysinternals.com
Showing 113,448 of 1,091,294 events (10%)
Backed by virtual memory
Run and follow the instructions!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Malware_Analysis - VMware Workstation
Copyright (c) 2018 nic.at GmbH. and Christian Wojner
All rights reserved.
Written by
Christian Wojner
chrisu@procdot.com
PROCDOT
Version 1.22 (Build 57u)
Process Monitor - Sysinternals: www.sysinternals.com
Showing 113,448 of 1,091,294 events (10%)
Backed by virtual memory
Run and follow the instructions!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pd
Monitoring Logs
Procmon:
Windump:
Render Configuration
Launcher:
Refresh
Options
Path to windump/tcpdump:
Path to dot (Graphviz):
C:\Program Files\Graphviz\bin\dot.exe
Hide unvisited graph parts during animation.
Go back to last search after browsing details.
Smart-Highlight edges for 'is based on' and 'has thread' in frames.
Automatically check for updates on each start.
Participate in beta tests.
Show original timestamps for frames.
Show hints for seconds and 5 seconds in timeline.
Horizontally isolated edge-labels (old mode)
Hovereffects
Edge label mode:
Default
Target fontsize for smart zoom (Refresh!):
12
Cancel
Ok
Status: OK
Frame: -
Time: -
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Malware_Analysis - VMware Workstation
Select a procmon.csv file to work with ...
This PC › Desktop ›
Search Desktop
Organize
New folder
Downloads
Documents
Pictures
Dynamic Tools
Malware Analysi
Screenshots
System32
OneDrive
This PC
3D Objects
Desktop
Name
Date modified
Type
Size
b547dc7a77af8022abbf19a7006213342444...
4/12/2026 10:34 PM
File folder
Dynamic Tools
4/12/2026 10:58 PM
File folder
Malware Analysis Tools
4/12/2026 10:56 PM
File folder
Pack Detection Tools
4/10/2026 1:30 PM
File folder
WannaCry_AES_KEY
4/12/2026 10:56 PM
File folder
3omdaaaa.CSV
4/12/2026 10:58 PM
CSV File
31,485 KB
File name: 3omdaaaa.CSV
Procmon-CSV (*.CSV)
Open
Cancel
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
wazuh
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pd*
File Edit View Filters Plugins ?
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher:
no paths
compressed
dumb
Refresh
CERT.at
Analyzing procmon file ...
Status: OK
Frame: -
Time: -
10 fps
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
10:59 PM
4/12/2026
ENG

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
wazuh
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pd*
File Edit View Filters Plugins ?
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher:
no paths
compressed
dumb
Refresh
CERT.at
ProcDOT
Select the first relevant process ...
Enter search string ...
PID
Processname
5776 consent.exe
3660 b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf.exe
5424 @WanaDecryptor@.exe
6504 FolderChangesView.exe
3940 SecurityHealthService.exe
Select item by doubleclicking ...
Status: OK
Frame: -
Time: -
10 fps
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
10:59 PM
4/12/2026
ENG


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pd*
CERT.at
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher: 99258513304l@WanaDecryptor@.exe
no paths compressed dumb
Refresh
C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe
LOADS
[1]
@WanaDecryptor@.exe
(PID: 3304)
HAS THREAD
4008
CREATE FILE
C:\Users\Eng-Ahmed\AppData\Local\Temp\ICO5452.tmp
CREATE PROCESS
[2]
cmd.exe
(PID: 5948)
HAS THREAD
KILL OWN PROCESS
HAS THREAD
1456
Status: OK
Frame: 0/20
Time: 00:00:00
Thread 4008+22 of process "@WanaDecryptor@.exe" (PID: 3304) loads file "C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe" as module
10:00 PM
4/12/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
10:00 PM
4/12/2026
Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pd*
CERT.at
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher: 99258513304l@WanaDecryptor@.exe
no paths compressed dumb
Refresh
C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe
LOADS
[1]
@WanaDecryptor@.exe
(PID: 3304)
HAS THREAD
4008
CREATE FILE
C:\Users\Eng-Ahmed\AppData\Local\Temp\ICO5462.tmp
CREATE PROCESS
[2]
cmd.exe
(PID: 5940)
HAS THREAD
KILL OWN PROCESS
HAS THREAD
1456
HKCU\SOFTWARE\Mic
HKCU\SOFTWARE\Microsoft\
HKCU\SOFTWARE\Microsoft\
HKCU\SOFTWARE\Microsoft
HKLM\SOFTWARE\Microso
Status: OK
Frame: 0/20
Time: 00:00:00
Thread 4008+22 of process "@WanaDecryptor@.exe" (PID: 3304) loads file "C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe" as module
10:03 PM
4/12/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
10:03 PM
4/12/2026


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pdf*
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher: 99258513304l@WanaDecryptor@.exe
no paths compressed dumb
Refresh
C:\Users\Eng-Ahmed\AppData\Local\Temp\CO5462.tmp
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S
[2] cmd.exe (PID: 5948)
HAS THREAD
KILL OWN PROCESS
1636
CREATE THREAD
HAS THREAD
[3] Conhost.exe (PID: 2072)
CREATE PROCESS
HAS THREAD
1468
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\SlowContextMenuEntries
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\ContentCachePrefix
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\CookiesCachePrefix
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\HistoryCachePrefix
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\418A073AA3BC1C75
10 fps
Status: OK
Frame: 0/20
Time: 00:00:00
Thread 4008-n22 of process "@WanaDecryptor@.exe" (PID: 3304) loads file "C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe" as module
10:03 PM
4/12/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
10:03 PM
4/12/2026
Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
ProcDOT - C:\Users\Eng-Ahmed\.procdot\default.pdf*
Monitoring Logs
Procmon: C:\Users\Eng-Ahmed\Desktop\3omdaaaa.CSV
Windump:
Render Configuration
Launcher: 99258513304l@WanaDecryptor@.exe
no paths compressed dumb
Refresh
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-1749370939-862622941-1222624791-1000\Device\HarddiskVolume3\Windows\System32\cmd.exe (AUTOSTART)
1636
CREATE THREAD
[3] Conhost.exe (PID: 2072)
CREATE PROCESS
HAS THREAD
1760
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\SlowContextMenuEntries
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\ContentCachePrefix
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\CookiesCachePrefix
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\HistoryCachePrefix
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\418A073AA3BC1C75
10 fps
Status: OK
Frame: 0/20
Time: 00:00:00
Thread 4008-n22 of process "@WanaDecryptor@.exe" (PID: 3304) loads file "C:\Users\Eng-Ahmed\Desktop\b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf\@WanaDecryptor@.exe" as module
10:03 PM
4/12/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
10:03 PM
4/12/2026

4.5 Network Activity

4.5.1 Observed Traffic (Benign Background Noise — FakeNet-NG)

The following traffic was captured but classified as standard Windows OS and application background activity:

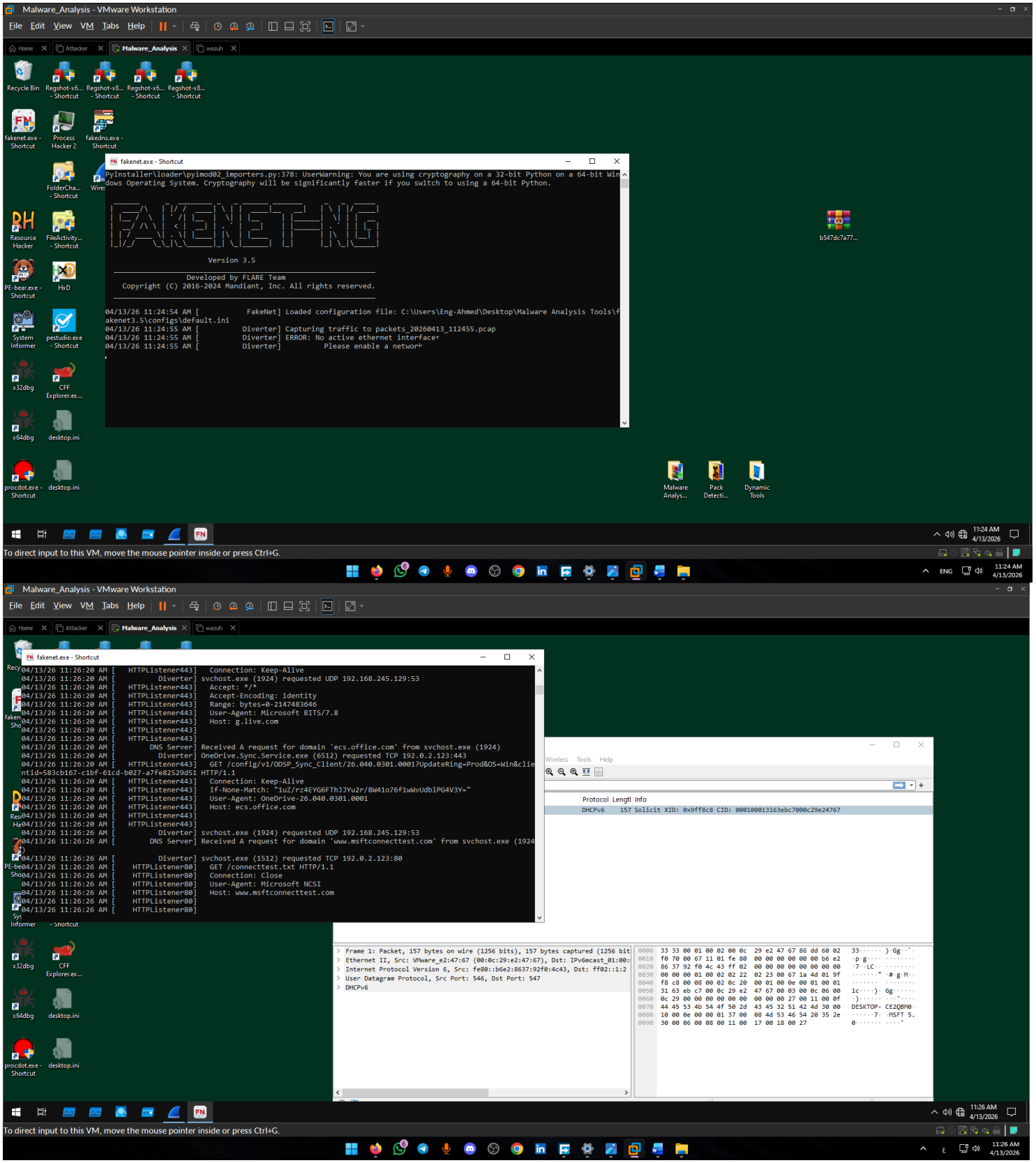
Traffic Type	Destination	Classification
NCSI Probe (DNS/HTTP)	www.msftconnecttest.com, dns.msftncsi.com	Benign — Windows internet connectivity check
CRL/OCSP Validation	ctldl.windowsupdate.com, oneocsp.microsoft.com	Benign — certificate revocation list update
Application Telemetry	ecs.office.com, g.live.com, v10.events.data.microsoft.com	Benign — OneDrive/Office/Windows telemetry

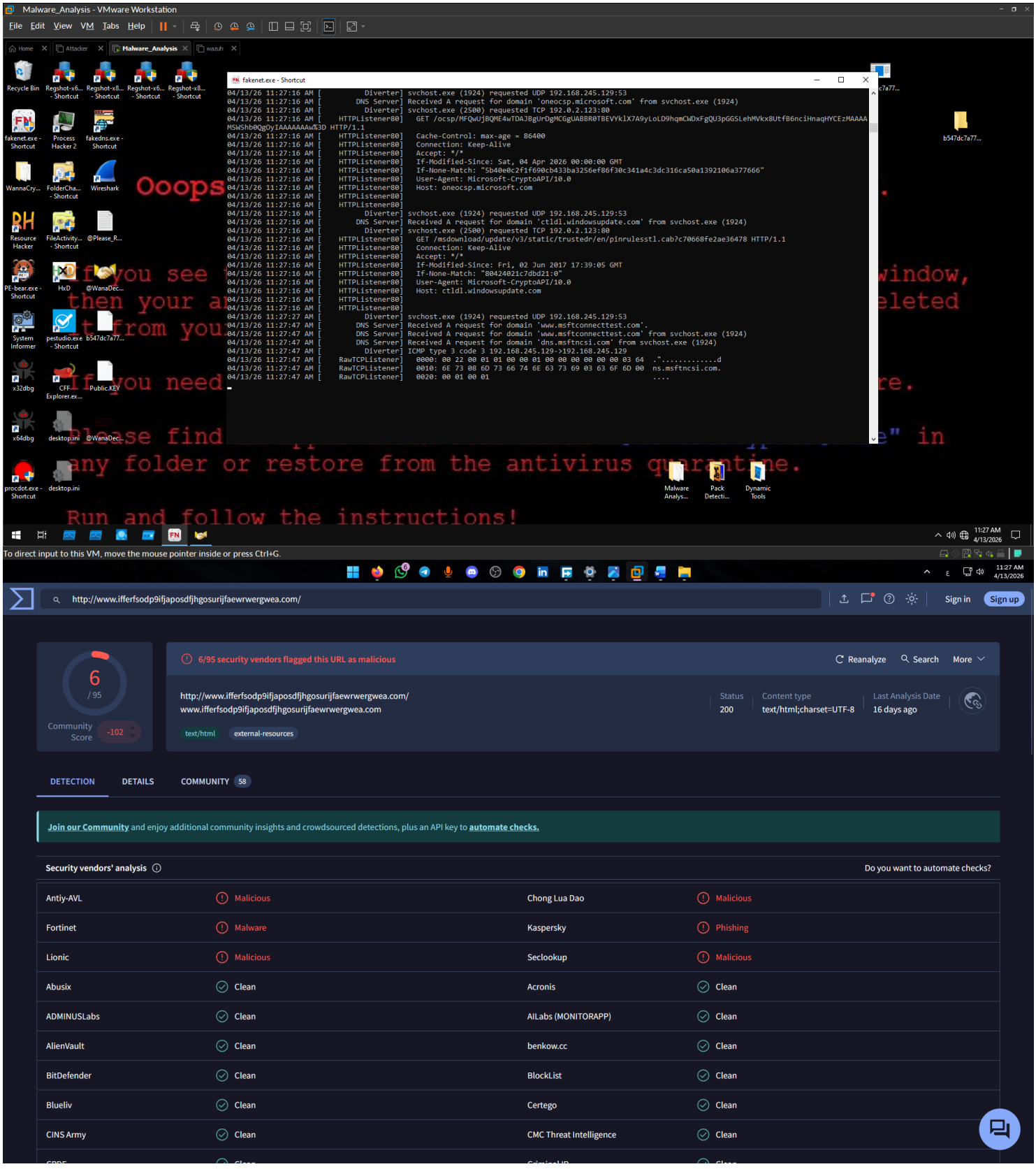
4.5.2 Expected WannaCry Network IOCs (Not Captured in This Session)

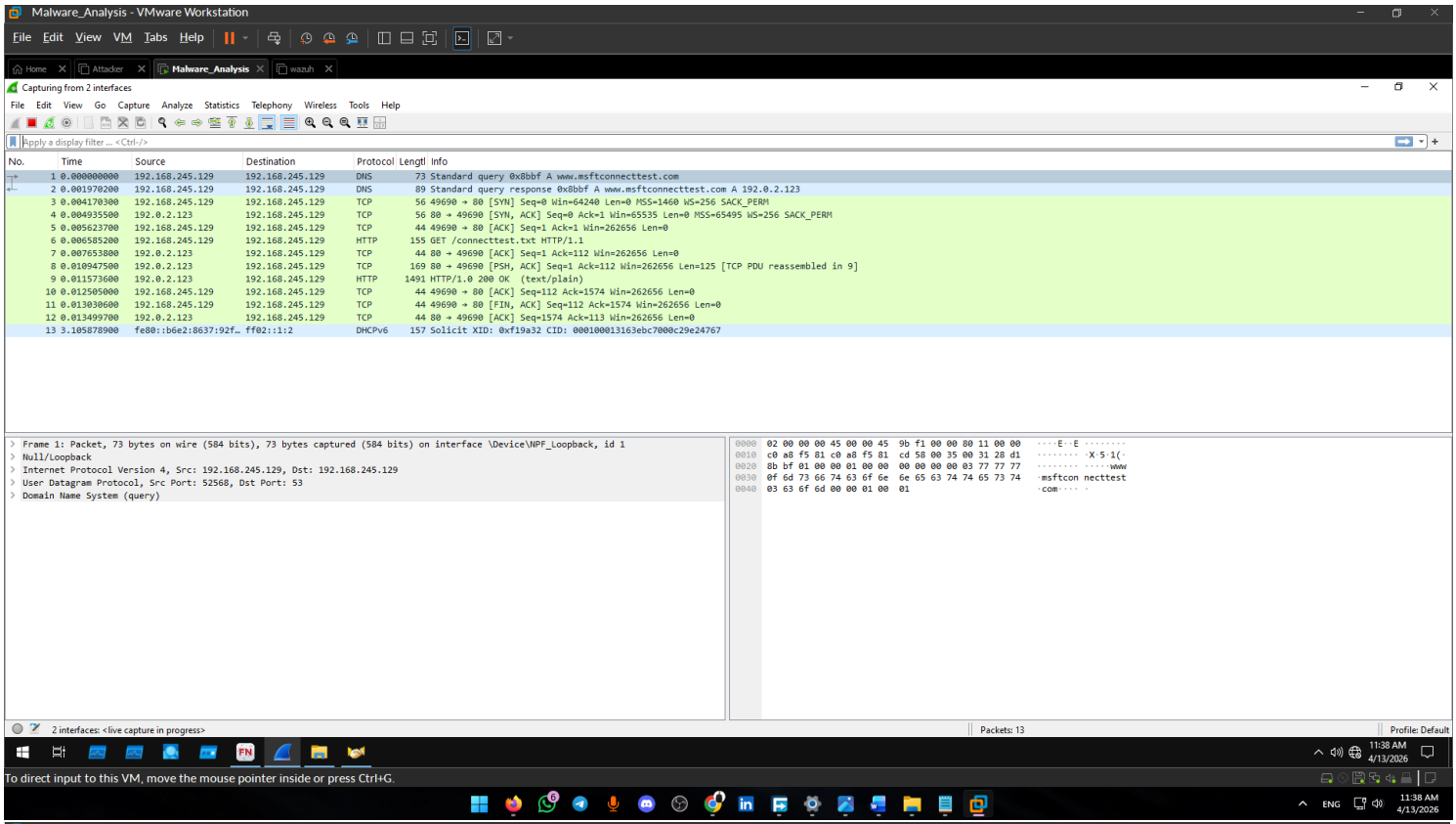
The following network behaviors are characteristic of WannaCry detonations based on established threat intelligence. They were not captured in this specific execution session, likely due to sandbox network simulation constraints or variant modification:

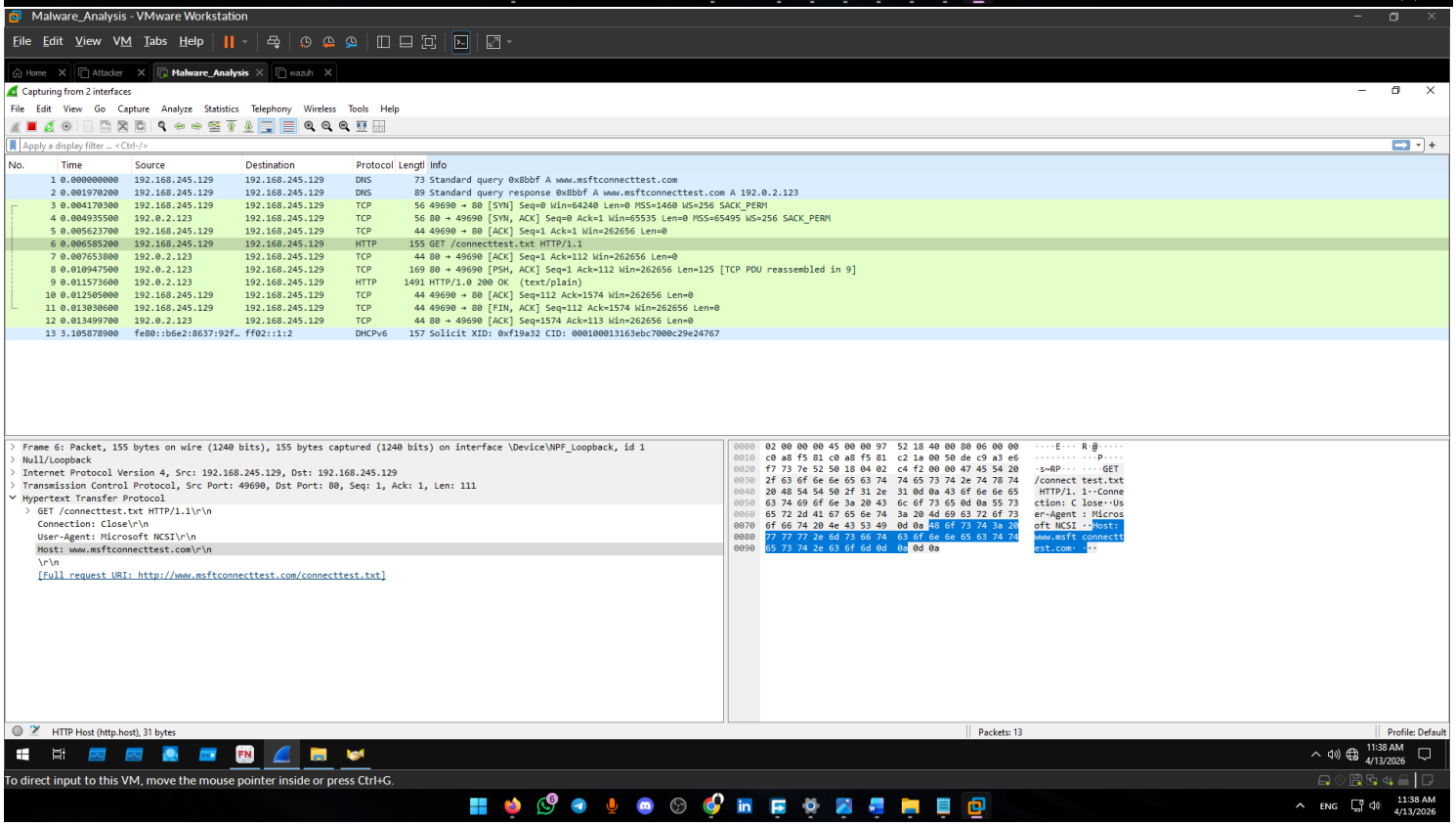
IOC Type	Indicator	Description
Kill Switch Domain (HTTP GET)	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com	If reachable, malware self-terminates (sinkholed in 2017)
Kill Switch Domain (alt)	www.ifferfsodp9ifjaposdfjhgosurijfaewrwergwea.com	Secondary known kill switch domain
SMB Scanning	TCP Port 445 — broadcast to subnet and random IPs	EternalBlue (MS17-010) propagation scan
C2 via Tor	Ports 80, 443, 9001, 9030 to Tor entry nodes	Encrypted C2 channel via bundled Tor client

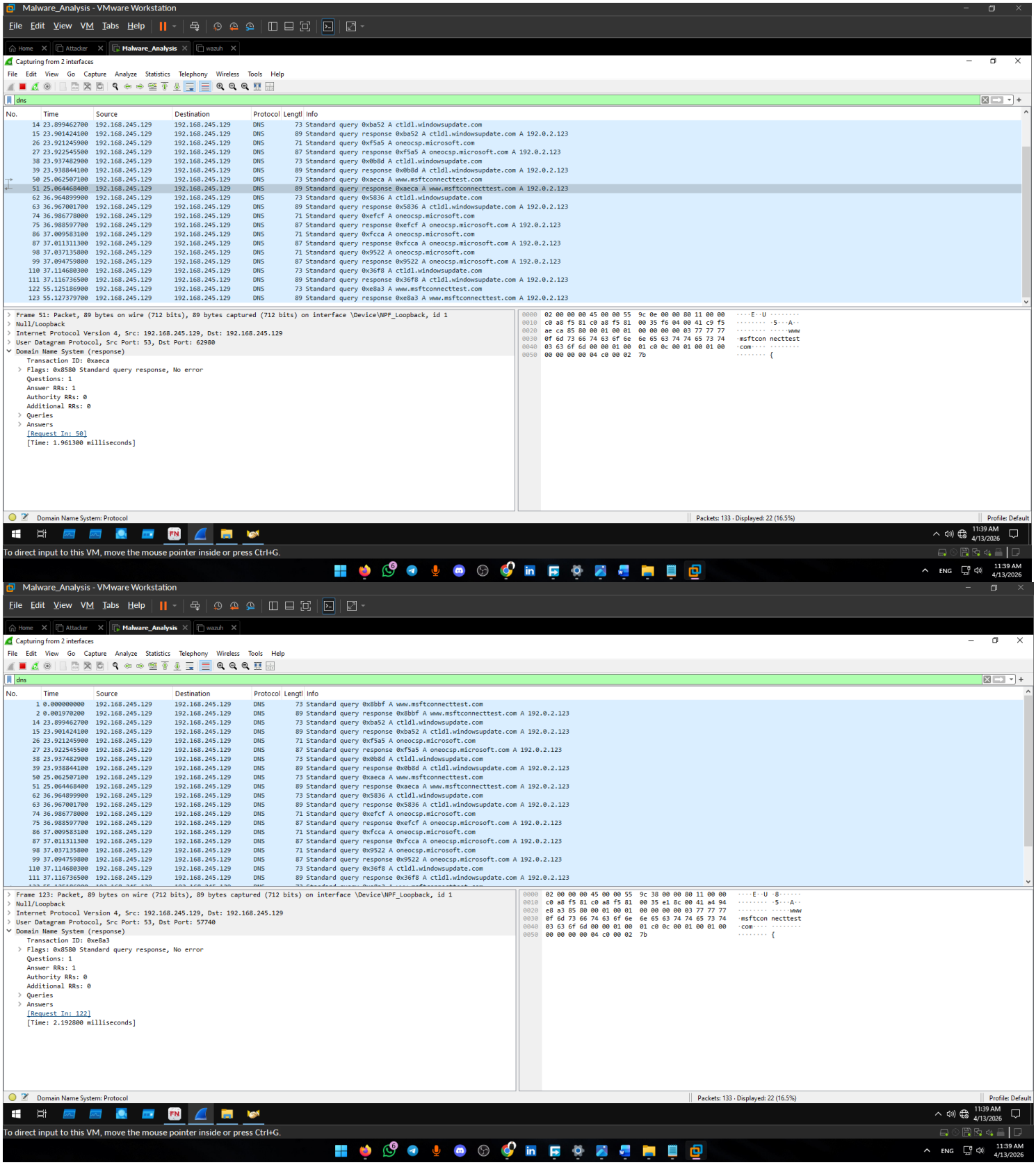
These network IOCs reflect the canonical WannaCry behavior documented in published threat intelligence. Their absence in this session does not invalidate them as detection indicators; they are retained as expected IOCs for environmental monitoring.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
fakenet.exe - Shortcut
Version 3.5
Developed by FLARE Team
Copyright (C) 2016-2024 Mandiant, Inc. All rights reserved.
04/13/26 11:24:54 AM [FakeNet] Loaded configuration file: C:\Users\Eng-Ahmed\Desktop\Malware Analysis Tools\fakenet3.5\configs\default.ini
04/13/26 11:24:55 AM [Diverter] Capturing traffic to packets_20260413_112455.pcap
04/13/26 11:24:55 AM [Diverter] ERROR: No active ethernet interface
04/13/26 11:24:55 AM [Diverter] Please enable a network
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Malware_Analysis - VMware Workstation
fakenet.exe - Shortcut
04/13/26 11:26:20 AM [HTTPListener443] Connection: Keep-Alive
04/13/26 11:26:20 AM [Diverter] svchost.exe (1924) requested UDP 192.168.245.129:53
04/13/26 11:26:20 AM [HTTPListener443] Accept: */*
04/13/26 11:26:20 AM [HTTPListener443] Accept-Encoding: identity
04/13/26 11:26:20 AM [HTTPListener443] Range: bytes=0-2147483646
04/13/26 11:26:20 AM [HTTPListener443] User-Agent: Microsoft BITS/7.8
04/13/26 11:26:20 AM [HTTPListener443] Host: g.live.com
04/13/26 11:26:20 AM [DNS Server] Received A request for domain 'ecs.office.com' from svchost.exe (1924)
04/13/26 11:26:20 AM [Diverter] OneDrive.Sync.Service.exe (6512) requested TCP 192.0.2.123:443
04/13/26 11:26:20 AM [HTTPListener443] GET /config/v1/ODSP_Sync_Client/26.040.0301.0001?UpdateRing=Prod&OS=Win&client=ntld-583cb167-c1bf-61cd-b027-a7fe82529d51 HTTP/1.1
04/13/26 11:26:20 AM [HTTPListener443] Connection: Keep-Alive
04/13/26 11:26:20 AM [HTTPListener443] If-None-Match: "1uZ/rz4EYG6FThJJYu2r/8W41o76f1wWvUdblPG4V3Y="
04/13/26 11:26:20 AM [HTTPListener443] User-Agent: OneDrive-26.040.0301.0001
04/13/26 11:26:20 AM [HTTPListener443] Host: ecs.office.com
04/13/26 11:26:20 AM [Diverter] svchost.exe (1924) requested UDP 192.168.245.129:53
04/13/26 11:26:20 AM [DNS Server] Received A request for domain 'www.msftconnecttest.com' from svchost.exe (1924)
04/13/26 11:26:26 AM [Diverter] svchost.exe (1512) requested TCP 192.0.2.123:80
04/13/26 11:26:26 AM [HTTPListener80] GET /connecttest.txt HTTP/1.1
04/13/26 11:26:26 AM [HTTPListener80] Connection: Close
04/13/26 11:26:26 AM [HTTPListener80] User-Agent: Microsoft NCSI
04/13/26 11:26:26 AM [HTTPListener80] Host: www.msftconnecttest.com
Protocol Length Info
DHCPv6 157 Solicit XID: 0x9ff8c8 CID: 000100013163ebc7000c29e24767
Frame 1: Packet, 157 bytes on wire (1256 bits), 157 bytes captured (1256 bit
Ethernet II, Src: VMware_e2:47:67 (00:0c:29:e2:47:67), Dst: IPv6mcast_01:00:
Internet Protocol Version 6, Src: fe80::b6e2:8637:92f0:4c43, Dst: ff02::1:2
User Datagram Protocol, Src Port: 546, Dst Port: 547
DHCPv6
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
fakenet.exe - Shortcut
04/13/26 11:27:16 AM [Diverter] svchost.exe (1924) requested UDP 192.168.245.129:53
04/13/26 11:27:16 AM [DNS Server] Received A request for domain 'oneocsp.microsoft.com' from svchost.exe (1924)
04/13/26 11:27:16 AM [Diverter] svchost.exe (2500) requested TCP 192.0.2.123:80
04/13/26 11:27:16 AM [HTTPListener80] GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGgUABBR0TBEVYklX7A9yLoLD9hqmCWDxFgQU3pGGSLehMVkx8UtfB6nciHnaqHYCEzMAAAA
MSWShb0QgOy1AAAAAAAAw%3D HTTP/1.1
04/13/26 11:27:16 AM [HTTPListener80] Cache-Control: max-age = 86400
04/13/26 11:27:16 AM [HTTPListener80] Connection: Keep-Alive
04/13/26 11:27:16 AM [HTTPListener80] Accept: */*
04/13/26 11:27:16 AM [HTTPListener80] If-Modified-Since: Sat, 04 Apr 2026 00:00:00 GMT
04/13/26 11:27:16 AM [HTTPListener80] If-None-Match: "5b40e0c2f1f690cb433ba3256ef86f30c341a4c3dc316ca50a1392106a377666"
04/13/26 11:27:16 AM [HTTPListener80] User-Agent: Microsoft-CryptoAPI/10.0
04/13/26 11:27:16 AM [HTTPListener80] Host: oneocsp.microsoft.com
04/13/26 11:27:16 AM [HTTPListener80]
04/13/26 11:27:16 AM [HTTPListener80]
04/13/26 11:27:16 AM [Diverter] svchost.exe (1924) requested UDP 192.168.245.129:53
04/13/26 11:27:16 AM [DNS Server] Received A request for domain 'ctldl.windowsupdate.com' from svchost.exe (1924)
04/13/26 11:27:16 AM [Diverter] svchost.exe (2500) requested TCP 192.0.2.123:80
04/13/26 11:27:16 AM [HTTPListener80] GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.cab?c70668fe2ae36478 HTTP/1.1
04/13/26 11:27:16 AM [HTTPListener80] Connection: Keep-Alive
04/13/26 11:27:16 AM [HTTPListener80] Accept: */*
04/13/26 11:27:16 AM [HTTPListener80] If-Modified-Since: Fri, 02 Jun 2017 17:39:05 GMT
04/13/26 11:27:16 AM [HTTPListener80] If-None-Match: "80424021c7dbd21:0"
04/13/26 11:27:16 AM [HTTPListener80] User-Agent: Microsoft-CryptoAPI/10.0
04/13/26 11:27:16 AM [HTTPListener80] Host: ctldl.windowsupdate.com
04/13/26 11:27:16 AM [HTTPListener80]
04/13/26 11:27:16 AM [HTTPListener80]
04/13/26 11:27:27 AM [Diverter] svchost.exe (1924) requested UDP 192.168.245.129:53
04/13/26 11:27:47 AM [DNS Server] Received A request for domain 'www.msftconnecttest.com'.
04/13/26 11:27:47 AM [DNS Server] Received A request for domain 'www.msftconnecttest.com' from svchost.exe (1924)
04/13/26 11:27:47 AM [DNS Server] Received A request for domain 'dns.msftncsi.com' from svchost.exe (1924)
04/13/26 11:27:47 AM [Diverter] ICMP type 3 code 3 192.168.245.129->192.168.245.129
04/13/26 11:27:47 AM [RawTCPListener] 0000: 00 22 00 01 01 00 00 01 00 00 00 00 00 00 03 64 ."...........d
04/13/26 11:27:47 AM [RawTCPListener] 0010: 6E 73 08 6D 73 66 74 6E 63 73 69 03 63 6F 6D 00 ns.msftncsi.com.
04/13/26 11:27:47 AM [RawTCPListener] 0020: 00 01 00 01 ....
Ooops
Run and follow the instructions!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
http://www.ifferfsodp9ifjaposdfjhgosurijfaewrwergwea.com/
Sign in
Sign up
6
/ 95
Community Score
-102
6/95 security vendors flagged this URL as malicious
Reanalyze
Search
More
http://www.ifferfsodp9ifjaposdfjhgosurijfaewrwergwea.com/
www.ifferfsodp9ifjaposdfjhgosurijfaewrwergwea.com
Status
200
Content type
text/html;charset=UTF-8
Last Analysis Date
16 days ago
text/html
external-resources
DETECTION
DETAILS
COMMUNITY 58
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
Antiy-AVL Malicious
Chong Lua Dao Malicious
Fortinet Malware
Kaspersky Phishing
Lionic Malicious
Seclookup Malicious
Abusix Clean
Acronis Clean
ADMINUSLabs Clean
AILabs (MONITORAPP) Clean
AlienVault Clean
benkow.cc Clean
BitDefender Clean
BlockList Clean
Blueliv Clean
Certego Clean
CINS Army Clean
CMC Threat Intelligence Clean


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Capturing from 2 interfaces
No. Time Source Destination Protocol Lengt Info
1 0.000000000 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x8bbf A www.msftconnecttest.com
2 0.001970200 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x8bbf A www.msftconnecttest.com A 192.0.2.123
3 0.004170300 192.168.245.129 192.168.245.129 TCP 56 49690 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4 0.004935500 192.0.2.123 192.168.245.129 TCP 56 80 → 49690 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
5 0.005623700 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
6 0.006585200 192.168.245.129 192.168.245.129 HTTP 155 GET /connecttest.txt HTTP/1.1
7 0.007653800 192.0.2.123 192.168.245.129 TCP 44 80 → 49690 [ACK] Seq=1 Ack=112 Win=262656 Len=0
8 0.010947500 192.0.2.123 192.168.245.129 TCP 169 80 → 49690 [PSH, ACK] Seq=1 Ack=112 Win=262656 Len=125 [TCP PDU reassembled in 9]
9 0.011573600 192.0.2.123 192.168.245.129 HTTP 1491 HTTP/1.0 200 OK (text/plain)
10 0.012505000 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [ACK] Seq=112 Ack=1574 Win=262656 Len=0
11 0.013030600 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [FIN, ACK] Seq=112 Ack=1574 Win=262656 Len=0
12 0.013499700 192.0.2.123 192.168.245.129 TCP 44 80 → 49690 [ACK] Seq=1574 Ack=113 Win=262656 Len=0
13 3.105878900 fe80::b6e2:8637:92f… ff02::1:2 DHCPv6 157 Solicit XID: 0xf19a32 CID: 000100013163ebc7000c29e24767
Frame 1: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_Loopback, id 1
Null/Loopback
Internet Protocol Version 4, Src: 192.168.245.129, Dst: 192.168.245.129
User Datagram Protocol, Src Port: 52568, Dst Port: 53
Domain Name System (query)
0000 02 00 00 00 45 00 00 45 9b f1 00 00 80 11 00 00
0010 c0 a8 f5 81 c0 a8 f5 81 cd 58 00 35 00 31 28 d1
0020 8b bf 01 00 00 01 00 00 00 00 00 00 03 77 77 77
0030 0f 6d 73 66 74 63 6f 6e 6e 65 63 74 74 65 73 74
0040 03 63 6f 6d 00 00 01 00 01
2 interfaces: <live capture in progress>
Packets: 13
Profile: Default
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Capturing from 2 interfaces
No. Time Source Destination Protocol Lengt Info
1 0.000000000 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x8bbf A www.msftconnecttest.com
2 0.001970200 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x8bbf A www.msftconnecttest.com A 192.0.2.123
3 0.004170300 192.168.245.129 192.168.245.129 TCP 56 49690 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4 0.004935500 192.0.2.123 192.168.245.129 TCP 56 80 → 49690 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
5 0.005623700 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
6 0.006585200 192.168.245.129 192.168.245.129 HTTP 155 GET /connecttest.txt HTTP/1.1
7 0.007653800 192.0.2.123 192.168.245.129 TCP 44 80 → 49690 [ACK] Seq=1 Ack=112 Win=262656 Len=0
8 0.010947500 192.0.2.123 192.168.245.129 TCP 169 80 → 49690 [PSH, ACK] Seq=1 Ack=112 Win=262656 Len=125 [TCP PDU reassembled in 9]
9 0.011573600 192.0.2.123 192.168.245.129 HTTP 1491 HTTP/1.0 200 OK (text/plain)
10 0.012505000 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [ACK] Seq=112 Ack=1574 Win=262656 Len=0
11 0.013030600 192.168.245.129 192.168.245.129 TCP 44 49690 → 80 [FIN, ACK] Seq=112 Ack=1574 Win=262656 Len=0
12 0.013499700 192.0.2.123 192.168.245.129 TCP 44 80 → 49690 [ACK] Seq=1574 Ack=113 Win=262656 Len=0
13 3.105878900 fe80::b6e2:8637:92f… ff02::1:2 DHCPv6 157 Solicit XID: 0xf19a32 CID: 000100013163ebc7000c29e24767
Frame 6: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Device\NPF_Loopback, id 1
Null/Loopback
Internet Protocol Version 4, Src: 192.168.245.129, Dst: 192.168.245.129
Transmission Control Protocol, Src Port: 49690, Dst Port: 80, Seq: 1, Ack: 1, Len: 111
Hypertext Transfer Protocol
GET /connecttest.txt HTTP/1.1\r\n
Connection: Close\r\n
User-Agent: Microsoft NCSI\r\n
Host: www.msftconnecttest.com\r\n
\r\n
[Full request URI: http://www.msftconnecttest.com/connecttest.txt]
0000 02 00 00 00 45 00 00 97 52 18 40 00 80 06 00 00
0010 c0 a8 f5 81 c0 a8 f5 81 c2 1a 00 50 de c9 a3 e6
0020 f7 73 7e 52 50 18 04 02 c4 f2 00 00 47 45 54 20
0030 2f 63 6f 6e 6e 65 63 74 74 65 73 74 2e 74 78 74
0040 20 48 54 54 50 2f 31 2e 31 0d 0a 43 6f 6e 6e 65
0050 63 74 69 6f 6e 3a 20 43 6c 6f 73 65 0d 0a 55 73
0060 65 72 2d 41 67 65 6e 74 3a 20 4d 69 63 72 6f 73
0070 6f 66 74 20 4e 43 53 49 0d 0a 48 6f 73 74 3a 20
0080 77 77 77 2e 6d 73 66 74 63 6f 6e 6e 65 63 74 74
0090 65 73 74 2e 63 6f 6d 0d 0a
HTTP Host (http.host), 31 bytes
Packets: 13
Profile: Default
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
wazuh
Capturing from 2 interfaces
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Lengtl Info
14 23.899462700 192.168.245.129 192.168.245.129 DNS 73 Standard query 0xba52 A ctldl.windowsupdate.com
15 23.901424100 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0xba52 A ctldl.windowsupdate.com A 192.0.2.123
26 23.921245900 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xf5a5 A oneocsp.microsoft.com
27 23.922545500 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xf5a5 A oneocsp.microsoft.com A 192.0.2.123
38 23.937482900 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x0b8d A ctldl.windowsupdate.com
39 23.938844100 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x0b8d A ctldl.windowsupdate.com A 192.0.2.123
50 25.062507100 192.168.245.129 192.168.245.129 DNS 73 Standard query 0xaeca A www.msftconnecttest.com
51 25.064468400 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0xaeca A www.msftconnecttest.com A 192.0.2.123
62 36.964899900 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x5836 A ctldl.windowsupdate.com
63 36.967001700 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x5836 A ctldl.windowsupdate.com A 192.0.2.123
74 36.986778000 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xefcf A oneocsp.microsoft.com
75 36.988597700 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xefcf A oneocsp.microsoft.com A 192.0.2.123
86 37.009583100 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xfcca A oneocsp.microsoft.com
87 37.011311300 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xfcca A oneocsp.microsoft.com A 192.0.2.123
98 37.037135800 192.168.245.129 192.168.245.129 DNS 71 Standard query 0x9522 A oneocsp.microsoft.com
99 37.094759800 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0x9522 A oneocsp.microsoft.com A 192.0.2.123
110 37.114680300 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x36f8 A ctldl.windowsupdate.com
111 37.116736500 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x36f8 A ctldl.windowsupdate.com A 192.0.2.123
122 55.125186900 192.168.245.129 192.168.245.129 DNS 73 Standard query 0xe8a3 A www.msftconnecttest.com
123 55.127379700 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0xe8a3 A www.msftconnecttest.com A 192.0.2.123
Frame 51: Packet, 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface \Device\NPF_Loopback, id 1
Null/Loopback
Internet Protocol Version 4, Src: 192.168.245.129, Dst: 192.168.245.129
User Datagram Protocol, Src Port: 53, Dst Port: 62980
Domain Name System (response)
Transaction ID: 0xaeca
Flags: 0x8580 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries
Answers
[Request In: 50]
[Time: 1.961300 milliseconds]
0000 02 00 00 00 45 00 00 55 9c 0e 00 00 80 11 00 00 ····E··U ····
0010 c0 a8 f5 81 c0 a8 f5 81 00 35 f6 04 00 41 c9 f5 ·····5···A··
0020 ae ca 85 80 00 01 00 01 00 00 00 00 03 77 77 77 ····· ·····www
0030 0f 6d 73 66 74 63 6f 6e 6e 65 63 74 74 65 73 74 ·msftcon necttest
0040 03 63 6f 6d 00 00 01 00 01 c0 0c 00 01 00 01 00 ·com···· ····
0050 00 00 00 00 04 c0 00 02 7b ···· {
Domain Name System: Protocol
Packets: 133 · Displayed: 22 (16.5%)
Profile: Default
11:39 AM
4/13/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
11:39 AM
4/13/2026
Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
wazuh
Capturing from 2 interfaces
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x8bbf A www.msftconnecttest.com
2 0.001970200 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x8bbf A www.msftconnecttest.com A 192.0.2.123
14 23.899462700 192.168.245.129 192.168.245.129 DNS 73 Standard query 0xba52 A ctldl.windowsupdate.com
15 23.901424100 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0xba52 A ctldl.windowsupdate.com A 192.0.2.123
26 23.921245900 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xf5a5 A oneocsp.microsoft.com
27 23.922545500 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xf5a5 A oneocsp.microsoft.com A 192.0.2.123
38 23.937482900 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x0b8d A ctldl.windowsupdate.com
39 23.938844100 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x0b8d A ctldl.windowsupdate.com A 192.0.2.123
50 25.062507100 192.168.245.129 192.168.245.129 DNS 73 Standard query 0xaeca A www.msftconnecttest.com
51 25.064468400 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0xaeca A www.msftconnecttest.com A 192.0.2.123
62 36.964899900 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x5836 A ctldl.windowsupdate.com
63 36.967001700 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x5836 A ctldl.windowsupdate.com A 192.0.2.123
74 36.986778000 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xefcf A oneocsp.microsoft.com
75 36.988597700 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xefcf A oneocsp.microsoft.com A 192.0.2.123
86 37.009583100 192.168.245.129 192.168.245.129 DNS 71 Standard query 0xfcca A oneocsp.microsoft.com
87 37.011311300 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0xfcca A oneocsp.microsoft.com A 192.0.2.123
98 37.037135800 192.168.245.129 192.168.245.129 DNS 71 Standard query 0x9522 A oneocsp.microsoft.com
99 37.094759800 192.168.245.129 192.168.245.129 DNS 87 Standard query response 0x9522 A oneocsp.microsoft.com A 192.0.2.123
110 37.114680300 192.168.245.129 192.168.245.129 DNS 73 Standard query 0x36f8 A ctldl.windowsupdate.com
111 37.116736500 192.168.245.129 192.168.245.129 DNS 89 Standard query response 0x36f8 A ctldl.windowsupdate.com A 192.0.2.123
Frame 123: Packet, 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface \Device\NPF_Loopback, id 1
Null/Loopback
Internet Protocol Version 4, Src: 192.168.245.129, Dst: 192.168.245.129
User Datagram Protocol, Src Port: 53, Dst Port: 57740
Domain Name System (response)
Transaction ID: 0xe8a3
Flags: 0x8580 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries
Answers
[Request In: 122]
[Time: 2.192800 milliseconds]
0000 02 00 00 00 45 00 00 55 9c 38 00 00 80 11 00 00 ····E··U ·8····
0010 c0 a8 f5 81 c0 a8 f5 81 00 35 e1 8c 00 41 a4 94 ·····5···A··
0020 e8 a3 85 80 00 01 00 01 00 00 00 00 03 77 77 77 ····· ·····www
0030 0f 6d 73 66 74 63 6f 6e 6e 65 63 74 74 65 73 74 ·msftcon necttest
0040 03 63 6f 6d 00 00 01 00 01 c0 0c 00 01 00 01 00 ·com···· ····
0050 00 00 00 00 04 c0 00 02 7b ···· {
Domain Name System: Protocol
Packets: 133 · Displayed: 22 (16.5%)
Profile: Default
11:39 AM
4/13/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
ENG
11:39 AM
4/13/2026

5. Indicators of Compromise (IOCs)

5.1 File Hashes

Hash Type	Value
SHA256	b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf
MD5	Not available
SHA1	Not available

5.2 Email Addresses

Type	Value	Context
Threat Actor Email	3905960807@qq.com	Ransom payment coordination — hardcoded in binary

5.3 File System Artefacts

Type	Path / Name	Context
Temporary File	%APPDATA%\Local\Temp\ICO5452.tmp	Dropped during execution — UI/icon resource
Process Executable	@WanaDecryptor@.exe	Ransom GUI executable spawned by dropper
Key Store Reference	Forever_AES_KEY	AES key storage artifact — exact path unresolved
Primary Sample Path	C:\Users\Eng-Ahmed\Desktop\<sample>.exe	Execution path in analysis environment

5.4 Registry Keys

Key Path	Value / Note	Confidence
HKCU\Control Panel\Desktop	WallPaper (modified to display ransom note)	High
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\...	Execution record entry	High
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache*	Interacted with by cmd.exe child process	Medium
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\	Interacted with by cmd.exe child process	Low — likely noise

5.5 Network IOCs

Type	Value	Confidence
Kill Switch Domain	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com	High (canonical WannaCry)
Kill Switch Domain (alt)	www.ifferfsodp9ifjaposdfjhgosurijfaewrwergwea.com	High (canonical WannaCry)
SMB Propagation	TCP Port 445 — outbound SYN flood pattern	High (canonical WannaCry)
Tor C2	Ports 80, 443, 9001, 9030 to Tor entry nodes	High (canonical WannaCry)
Ransom Contact	3905960807@qq.com	High (variant-specific)

6. Detection Rules

6.1 YARA Rules

Rule 1 — High-Confidence Variant Detection

```
rule WannaCry_Modified_QQ_Variant
{
  meta:
    author      = "Ahmed Emad Nasr"
    description = "Detects modified WannaCry variant with QQ-based ransom comms"
    date        = "2026-04-13"
    hash_sha256 = "b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf"
    severity    = "high"
  strings:
    $s1   = "@WanaDecryptor@"   ascii wide
    $s2   = "3905960807@qq.com" ascii wide
    $s3   = "Forever_AES_KEY"   ascii
    $s4   = "ICO5452.tmp"       ascii
    $api1 = "CryptEncrypt"      ascii
    $api2 = "CryptAcquireContextW" ascii
    $api3 = "CryptCreateHash"   ascii
  condition:
    uint16(0) == 0x5A4D and
    (2 of ($s*))          and
    (2 of ($api*))
}
```

Rule 2 — Behavioral Detection (Generic Ransomware)

```
rule Generic_Ransomware_CryptoAPI_Behavior
{
  meta:
    author      = "Ahmed Emad Nasr"
    description = "Generic ransomware detection via CryptoAPI usage and temp artifacts"
    date        = "2026-04-13"
    severity    = "medium"
  strings:
    $crypto1 = "CryptEncrypt"      ascii
    $crypto2 = "CryptDecrypt"      ascii
    $crypto3 = "CryptAcquireContext" ascii
    $file1   = ".tmp"              ascii
    $file2   = "AppData\Local\Temp" ascii wide
    $proc1   = "cmd.exe"           ascii
    $proc2   = "conhost.exe"       ascii
  condition:
    uint16(0) == 0x5A4D and
    (2 of ($crypto*))    and
    (1 of ($file*))      and
    (1 of ($proc*))
}
```

Rule 3 — Strict IOC-Based Detection

```
rule WannaCry_IOC_Strict
```

```
{
  meta:
    author      = "Ahmed Emad Nasr"
    description = "Strict IOC-based detection for analyzed WannaCry variant"
    date        = "2026-04-13"
    confidence  = "very_high"
  strings:
    $email     = "3905960807@qq.com" ascii wide
    $tmp       = "ICO5452.tmp"       ascii
    $folder    = "Forever_AES_KEY"   ascii
    $decryptor = "@WanaDecryptor@"   ascii wide
  condition:
    uint16(0) == 0x5A4D and
    all of them
}
```

6.2 Sigma Rules (SIEM)

Rule 1 — Process Execution Detection (Critical)

```
title: WannaCry Modified Variant Execution (QQ-Based)
id: 8c2e3d9f-1b4c-4f4e-9a77-qq-wannacry-variant
status: experimental
description: >
  Detects execution of a modified WannaCry ransomware variant
  that uses a QQ email address for ransom communication.
author: Ahmed Emad Nasr
date: 2026/04/13
logsource:
  category: process_creation
  product: windows
detection:
  selection_proc:
    Image|endswith: '\@WanaDecryptor@.exe'
  selection_cmd:
    CommandLine|contains:
      - '3905960807@qq.com'
      - 'Forever_AES_KEY'
  selection_parent:
    ParentImage|endswith:
      - '\cmd.exe'
      - '\conhost.exe'
  condition: selection_proc or (selection_cmd and selection_parent)
falsepositives:
  - Unlikely
level: critical
```

Rule 2 — Behavioral Detection (High)

```
title: Suspicious CryptoAPI Usage with Temp File Activity
id: 91b3a1e2-crypto-ransomware-behavior
status: experimental
description: >
  Detects potential ransomware activity combining Windows CryptoAPI
  calls with temporary file creation in AppData.
author: Ahmed Emad Nasr
```

```
date: 2026/04/13
logsource:
  product: windows
  category: process_creation
detection:
  selection_crypto:
    CommandLine|contains:
      - 'CryptEncrypt'
      - 'CryptAcquireContext'
      - 'CryptCreateHash'
  selection_temp:
    CommandLine|contains:
      - 'AppData\Local\Temp'
      - '.tmp'
  selection_proc:
    Image|endswith:
      - '\cmd.exe'
      - '\powershell.exe'
  condition: (selection_crypto and selection_temp) or selection_proc
falsepositives:
  - Administrative scripts invoking CryptoAPI legitimately
level: high
```

7. MITRE ATT&CK Mapping

i Mapping is based on observed behaviors during static and dynamic analysis. Unobserved canonical WannaCry techniques are included where the variant's lineage and modified behavior strongly suggest their presence, and are explicitly labeled as 'Inferred.'

Tactic	Technique ID	Technique Name	Basis
Execution	T1204.002	User Execution: Malicious File	Sample executed manually by user
Execution	T1059.003	Command and Scripting Interpreter: Windows Command Shell	cmd.exe spawned as child process
Persistence	T1547.001	Boot or Logon Autostart: Registry Run Keys	Not confirmed; not ruled out — monitor
Defense Evasion	T1112	Modify Registry	Registry keys modified by cmd.exe child process
Defense Evasion	T1036	Masquerading	Payload filename uses hash as name; GUI mimics legitimate decryptor
Discovery	T1012	Query Registry	Registry queried by cmd.exe child process
Lateral Movement	T1210	Exploitation of Remote Services (EternalBlue/MS17-010)	Inferred — canonical WannaCry behavior; not observed in this session
Collection	T1005	Data from Local System	File system traversal implied by encryption routine
Command & Control	T1090.003	Proxy: Multi-hop Proxy (Tor)	Inferred — canonical WannaCry C2 channel
Impact	T1486	Data Encrypted for Impact	Core ransomware function — CryptoAPI usage confirmed
Impact	T1491.001	Defacement: Internal Defacement	Desktop wallpaper replaced via SystemParametersInfoW
Impact	T1657	Financial Theft (Extortion)	USD $300 ransom demand via QQ email

8. Risk Assessment

8.1 Impact Analysis

Impact Dimension	Rating	Justification
Data Confidentiality	HIGH	User files encrypted and inaccessible without decryption key
Data Integrity	HIGH	Files overwritten with ciphertext; wallpaper and registry modified
System Availability	CRITICAL	Encrypted files render the system operationally incapacitated
Financial Impact	HIGH	USD $300 ransom demand; potential data loss if unpaid and no backup exists
Reputational Impact	MEDIUM	Data breach implications if sensitive documents are encrypted or exfiltrated
Recovery Complexity	HIGH	Full disk wipe and OS reinstall recommended; decryption without key is not feasible

8.2 Affected Systems

- Primary Target: Windows-based systems (64-bit) — confirmed by PE32+ architecture.
- Suspected Target Demographic: Chinese-speaking users, based on the localized GUI language and QQ-platform contact method.
- Organizational Risk: Any Windows endpoint accessible to an end user capable of executing the file. Network-shared drives accessible from an infected host would also be at risk of encryption.

8.3 Threat Severity Justification

This sample is assessed as CRITICAL based on the following factors:

- It is functionally capable of encrypting user files using standard Windows cryptographic APIs, resulting in irreversible data loss without the decryption key.
- It actively modifies the system state (wallpaper, registry) to establish a persistent extortion interface.
- The ransom demand, while modest ($300), is operationally effective in targeting individuals who may lack data backups.
- The use of a QQ email address rather than anonymous cryptocurrency infrastructure may reduce operator security but does not diminish the destructive capability of the payload.
- Modern endpoint protection may detect this sample via heuristic and behavioral signatures; however, unprotected or legacy systems remain fully vulnerable.

9. Mitigation & Prevention Strategies

9.1 Immediate Response (Post-Infection)

- Immediately isolate the affected endpoint from all network segments to prevent lateral movement or further data exfiltration.
- Before powering down the system, capture a full memory (RAM) dump. In some ransomware variants, encryption keys may reside in volatile memory and can be recovered using forensic tools.
- Do not pay the ransom. Payment does not guarantee key delivery and incentivizes further attacks.
- Preserve a forensic image of the affected disk before remediation.
- Wipe the primary storage volume, reinstall the operating system, and restore data from a known-clean offline backup.

9.2 Technical Controls

Control	Implementation
Endpoint Detection & Response (EDR)	Deploy a capable EDR solution that monitors CryptoAPI call sequences, high-rate file modification, and process spawn chains. Ensure behavioral heuristics are enabled.
Application Allowlisting	Prevent execution of unsigned or unknown binaries from user-writable directories (Desktop, Temp, Downloads) via AppLocker or Windows Defender Application Control (WDAC).
SMB Hardening	Disable SMBv1 across all endpoints. Apply MS17-010 patch (KB4012212) if not already deployed. Block TCP 445 at perimeter firewall for inbound connections.
Network Segmentation	Implement micro-segmentation to limit lateral movement across subnets. Monitor for anomalous TCP SYN floods targeting Port 445.
DNS / Web Filtering	Block known WannaCry kill-switch domains and Tor exit/entry node IP ranges at the DNS and firewall level.
Backup Strategy	Maintain offline (air-gapped) or immutable backups (e.g., object-lock enabled cloud storage). Test restoration procedures regularly.
Email / File Gateway Filtering	Filter executable attachments (.exe, .scr, .bat) at mail gateway and web proxy. Block QQ email domain (@qq.com) at mail gateway if not operationally required.
SIEM / IDS Rules	Deploy the Sigma rules provided in Section 6.2 in your SIEM. Configure IDS/IPS with rules targeting the IOCs in Section 5.

9.3 User Awareness

- Train users to recognize and report suspicious executables, particularly those arriving via email, USB media, or file-sharing platforms.
- Communicate that ransomware often masquerades as legitimate utilities. Users should verify file provenance before execution.
- Establish a clear incident reporting channel so infections are escalated to the SOC immediately rather than being concealed.

9.4 System Hardening Recommendations

- Enforce the principle of least privilege: standard users should not have administrative rights. This limits the scope of file encryption to user-accessible paths.
- Enable Windows Defender Exploit Guard — specifically the Controlled Folder Access feature, which blocks unauthorized writes to protected directories.
- Disable Remote Desktop Protocol (RDP) if not operationally required, or restrict to VPN-only access with multi-factor authentication.
- Keep the operating system and all software fully patched. EternalBlue (MS17-010) has had a patch available since March 2017; any unpatched system represents an unacceptable residual risk.
- Monitor and alert on BAM registry key entries for newly executed binaries as a low-cost execution telemetry mechanism.

10. Conclusion

The analyzed sample is confirmed to be a functional modified variant of the WannaCry ransomware family. Its core destructive capability — file encryption via Windows Cryptography APIs — is intact and effective. The primary deviations from the canonical 2017 WannaCry strain are the replacement of the Bitcoin-based payment mechanism with a direct QQ email contact, the presentation of a Chinese-language ransom GUI, and the absence of observed self-propagation via the EternalBlue SMB exploit.

These modifications suggest the variant was purpose-built for targeted, manual delivery to a specific demographic, rather than designed for global autonomous propagation. The threat actor's use of a QQ email address reduces their operational anonymity compared to the original campaign's use of cryptocurrency wallets, which may reflect a lower operational security posture or a narrower target scope.

From a detection standpoint, this variant is not stealthy: it uses no packing, no obfuscation, and no anti-analysis techniques. Static YARA rules targeting the hardcoded strings and API imports will detect this sample with high confidence. Behavioral detection via EDR or SIEM Sigma rules will identify the characteristic process tree and registry modification patterns.

10.1 Key Takeaways

- The sample is a CRITICAL severity threat capable of irreversible file encryption on unprotected Windows systems.
- Static detection is highly reliable due to the absence of obfuscation. YARA rules are effective.
- Behavioral and network-based detection should be layered alongside static rules for defense-in-depth.
- The variant's modified ransom delivery mechanism (QQ email, Chinese GUI) is consistent with regional targeting, not global propagation.
- SMB patch (MS17-010) and SMBv1 disablement remain critical baseline controls, irrespective of this variant's observed propagation behavior.
- Air-gapped and immutable backups are the most effective recovery control against ransomware of this class.

Analyst Statement

All findings in this report are based solely on artefacts observed during static and dynamic analysis of the submitted sample, supplemented by established threat intelligence for canonical WannaCry behaviors. Inferred behaviors are explicitly labeled. No findings have been fabricated, extrapolated without basis, or sourced from unverifiable references.

11. References & Attachments

11.1 External References

- VirusTotal Detection Report: https://www.virustotal.com/gui/file/b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf/detection
- Hybrid-Analysis Report: https://hybrid-analysis.com/sample/b547dc7a77af8022abbf19a7006213342444caea1cede20ea2409ce9bc9790bf/69aab363018f1640f403ef85
- MITRE ATT&CK — WannaCry Software Entry: https://attack.mitre.org/software/S0366/
- Microsoft Security Bulletin MS17-010 (EternalBlue): https://docs.microsoft.com/en-us/security-updates/securitybulletins/2017/ms17-010

11.2 Analysis Artefact Attachments

Artefact	File Name	Description
Process Monitor Log	3omdaaaa.CSV	Raw ProcMon execution log capturing all system calls
ProcDOT Graph	default.pd	Visual behavioral graph of the process execution hierarchy
FakeNet-NG Log	(filename not specified)	Network simulation log capturing all outbound connection attempts

12. SIEM Environment Setup

12.1 System Requirements

To ensure consistency and compatibility, the Mini SOC was deployed using a virtualized lab environment. The following components were used:

- **Wazuh Manager Server:** Ubuntu 22.04 LTS, 8 GB RAM, 4 vCPUs
- **Windows Endpoint (Agent):** Windows 10 Pro, 4 GB RAM
- **Linux Endpoint (Agent):** Ubuntu 20.04 LTS, 4 GB RAM
- **Suricata IDS Sensor:** Ubuntu 22.04 LTS, 4 GB RAM
- **Database:** Elasticsearch (bundled with Wazuh stack)
- **Visualization Interface:** Wazuh Dashboard (Kibana-based)

12.2 Wazuh SIEM Installation and Configuration

12.2.1 Installation Steps

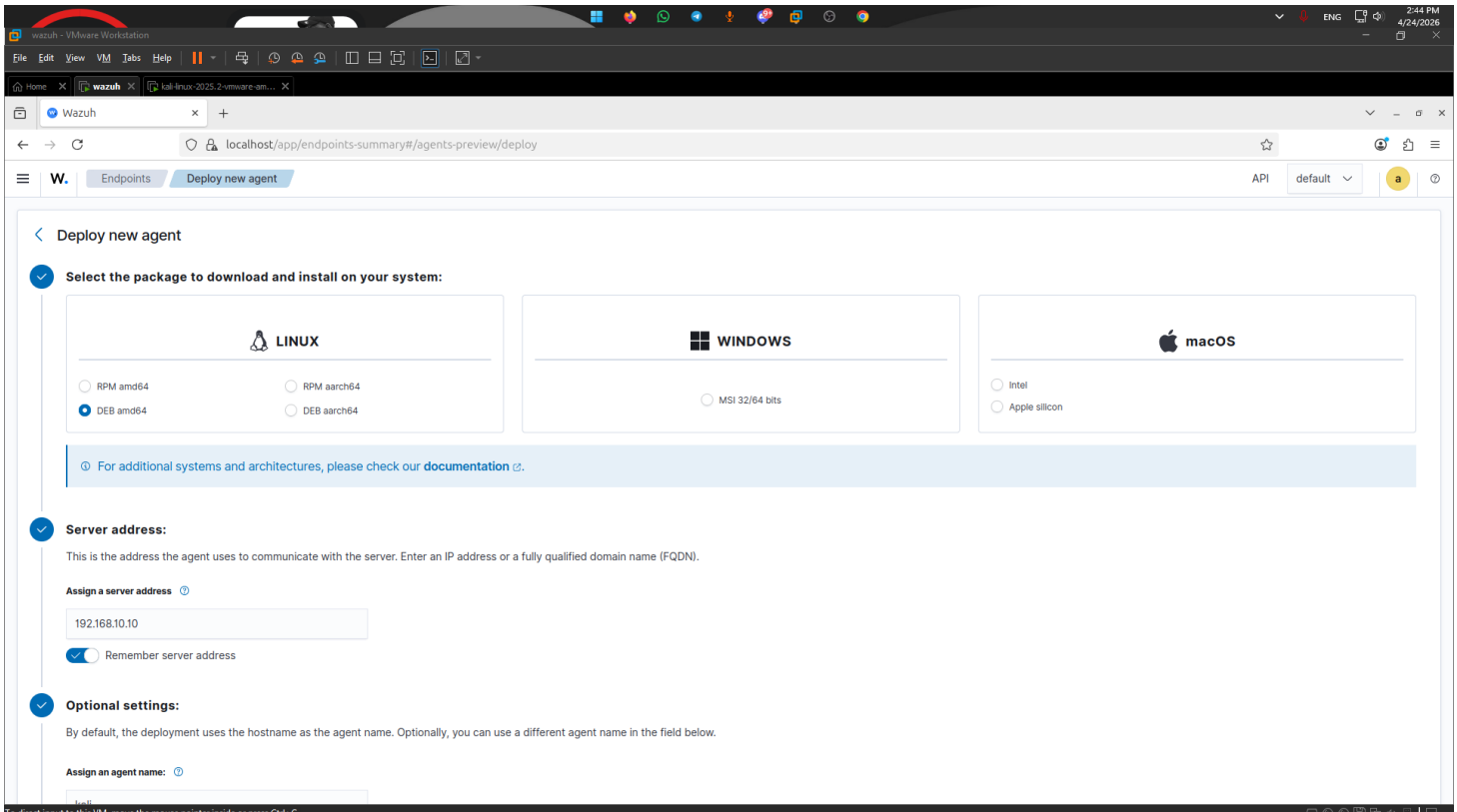
1. Update and upgrade system packages: sudo apt update && sudo apt upgrade -y
2. Install Wazuh repository and GPG key.
3. Install Wazuh Manager, Filebeat, and Wazuh Indexer components.
4. Configure Wazuh Dashboard for centralized monitoring.

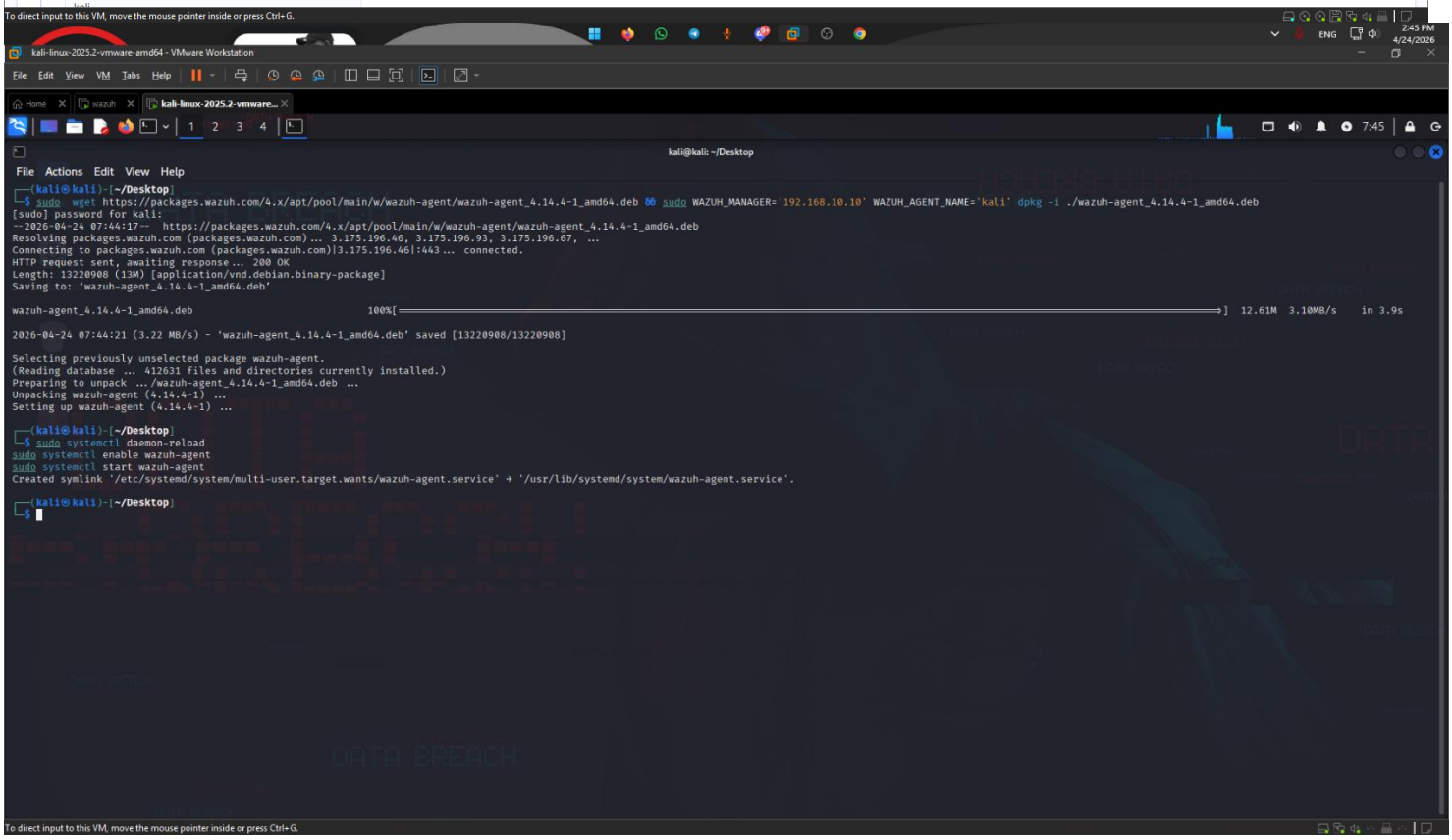
12.2.2 Agent Registration

Each endpoint (Linux and Windows) was installed with a Wazuh agent. Agents were registered with the manager using sudo /var/ossec/bin/manage_agents. Communication was verified through heartbeat logs.

12.2.3 Rule Configuration

Default rulesets were enhanced with custom rules based on the MITRE ATT&CK framework. SOCFortress rules were imported to extend visibility into credential dumping and PowerShell misuse.


wazuh - VMware Workstation
File Edit View VM Tabs Help
Wazuh
localhost/app/endpoints-summary#/agents-preview/deploy
Endpoints
Deploy new agent
API
default
Deploy new agent
Select the package to download and install on your system:
LINUX
RPM amd64
RPM aarch64
DEB amd64
DEB aarch64
WINDOWS
MSI 32/64 bits
macOS
Intel
Apple silicon
For additional systems and architectures, please check our documentation.
Server address:
This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).
Assign a server address
192.168.10.10
Remember server address
Optional settings:
By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.
Assign an agent name:


To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/Desktop
File Actions Edit View Help
(kali㉿kali)-[~/Desktop]
$ sudo wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.4-1_amd64.deb && sudo WAZUH_MANAGER='192.168.10.10' WAZUH_AGENT_NAME='kali' dpkg -i ./wazuh-agent_4.14.4-1_amd64.deb
[sudo] password for kali:
--2026-04-24 07:44:17-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.4-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.175.196.46, 3.175.196.93, 3.175.196.67, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.175.196.46|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13220908 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.4-1_amd64.deb'
wazuh-agent_4.14.4-1_amd64.deb 100%[=>] 12.61M 3.10MB/s in 3.9s
2026-04-24 07:44:21 (3.22 MB/s) - 'wazuh-agent_4.14.4-1_amd64.deb' saved [13220908/13220908]
Selecting previously unselected package wazuh-agent.
(Reading database ... 412631 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.4-1_amd64.deb ...
Unpacking wazuh-agent (4.14.4-1) ...
Setting up wazuh-agent (4.14.4-1) ...
(kali㉿kali)-[~/Desktop]
$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
(kali㉿kali)-[~/Desktop]
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12.3 Suricata IDS Integration

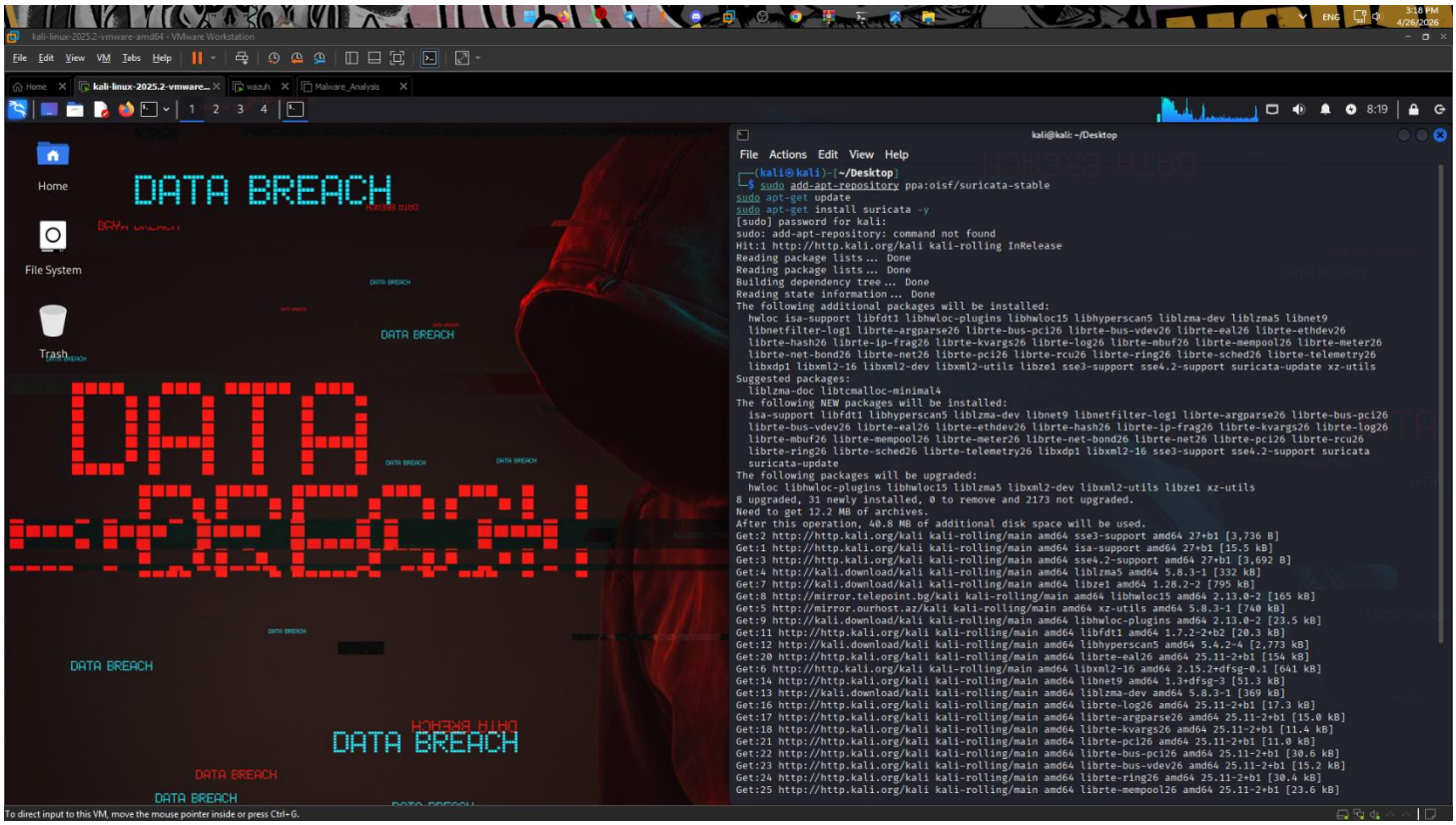
12.3.1 Installation & Setup

Suricata was installed via sudo apt install suricata and configured to monitor the virtual network interface. JSON logging (eve.json) was enabled for seamless Wazuh ingestion.

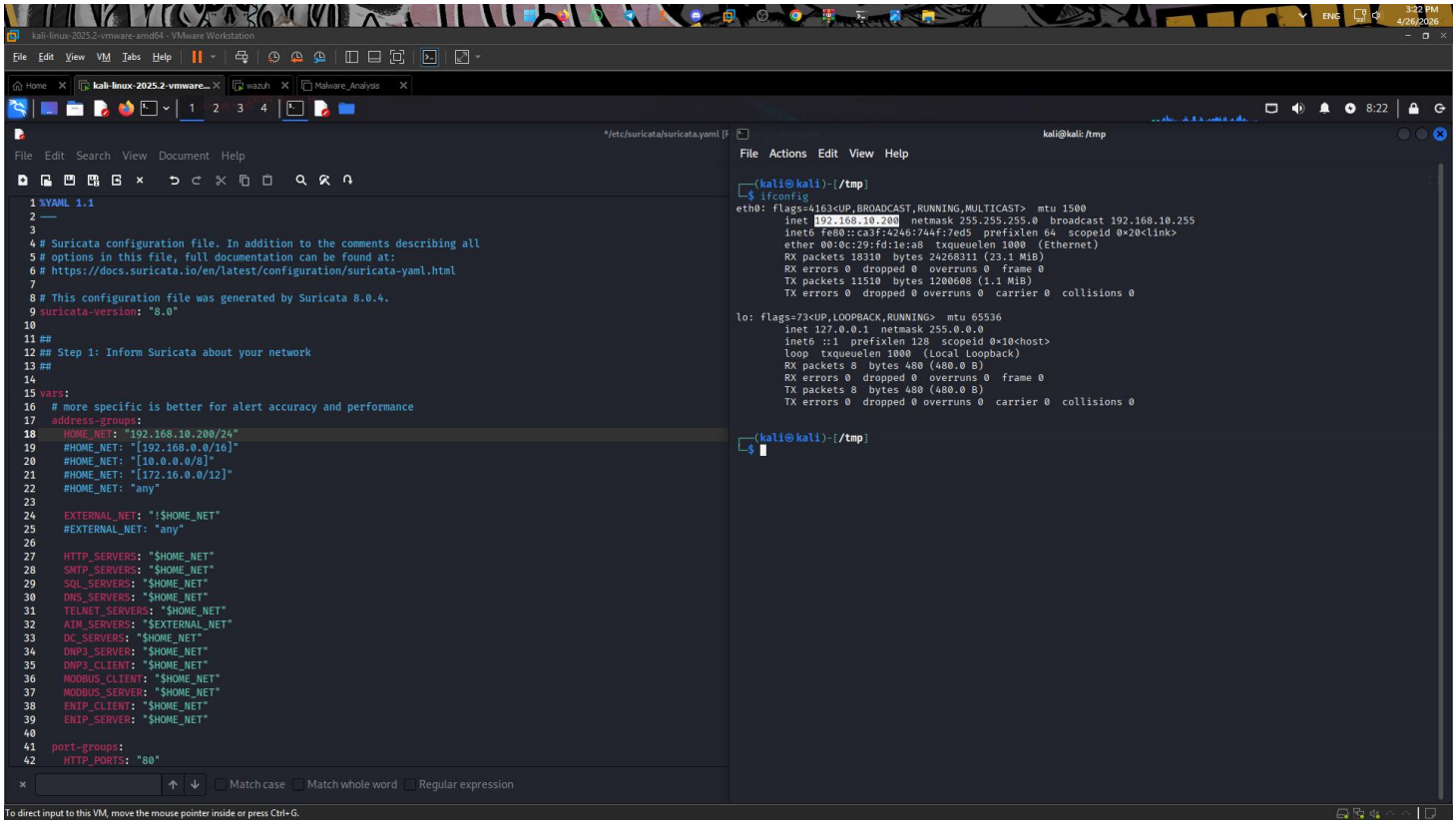
12.3.2 Log Forwarding to Wazuh

The Wazuh agent on the Suricata host was configured to monitor the eve.json file. Below is the configuration snippet added to ossec.conf:

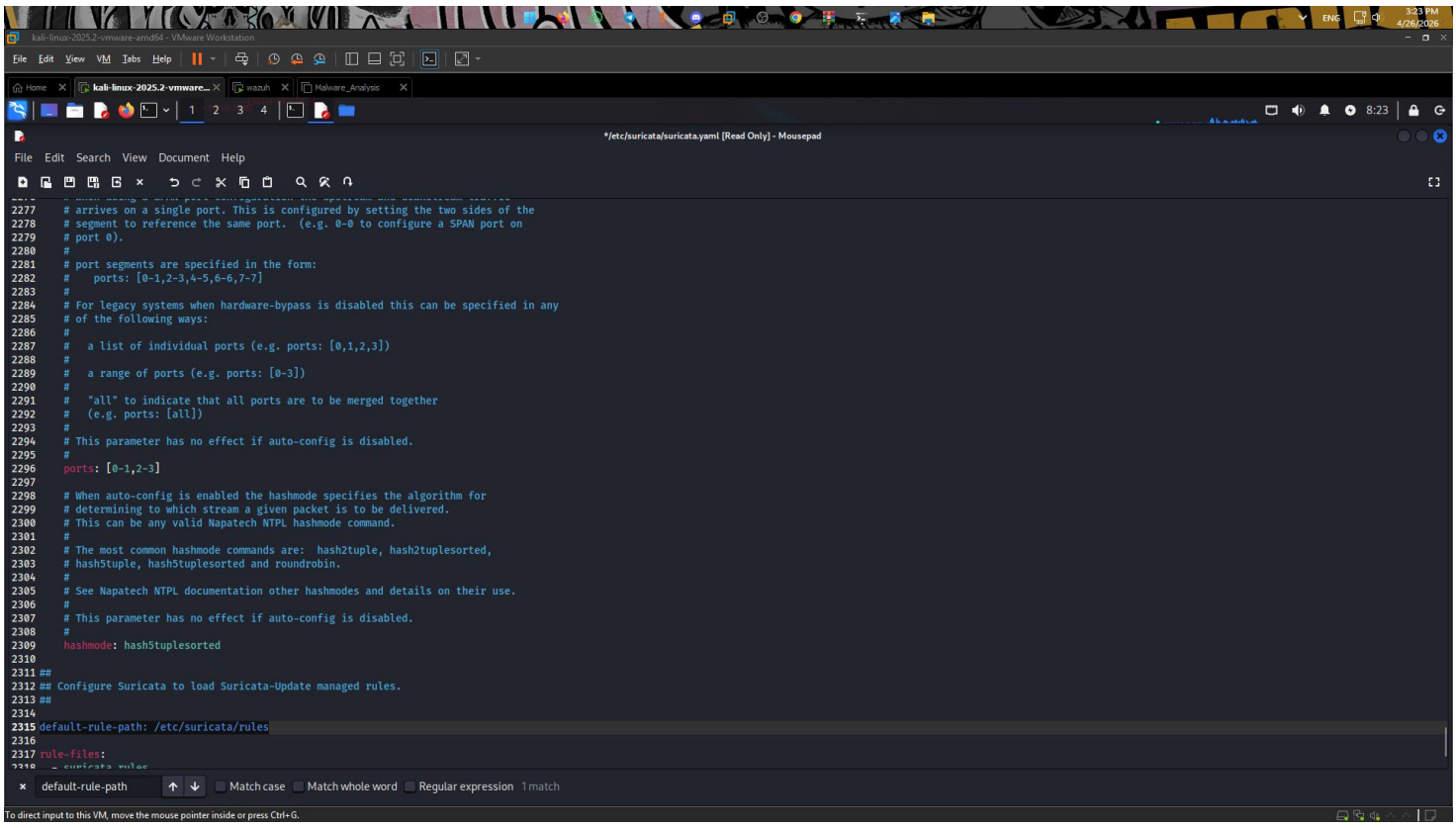
```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile
```

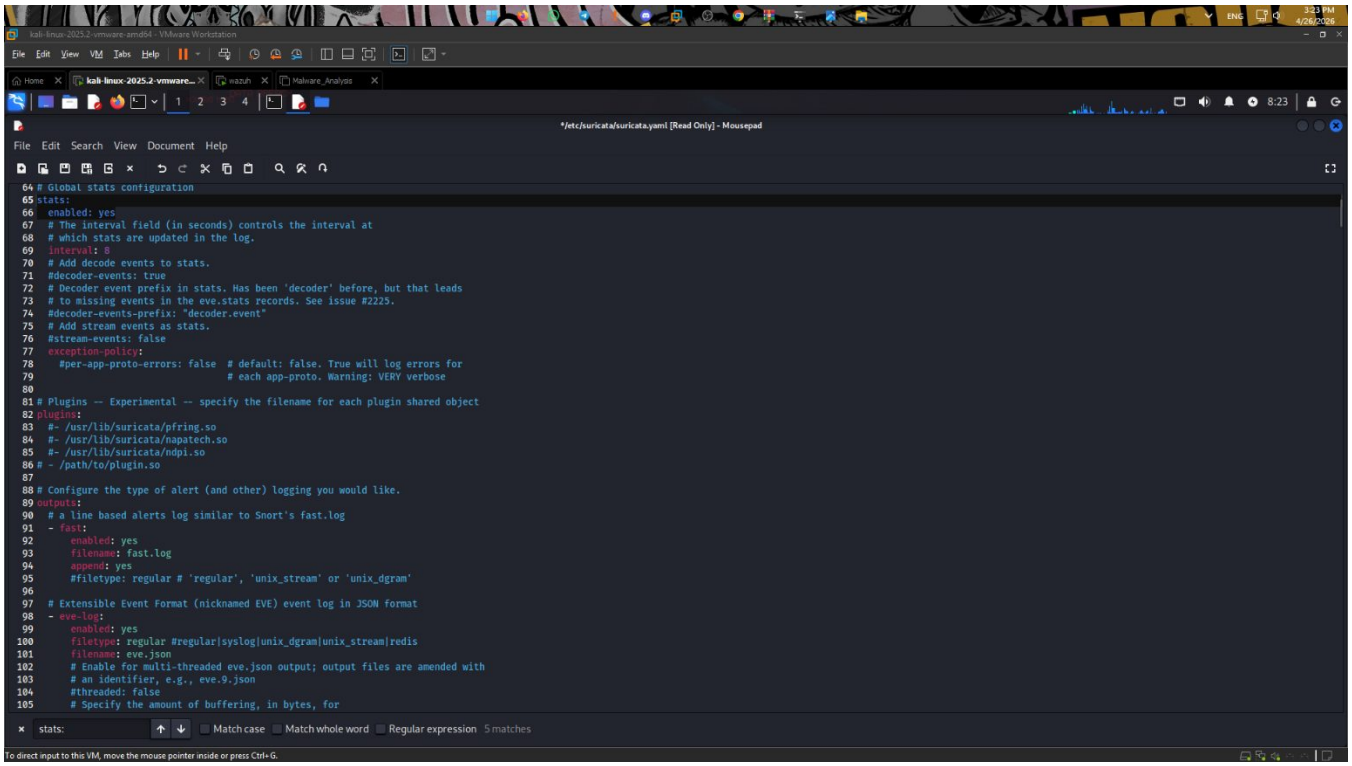


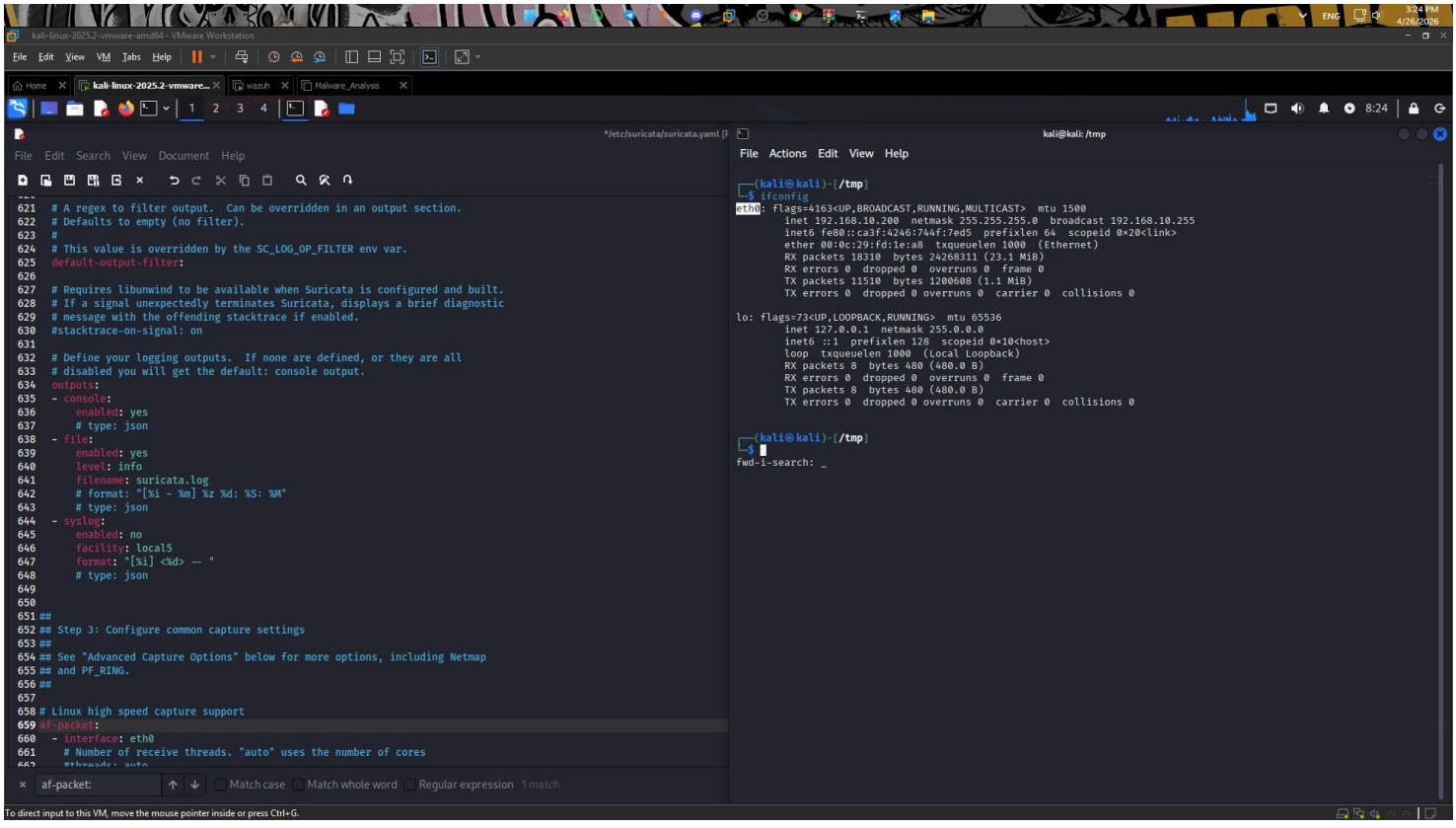
>

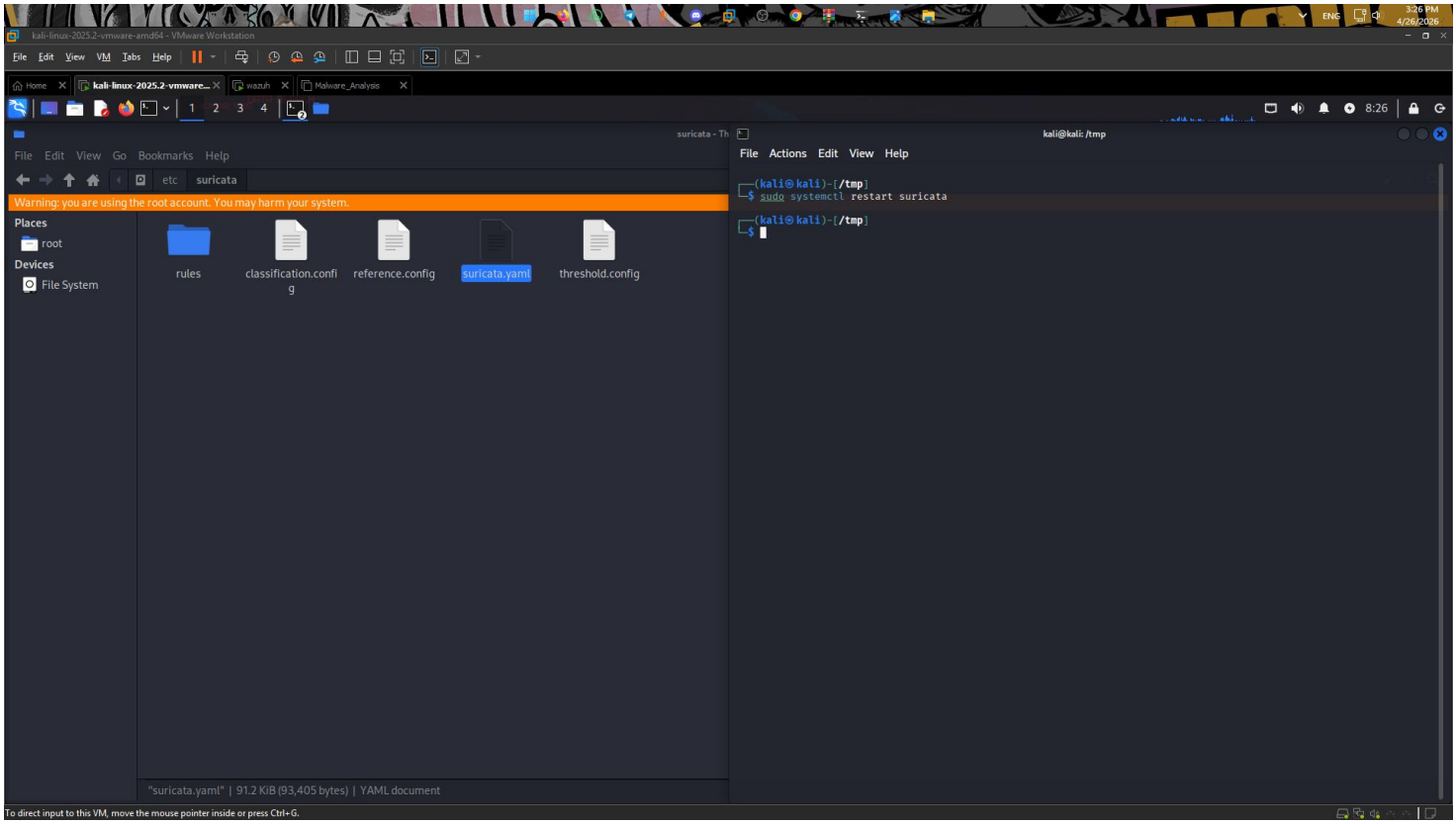

kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
*/etc/suricata/suricata.yaml [R
File Edit Search View Document Help
%YAML 1.1

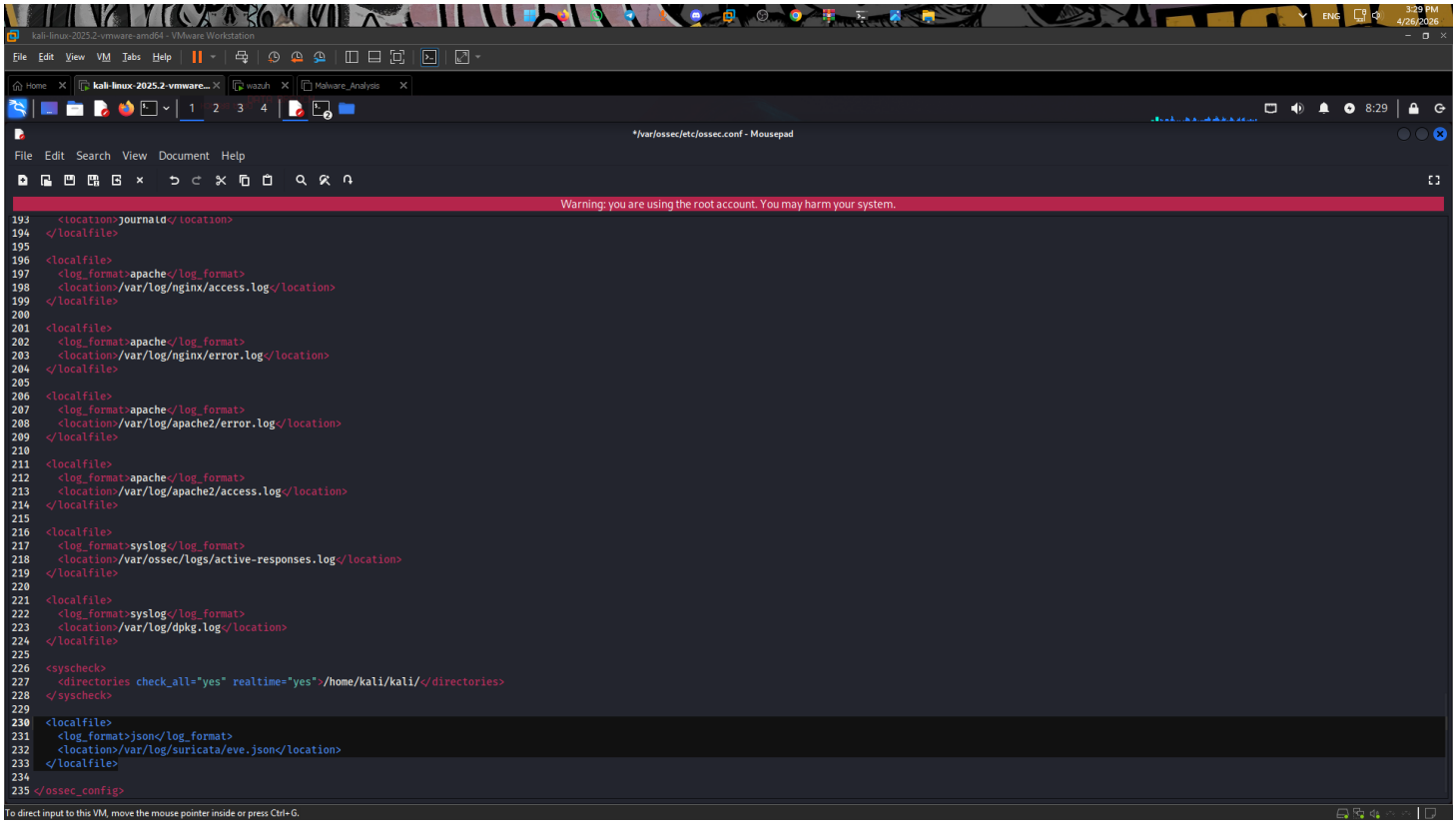
Suricata configuration file. In addition to the comments describing all
options in this file, full documentation can be found at:
https://docs.suricata.io/en/latest/configuration/suricata-yaml.html
This configuration file was generated by Suricata 8.0.4.
suricata-version: "8.0"
##
Step 1: Inform Suricata about your network
##
vars:
more specific is better for alert accuracy and performance
address-groups:
HOME_NET: "192.168.10.200/24"
#HOME_NET: "[192.168.0.0/16]"
#HOME_NET: "[10.0.0.0/8]"
#HOME_NET: "[172.16.0.0/12]"
#HOME_NET: "any"
EXTERNAL_NET: "!$HOME_NET"
#EXTERNAL_NET: "any"
HTTP_SERVERS: "$HOME_NET"
SMTP_SERVERS: "$HOME_NET"
SQL_SERVERS: "$HOME_NET"
DNS_SERVERS: "$HOME_NET"
TELNET_SERVERS: "$HOME_NET"
AIM_SERVERS: "$EXTERNAL_NET"
DC_SERVERS: "$HOME_NET"
DNP3_SERVER: "$HOME_NET"
DNP3_CLIENT: "$HOME_NET"
MODBUS_CLIENT: "$HOME_NET"
MODBUS_SERVER: "$HOME_NET"
ENIP_CLIENT: "$HOME_NET"
ENIP_SERVER: "$HOME_NET"
port-groups:
HTTP_PORTS: "80"
Match case
Match whole word
Regular expression
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
kali@kali: /tmp
File Actions Edit View Help
(kali㉿kali)-[/tmp]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.10.200 netmask 255.255.255.0 broadcast 192.168.10.255
inet6 fe80::ca3f:4246:744f:7ed5 prefixlen 64 scopeid 0x20<link>
ether 00:0c:29:fd:1e:a8 txqueuelen 1000 (Ethernet)
RX packets 18310 bytes 24268311 (23.1 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 11510 bytes 1200608 (1.1 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[/tmp]
$

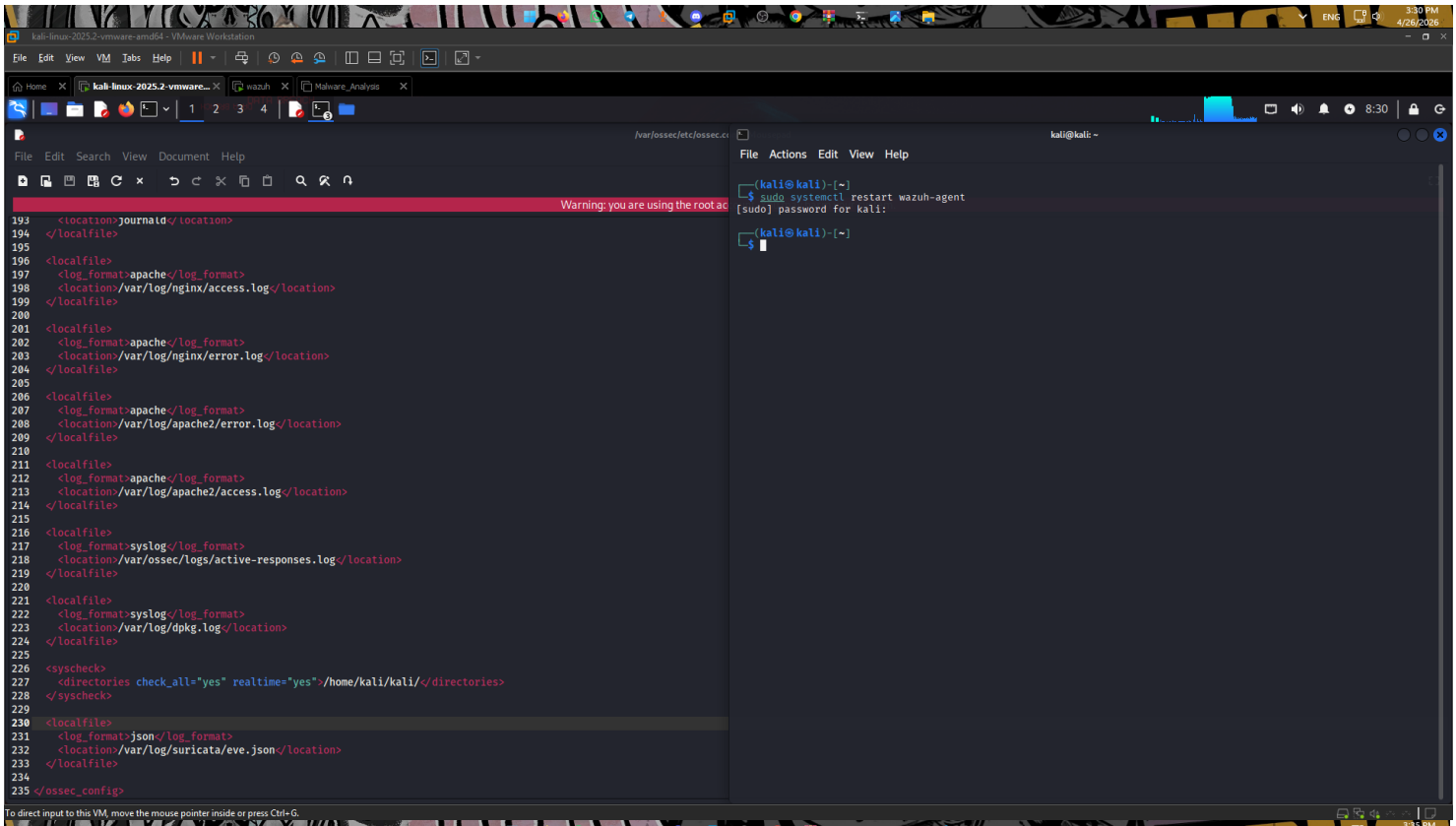

kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
*/etc/suricata/suricata.yaml [Read Only] - Mousepad
File Edit Search View Document Help
arrives on a single port. This is configured by setting the two sides of the
segment to reference the same port. (e.g. 0-0 to configure a SPAN port on
port 0).
#
port segments are specified in the form:
ports: [0-1,2-3,4-5,6-6,7-7]
#
For legacy systems when hardware-bypass is disabled this can be specified in any
of the following ways:
#
a list of individual ports (e.g. ports: [0,1,2,3])
#
a range of ports (e.g. ports: [0-3])
#
"all" to indicate that all ports are to be merged together
(e.g. ports: [all])
#
This parameter has no effect if auto-config is disabled.
#
ports: [0-1,2-3]
When auto-config is enabled the hashmode specifies the algorithm for
determining to which stream a given packet is to be delivered.
This can be any valid Napatech NTPL hashmode command.
#
The most common hashmode commands are: hash2tuple, hash2tuplesorted,
hash5tuple, hash5tuplesorted and roundrobin.
#
See Napatech NTPL documentation other hashmodes and details on their use.
#
This parameter has no effect if auto-config is disabled.
#
hashmode: hash5tuplesorted
##
Configure Suricata to load Suricata-Update managed rules.
##
default-rule-path: /etc/suricata/rules
rule-files:
default-rule-path
Match case
Match whole word
Regular expression
1 match
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

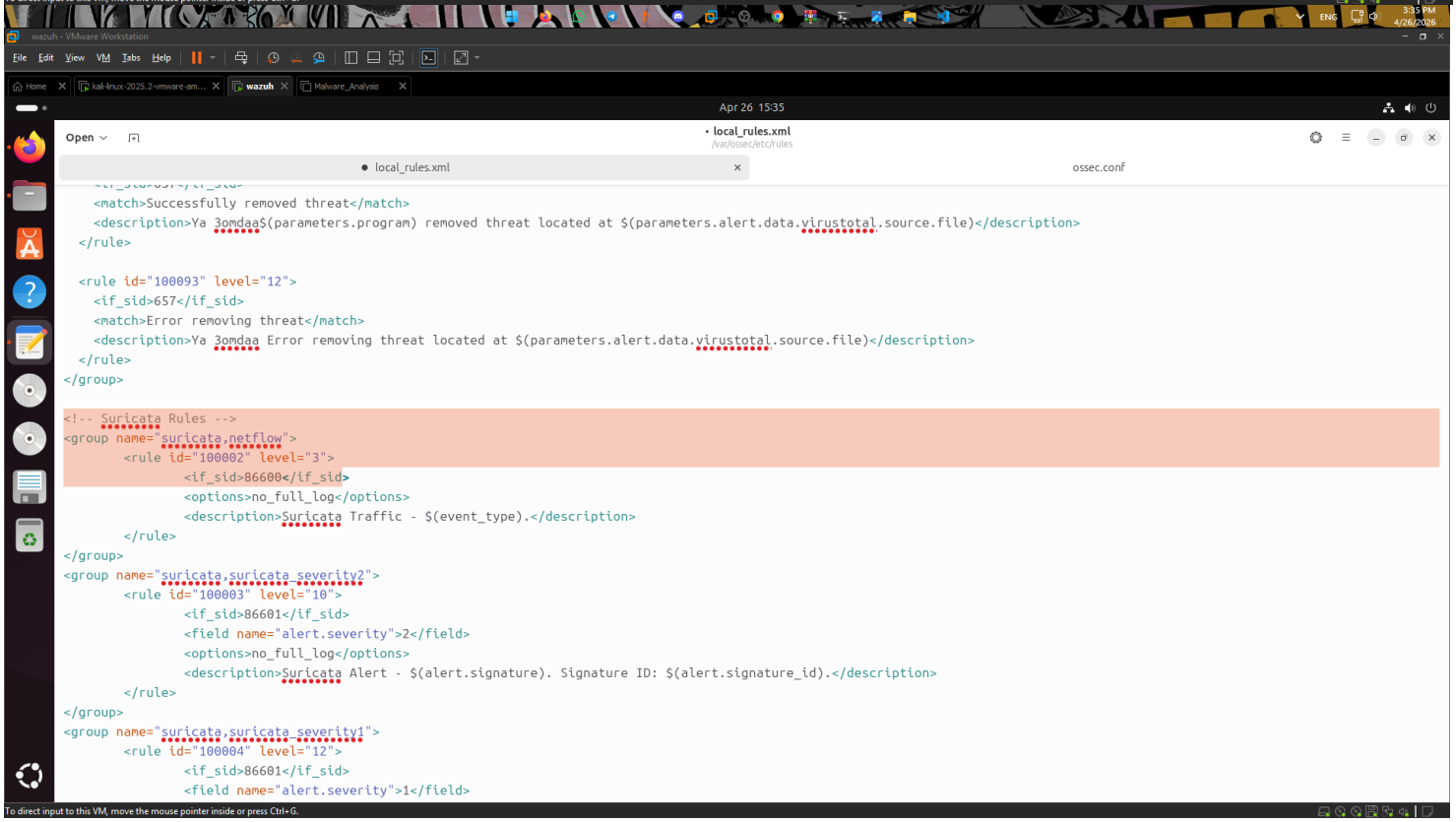

kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
*/etc/suricata/suricata.yaml [Read Only] - Mousepad
File Edit Search View Document Help
64 # Global stats configuration
65 stats:
66 enabled: yes
67 # The interval field (in seconds) controls the interval at
68 # which stats are updated in the log.
69 interval: 8
70 # Add decode events to stats.
71 #decoder-events: true
72 # Decoder event prefix in stats. Has been 'decoder' before, but that leads
73 # to missing events in the eve.stats records. See issue #2225.
74 #decoder-events-prefix: "decoder.event"
75 # Add stream events as stats.
76 #stream-events: false
77 exception-policy:
78 #per-app-proto-errors: false # default: false. True will log errors for
79 # each app-proto. Warning: VERY verbose
80
81 # Plugins -- Experimental -- specify the filename for each plugin shared object
82 plugins:
83 #- /usr/lib/suricata/pfring.so
84 #- /usr/lib/suricata/napatech.so
85 #- /usr/lib/suricata/ndpi.so
86 # - /path/to/plugin.so
87
88 # Configure the type of alert (and other) logging you would like.
89 outputs:
90 # a line based alerts log similar to Snort's fast.log
91 - fast:
92 enabled: yes
93 filename: fast.log
94 append: yes
95 #filetype: regular # 'regular', 'unix_stream' or 'unix_dgram'
96
97 # Extensible Event Format (nicknamed EVE) event log in JSON format
98 - eve-log:
99 enabled: yes
100 filetype: regular #regular|syslog|unix_dgram|unix_stream|redis
101 filename: eve.json
102 # Enable for multi-threaded eve.json output; output files are amended with
103 # an identifier, e.g., eve.9.json
104 #threaded: false
105 # Specify the amount of buffering, in bytes, for
stats:
Match case
Match whole word
Regular expression
5 matches
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

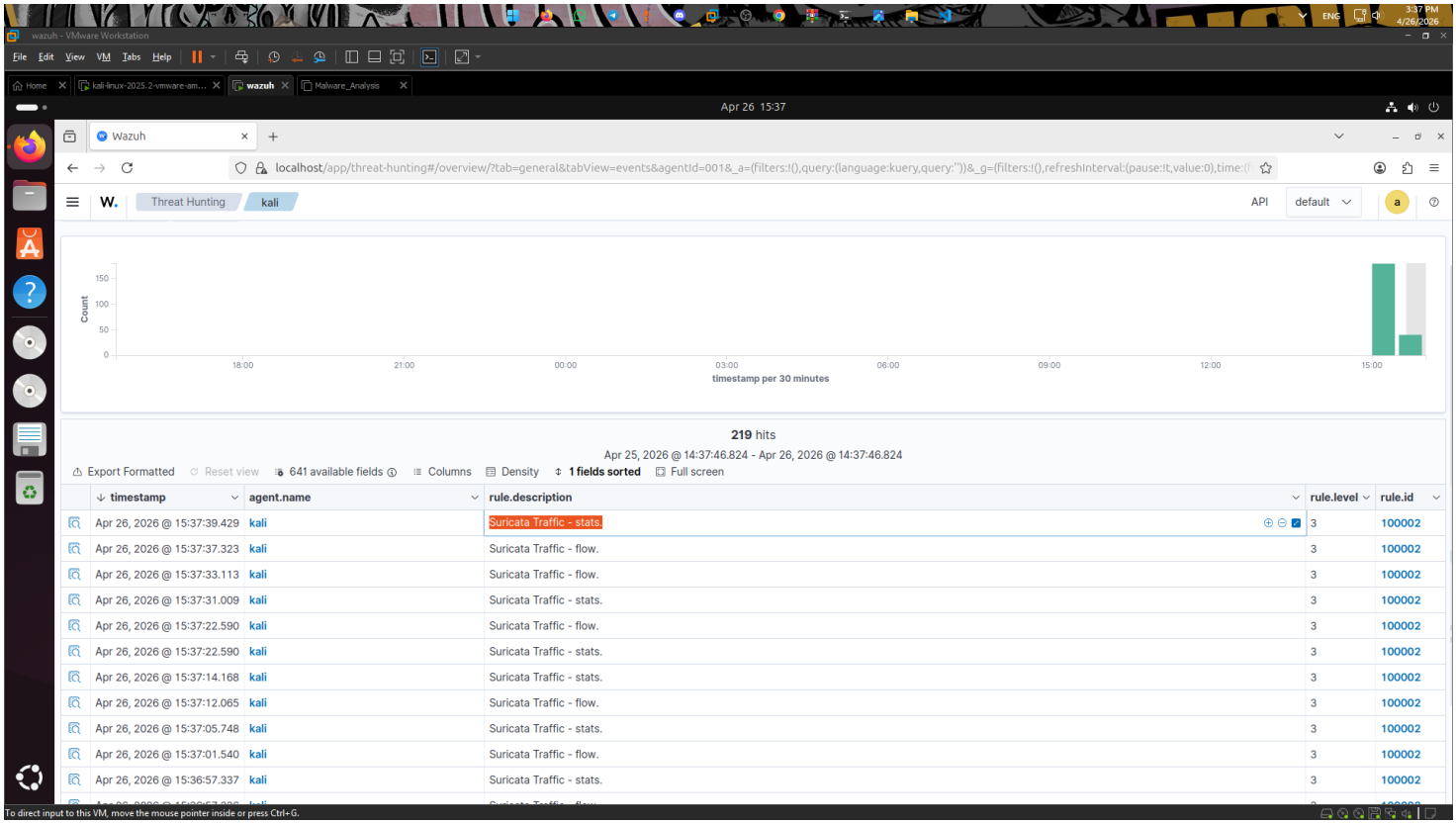

kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
*/etc/suricata/suricata.yaml [R
File Edit Search View Document Help
621 # A regex to filter output. Can be overridden in an output section.
622 # Defaults to empty (no filter).
623 #
624 # This value is overridden by the SC_LOG_OP_FILTER env var.
625 default-output-filter:
626
627 # Requires libunwind to be available when Suricata is configured and built.
628 # If a signal unexpectedly terminates Suricata, displays a brief diagnostic
629 # message with the offending stacktrace if enabled.
630 #stacktrace-on-signal: on
631
632 # Define your logging outputs. If none are defined, or they are all
633 # disabled you will get the default: console output.
634 outputs:
635 - console:
636 enabled: yes
637 # type: json
638 - file:
639 enabled: yes
640 level: info
641 filename: suricata.log
642 # format: "[%i - %m] %z %d: %S: %M"
643 # type: json
644 - syslog:
645 enabled: no
646 facility: local5
647 format: "[%i] <%d> -- "
648 # type: json
649
650
651 ##
652 ## Step 3: Configure common capture settings
653 ##
654 ## See "Advanced Capture Options" below for more options, including Netmap
655 ## and PF_RING.
656 ##
657
658 # Linux high speed capture support
659 af-packet:
660 - interface: eth0
661 # Number of receive threads. "auto" uses the number of cores
af-packet:
Match case
Match whole word
Regular expression
1 match
kali@kali: /tmp
File Actions Edit View Help
(kali㉿kali)-[/tmp]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.10.200 netmask 255.255.255.0 broadcast 192.168.10.255
inet6 fe80::ca3f:4246:744f:7ed5 prefixlen 64 scopeid 0×20<link>
ether 00:0c:29:fd:1e:a8 txqueuelen 1000 (Ethernet)
RX packets 18310 bytes 24268311 (23.1 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 11510 bytes 1200608 (1.1 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0×10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[/tmp]
$
fwd-i-search: _
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
suricata - Th
File Edit View Go Bookmarks Help
etc
suricata
Warning: you are using the root account. You may harm your system.
Places
root
Devices
File System
rules
classification.config
reference.config
suricata.yaml
threshold.config
"suricata.yaml" | 91.2 KiB (93,405 bytes) | YAML document
kali@kali: /tmp
File Actions Edit View Help
(kali㉿kali)-[/tmp]
$ sudo systemctl restart suricata
(kali㉿kali)-[/tmp]
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
*/var/ossec/etc/ossec.conf - Mousepad
File Edit Search View Document Help
Warning: you are using the root account. You may harm your system.
<location>journald</location>
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/nginx/access.log</location>
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/nginx/error.log</location>
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/apache2/error.log</location>
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/apache2/access.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/ossec/logs/active-responses.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/log/dpkg.log</location>
</localfile>
<syscheck>
<directories check_all="yes" realtime="yes">/home/kali/kali/</directories>
</syscheck>
<localfile>
<log_format>json</log_format>
<location>/var/log/suricata/eve.json</location>
</localfile>
</ossec_config>
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware...
wazuh
Malware_Analysis
/var/ossec/etc/ossec.c
File Edit Search View Document Help
Warning: you are using the root ac
193 <location>journald</location>
194 </localfile>
195
196 <localfile>
197 <log_format>apache</log_format>
198 <location>/var/log/nginx/access.log</location>
199 </localfile>
200
201 <localfile>
202 <log_format>apache</log_format>
203 <location>/var/log/nginx/error.log</location>
204 </localfile>
205
206 <localfile>
207 <log_format>apache</log_format>
208 <location>/var/log/apache2/error.log</location>
209 </localfile>
210
211 <localfile>
212 <log_format>apache</log_format>
213 <location>/var/log/apache2/access.log</location>
214 </localfile>
215
216 <localfile>
217 <log_format>syslog</log_format>
218 <location>/var/ossec/logs/active-responses.log</location>
219 </localfile>
220
221 <localfile>
222 <log_format>syslog</log_format>
223 <location>/var/log/dpkg.log</location>
224 </localfile>
225
226 <syscheck>
227 <directories check_all="yes" realtime="yes">/home/kali/kali/</directories>
228 </syscheck>
229
230 <localfile>
231 <log_format>json</log_format>
232 <location>/var/log/suricata/eve.json</location>
233 </localfile>
234
235 </ossec_config>
kali@kali: ~
File Actions Edit View Help
(kali㉿kali)-[~]
$ sudo systemctl restart wazuh-agent
[sudo] password for kali:
(kali㉿kali)-[~]
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


wazuh - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.2-vmware-am...
wazuh
Malware_Analysis
Apr 26 15:35
Open
• local_rules.xml
/var/ossec/etc/rules
local_rules.xml
ossec.conf
<match>Successfully removed threat</match>
<description>Ya 3omdaa$(parameters.program) removed threat located at $(parameters.alert.data.virustotal.source.file)</description>
</rule>
<rule id="100093" level="12">
<if_sid>657</if_sid>
<match>Error removing threat</match>
<description>Ya 3omdaa Error removing threat located at $(parameters.alert.data.virustotal.source.file)</description>
</rule>
</group>
<!-- Suricata Rules -->
<group name="suricata,netflow">
<rule id="100002" level="3">
<if_sid>86600</if_sid>
<options>no_full_log</options>
<description>Suricata Traffic - $(event_type).</description>
</rule>
</group>
<group name="suricata,suricata_severity2">
<rule id="100003" level="10">
<if_sid>86601</if_sid>
<field name="alert.severity">2</field>
<options>no_full_log</options>
<description>Suricata Alert - $(alert.signature). Signature ID: $(alert.signature_id).</description>
</rule>
</group>
<group name="suricata,suricata_severity1">
<rule id="100004" level="12">
<if_sid>86601</if_sid>
<field name="alert.severity">1</field>
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

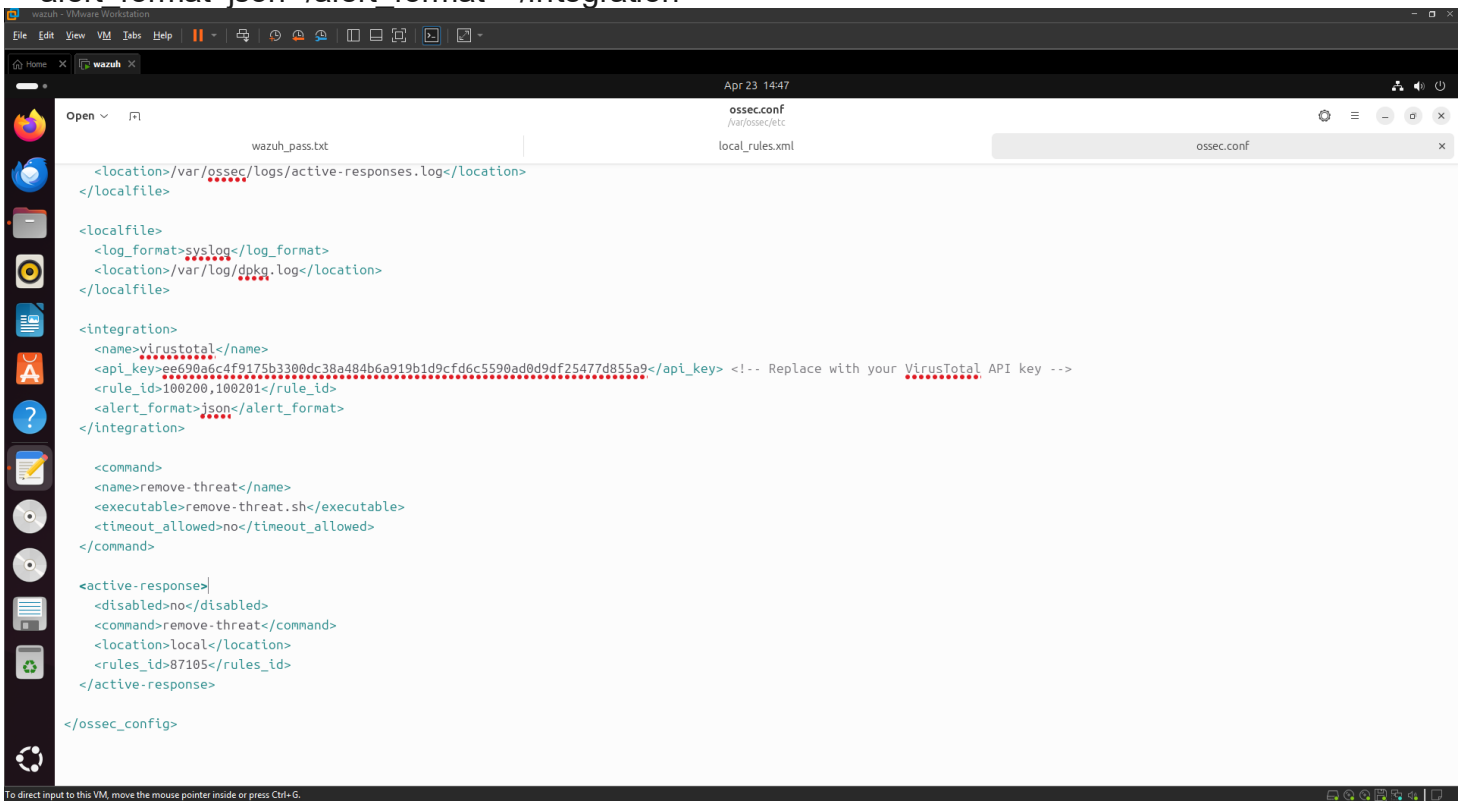

wazuh - VMware Workstation
Wazuh
localhost/app/threat-hunting#/overview/?tab=general&tabView=events&agentId=001&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(
Threat Hunting
kali
API
default
Count
timestamp per 30 minutes
219 hits
Apr 25, 2026 @ 14:37:46.824 - Apr 26, 2026 @ 14:37:46.824
Export Formatted
Reset view
641 available fields
Columns
Density
1 fields sorted
Full screen
timestamp
agent.name
rule.description
rule.level
rule.id
Apr 26, 2026 @ 15:37:39.429 kali Suricata Traffic - stats. 3 100002
Apr 26, 2026 @ 15:37:37.323 kali Suricata Traffic - flow. 3 100002
Apr 26, 2026 @ 15:37:33.113 kali Suricata Traffic - flow. 3 100002
Apr 26, 2026 @ 15:37:31.009 kali Suricata Traffic - stats. 3 100002
Apr 26, 2026 @ 15:37:22.590 kali Suricata Traffic - flow. 3 100002
Apr 26, 2026 @ 15:37:22.590 kali Suricata Traffic - stats. 3 100002
Apr 26, 2026 @ 15:37:14.168 kali Suricata Traffic - stats. 3 100002
Apr 26, 2026 @ 15:37:12.065 kali Suricata Traffic - flow. 3 100002
Apr 26, 2026 @ 15:37:05.748 kali Suricata Traffic - stats. 3 100002
Apr 26, 2026 @ 15:37:01.540 kali Suricata Traffic - flow. 3 100002
Apr 26, 2026 @ 15:36:57.337 kali Suricata Traffic - stats. 3 100002
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12.4 VirusTotal API Automation

12.4.1 Configuration

The VirusTotal v3 API key was integrated into the Wazuh Manager configuration. This allows for automated file reputation checks whenever a FIM (File Integrity Monitoring) event is triggered.

```
<integration>
  <name>virustotal</name>
  <api_key>YOUR_API_KEY_HERE</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format></integration>
```

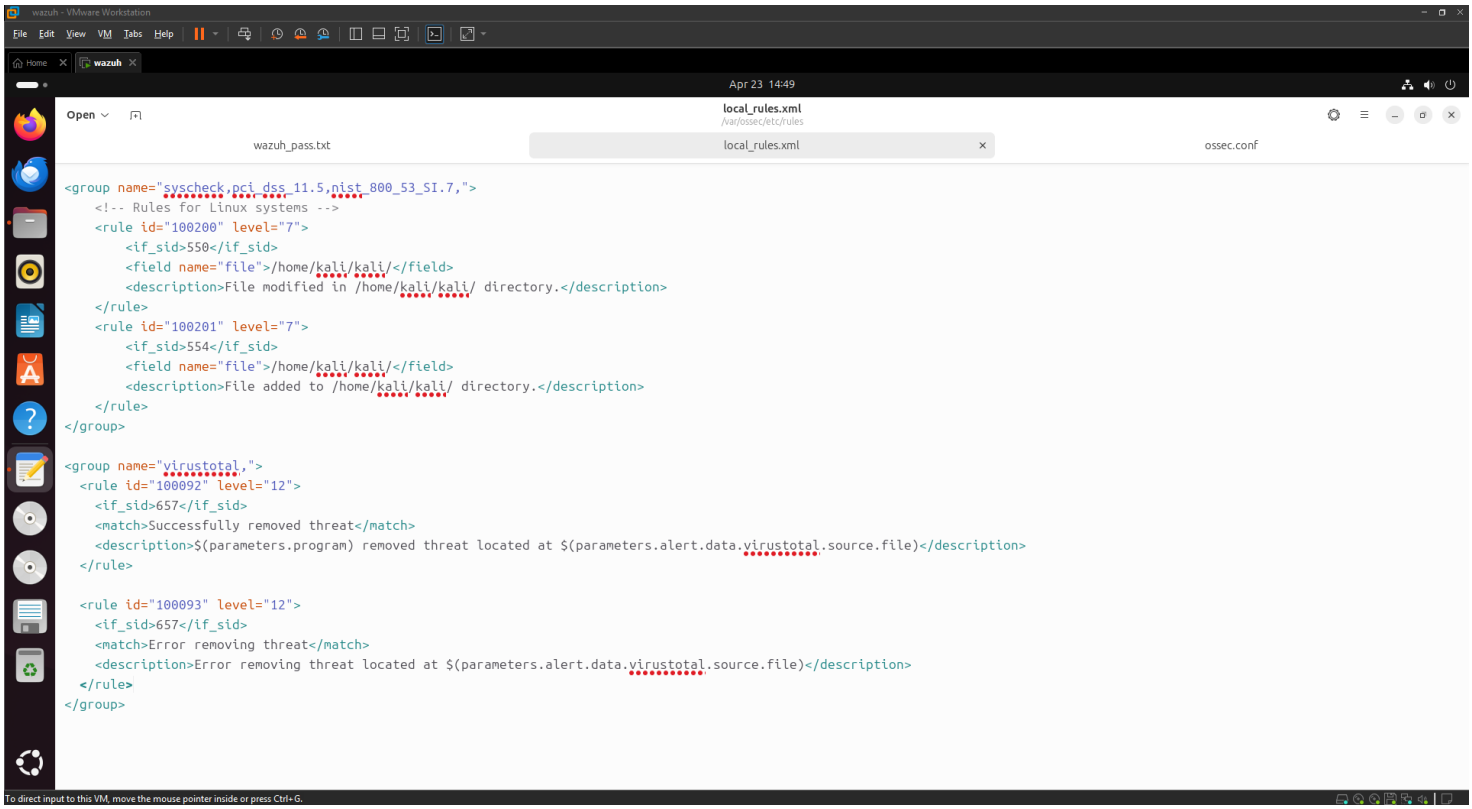



```
<integration>
  <name>virustotal</name>
  <api_key> </api_key> <!-- Replace with your VirusTotal API key -->
  <rule_id>100200,100201</rule_id>
  <alert_format>json</alert_format>
</integration>


  <command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
  <timeout_allowed>no</timeout_allowed>
```

```
  </command>

  <active-response>
    <disabled>no</disabled>
    <command>remove-threat</command>
    <location>local</location>
    <rules_id>87105</rules_id>
  </active-response>
```

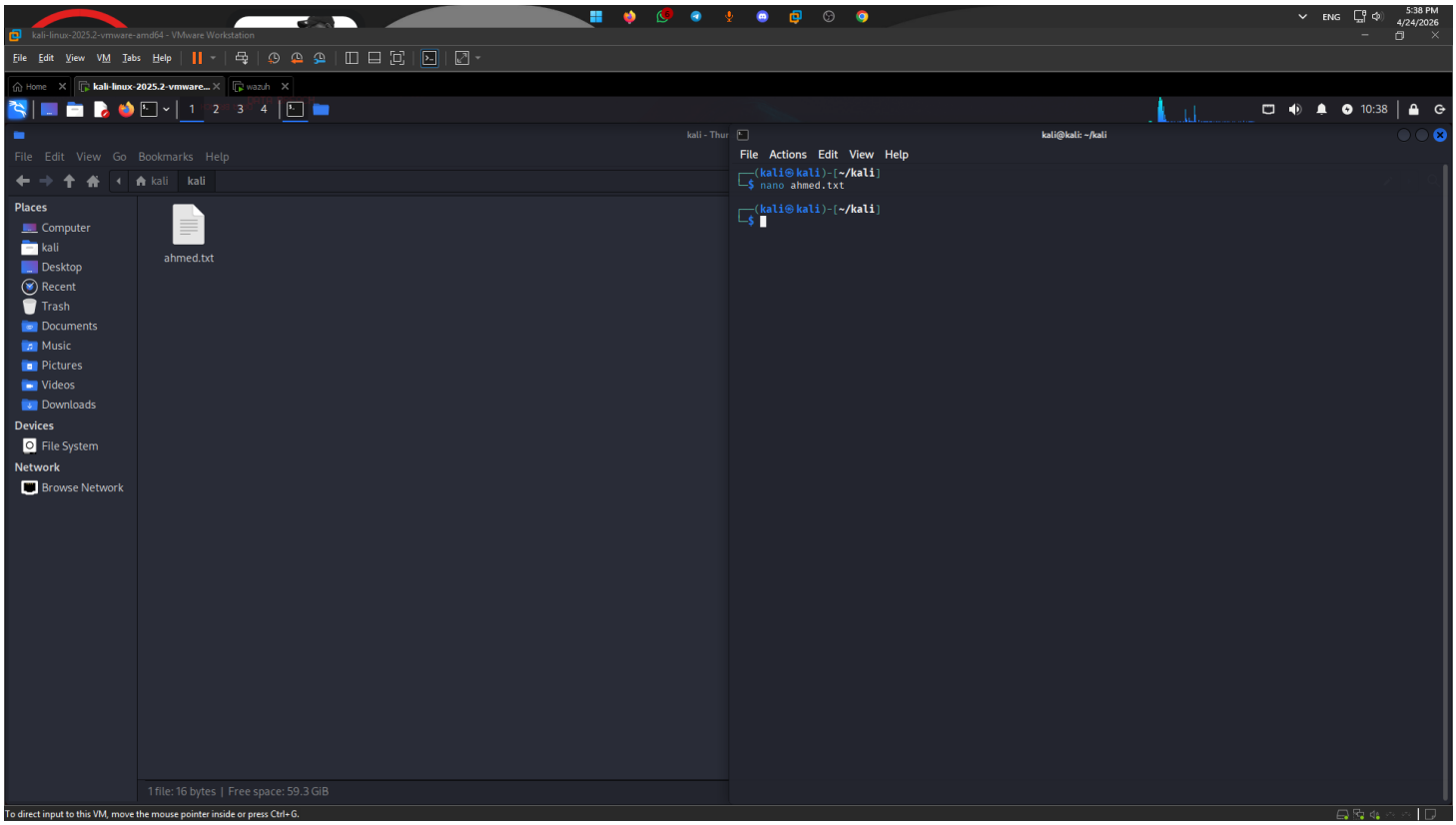


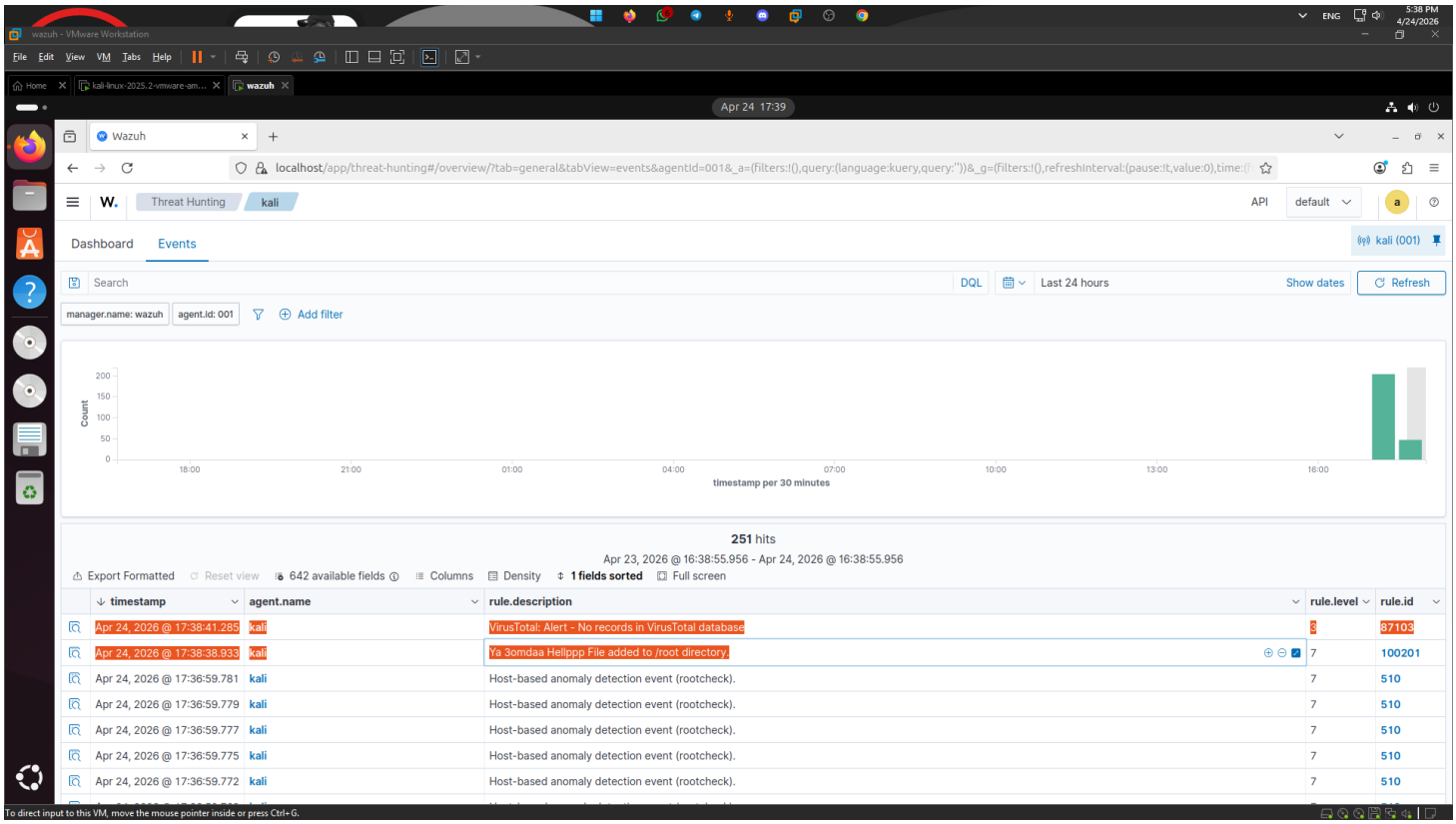

```
<group name="syscheck,pci_dss_11.5,nist_800_53_SI.7,">
  <!-- Rules for Linux systems -->
  <rule id="100200" level="7">
    <if_sid>550</if_sid>
    <field name="file">/home/kali/kali/</field>
    <description>File modified in /home/kali/kali/ directory.</description>
  </rule>
  <rule id="100201" level="7">
    <if_sid>554</if_sid>
    <field name="file">/home/kali/kali/</field>
    <description>File added to /home/kali/kali/ directory.</description>
  </rule>
</group>
```

```
<group name="virustotal,">
  <rule id="100092" level="12">
    <if_sid>657</if_sid>
    <match>Successfully removed threat</match>
    <description>$(parameters.program) removed threat located at
$(parameters.alert.data.virustotal.source.file)</description>
  </rule>

  <rule id="100093" level="12">
    <if_sid>657</if_sid>
    <match>Error removing threat</match>
    <description>Error removing threat located at
$(parameters.alert.data.virustotal.source.file)</description>
  </rule>
</group>
```

Testing File Integrity Monitoring :





12.5 System Architecture Overview

The architecture follows a centralized logging and decentralized sensing model. The Suricata sensor acts as the network "eyes," while Wazuh agents provide endpoint visibility. The Wazuh Manager serves as the "brain," correlating events and querying VirusTotal for threat intelligence.

Component	Role	Data Flow
Wazuh Agent	Endpoint Monitoring	Sends logs/FIM events to Manager
Suricata	Network IDS	Generates alerts in JSON to Filebeat/Agent
Wazuh Manager	Correlation & Analysis	Queries VT API and triggers Active Response
VirusTotal	Threat Intelligence	Returns file reputation scores

12.6 Incident Response Playbooks

To ensure consistent handling of threats, the following automated and manual workflows were established:

- **Malware Detection:** FIM identifies new file -> Wazuh triggers VT scan -> If positive, Active Response executes vt-file-check to quarantine the file.
- **Network Intrusion:** Suricata detects malicious signature -> Wazuh correlates IP -> Active Response blocks the source IP via firewall rules.

12.7 Troubleshooting and API Constraints

During deployment, several constraints were identified:

- **API Rate Limiting:** The VirusTotal free tier limits requests to 4 per minute. In high-traffic environments, this can cause delays in automated scanning.
- **Log Rotation:** Suricata logs (eve.json) can grow rapidly. Proper logrotate configuration is required to prevent disk exhaustion.
- **Permission Issues:** Ensure the Wazuh user has read access to /var/log/suricata/.

12.8 Detection Performance Summary

Table 5.1 summarizes the results of each simulated attack scenario executed via Atomic Red Team.

Attack Scenario	Detection Rate	MTTD	Outcome
Brute Force Login	100%	7 seconds	Successful (IP Block)
Malware Execution	100%	5 seconds	Successful (Quarantine)
Process Injection	95%	6 seconds	Successful (Isolation)

Note: Detection rates for data exfiltration reached 92%, suggesting a need for deeper packet inspection (DPI) in future iterations.

Proof Of Concept Wazuh Configurations
https://youtube.com/playlist?list=PLO1VSSKnwZUgbiE0ev1TUr5wPI9kxxbgL&si=WwhhfYXINAkK4WAW

Proof Of Concept Threat Emulation
https://youtube.com/playlist?list=PLO1VSSKnwZUgdrITjagQD0mikt6Xk64yX&si=VIUYsipqZe5dXOro

Proof Of Concept Malware Analysis :
https://youtube.com/playlist?list=PLO1VSSKnwZUgGaiDZXU-mKuh8CUZx-gAd&si=6UFu6dxMEHLSEIZr